*BY ORDER OF THE COMMANDER*
*NEVADA TEST AND TRAINING RANGE(ACC)*

*AIR FORCE MANUAL 13-212 VOLUME 1*

*AIR COMBAT COMMAND*
*Supplement*

*NEVADA TEST AND TRAINING RANGE*
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*RANGE PLANNING AND OPERATIONS*

**COMPLIANCE WITH THIS PUBLICATION IS MANDATORY**

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This addendum complements AFMAN 13-212, Volume 1, *Range Planning and Operations*, and complements that manual as well as AFMAN 13-212V1 ACC Supplement; it supersedes AFMAN13- 212V1_ACCSUP_NTTRSUP (23 Jul 20). It governs operations on the Nevada Test and Training Range (NTTR) and provides guidance for all range operations and applies to all range users. This Addendum requires the collection and maintenance of information protected by the Privacy Act (PA) of 1974. The authority to collect and/or maintain the records prescribed in this instruction is 37 USC 301a. Privacy Act system notice F036 AF PC C, Military Personnel Records System, applies. All applicable Air Force Manuals (AFMAN), Federal Aviation Administration (FAA) Orders, and Federal Aviation Regulations (FAR) apply. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Manual (AFMAN) 33-363, *Management of Records*, and disposed of IAW Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS) located at **https://www.my.af.mil/gcss- af61a/afrims/afrims/**. There are no waiverable provisions within this publication. Send comments and/or change recommendations via AF Form 847, *Recommendation for Change of Publication*, to NTTR/DO,

3770 Duffer Dr., Nellis AFB NV 89191-7001. Range users and sponsoring agencies may amplify this Addendum with more restrictive guidance, as necessary, in order to provide direction to their organizations.

***SUMMARY OF CHANGES.***

This publication has been significantly revised and must be reviewed in its entirety. Changes include but are not limited to: the southern range re-draw, NTTR fire recall procedures, incorporating numerous range safety amendments, and updates to the NTTR Business Management Plan.

# Chapter 1

# RESPONSIBILITIES

**1.1. General Information.** The Nevada Test and Training Range (NTTR) is designated a Major Range and Test Facility Base (MRTFB) resource by Department of Defense (DoD) Directive 3200.11, Major Range and Test Facility Base. The range is the United States' largest and most sophisticated instrumented battlespace for testing and training for US military and coalition forces. The range comprises 2.9 million acres of ground space and approximately 12,000 square nautical miles (NM) of airspace. Headquarters (HQ) NTTR (a named USAF unit responsible for managing the range) manages the range, working in close collaboration with the entire spectrum of range users. HQ NTTR provides an integrated battlespace, supporting air, space, and cyberspace testing, advanced training, and tactics development via live, virtual, and constructive environments.

1.1.1. IAW FAA Order JO 7400.8D, the Nevada National Security Site (NNSS) Nevada Field Office (NFO) has been designated as the Using Agency of R-4808, which overlies the NNSS. Additionally, IAW the 1981 Memorandum of Understanding between the Department of the Air Force (DAF) and Department of Energy (DOE), the agencies agree to the cooperative use of the land and airspace of the NNSS. NTTR users who wish to use R-4808 airspace will coordinate their desires through the NTTR Scheduling Process to obtain NNSS/NFO and DOE approval.

1.1.1.1 IAW FAA Order JO 7400.8D, the USAF Warfare Center has been designated as the Using Agency for R-4806, R-4807, and R-4809.

1.1.1.2. The lands that comprise the geographical NTTR were last renewed as a congressional withdrawal under the Military Lands Withdrawal Act of 1999 (PL 106-65, ) as amended by subsection 2862(d) of the National Defense Authorization Act For Fiscal Year 2024 (hear in after “FY24 NDAA”)(Pub. L. 118-283). Off range there are major communication sites to include Angel Peak, Highland Peak, Mt Ella and Mt Irish.

1.1.1.3. Under a 2019 DAF permit, the NNSA/Sandia Field Office (SFO) has been granted non-exclusive use, operation, and occupancy of specified portion of the NTTR. The real property defined is referred as Premises with a Coordinated Use Area. A 13 Nov 19 Operating Agreement (OA) defines SFO combined operations and AF support. SFO supports Joint Test Assembly (JTA) testing for the nuclear stockpile stewardship mission. NNSA/NA-10 and AF/A10 have a MOA titled the Joint Testing and Assessment of The Nuclear Weapons Stockpile, dated 13 Aug 12, defining JTA testing on the NTTR.

1.1.2. Authority over the land and airspace which encompasses the NTTR is delegated to three organizations on Nellis AFB:

1.1.2.1. The NTTR Commander is designated the Range Operating Authority (ROA) and is responsible for leading overall range management. This includes scheduling, operating and maintaining range equipment, and maintaining the safety of personnel.

1.1.2.2. The Commander, 99th Air Base Wing (99 ABW) is designated the Installation Command Authority (ICA). The 99 Civil Engineer Squadron (99 CES) provides real property management as the Base Civil Engineer (BCE), as well as Explosive Ordnance Disposal (EOD) support to the range.

1.1.2.3. The 57th Operations Support Squadron Airspace Management Office (57 OSS/OSOP) is responsible for airspace management.

**1.2. HQ NTTR Organization.**

1.2.1. Commander, NTTR (NTTR/CC). NTTR/CC was designated the Range Operating Authority (ROA) for the NTTR in AFMAN13-212V1, para 2.9. The ROA is responsible for leading overall range management. This includes operating and maintaining range equipment, maintaining safety of personnel and material resources. The scope of the geographical range area includes lands internal to the NTTR boundary and several off- range communication sites.

1.2.2. The ROA coordinates the NTTR's Special Use Airspace with the 57 WG. The ROA manages range activities for Air Combat Command (ACC) as institutionally funded and as project support as an MRTFB through user funded means.

1.2.3. HQ NTTR Organization.

1.2.3.1. NTTR Plans and Programs Office (NTTR/XP) is responsible for advocating for long-term range requirements in development of the Program Objective Memorandum (POM) with HQ ACC. NTTR/XP ensures the NTTR is able to meet long-term test and training requirements as development and integration by working with multiple using activities and higher headquarters. NTTR/XP serves as the range liaison for agreements, land use, encroachment and environmental with 99 ABW units and off-base organizations. NTTR/XP works closely with state, county and federal agencies and notably focuses range coordination with the Bureau of Land Management (BLM) and the US Fish and Wildlife Service (USFWS).

1.2.3.2. NTTR Security Office (NTTR/SD) is responsible for NTTR collateral security, access control, and SCI access. NTTR Advanced Program Office (NTTR/AP) is responsible for Advanced program security. NTTR/AP establishes procedures and assigns responsibilities governing advanced program matters and coordinates with ACC/A58Z and other AP offices.

1.2.3.3. NTTR Safety Office (NTTR/SE) is responsible for overall range safety policies, developing Range Safety Assessments (RSA) for unique and emerging range uses, supporting the Weapon Danger Zone (WDZ)/Surface Danger Zone (SDZ) processes, managing laser, electromagnetic frequency and ionizing radiological safety, conducting safety review boards (SRB), and leading NTTR safety investigations.

1.2.3.4. NTTR Financial Management Office (NTTR/FM) is responsible for NTTR budgetary activities. NTTR/FM manages the Job Order Cost Accounting System (JOCAS) II interface and prepares wing-level budget documents. NTTR/FM assigns a Job Order Number (JON) that corresponds to the user's specific activity. Project coordination and funding (if applicable) will be conducted IAW the NTTR Business Management Process (BMP). NTTR/FM is the OPR for the NTTR Products and Services Catalog.

1.2.3.5. NTTR Program Management Office (NTTR/PM) is responsible for program management of the Range Support Services (RSS) contract across multiple sites (to include NTTR and Creech AFB). NTTR/PM oversees all aspects of RSS contract execution, including cost, schedule, performance, and proposals estimate coordination. NTTR/PM serves as the primary conduit to both the Contracting Officer and the RSS contractor.

1.2.4. NTTR Range Support Directorate (NTTR/RS) is responsible for program management of the Joint Range Technical Services Contract (JTECH) II contract. NTTR/RS functions as the primary conduit between the Contracting Officer and the JTECH II contractor for NTTR JTECH functions, while the JPMO at Edwards AFB serves as the PM for the contract. NTTR/RS also manages NTTR Threat Systems (radio callsign ROULETTE), Air Combat Training System P5 pods, debriefing systems, communications, engineering support, and (through the NTTR Spectrum Management Office (SMO)) radio frequency (RF) spectrum management (IAW AFI 17-220 and/or CJCSM 3212.02). Threat scheduling is managed by the JT4 Threat Schedulers (NTTR Resource Scheduling).

1.2.5. NTTR Mission Support Directorate (NTTR/MS) is responsible for logistics support (supply, transportation, and fuel) and civil engineering management (operations, construction, and infrastructure) on the NTTR. NTTR/MS executes oversight and evaluation of contractor operations for the RSS contract.

1.2.6. NTTR Operations Directorate (NTTR/DO) is responsible for daily range management, planning, and maintenance management for the NTTR and is the focal point for operational requirements and flight safety (including airspace deviation processing). The Director of Range Operations leads NTTR/DO and is designated as the Range Operations Officer (ROO). The ROO serves as the Scheduling Authority (SA) for flight and ground operations on NTTR. The ROO is the NTTR deviation and waiver authority for this Addendum. Key organizational sections include:

1.2.6.1. NTTR Projects Office provides a procedural framework for the planning of range support to range events that are managed by HQ NTTR. As the HQ NTTR lead for the Business Management Process (BMP), the NTTR Projects Office provides the initial entry point for use of the NTTR. Project Managers support range users by serving as the primary point of contact (POC) between users and all support agencies participating in the NTTR BMP.

1.2.6.2. NTTR Range and Resource Scheduling schedules activities on the NTTR and monitors range utilization rates. Range Scheduling utilizes the Center Scheduling Enterprise (CSE) software to collaborate with a wide variety of range stakeholders in scheduling NTTR activities.

1.2.6.3. NTTR Target Management Office oversees target configuration, distribution, maintenance, remediation and ongoing use of range targets, including two Cluster Bomb Unit (CBU) targets and one Depleted Uranium (DU) strafe target. Additionally, NTTR Target Management Office collaborates with the range support contractor to manage multiple target storage yards, including multiple DU storage yards. Finally, Target Management Office supports the WDZ/SDZ processes.

1.2.6.4. NTTR Range Monitoring (i.e., BLACKJACK) is a contractor-staffed, range safety monitoring section that provides real-time range scheduling (day prior and day of activity), ground party access approval for scheduled ground activities, personnel recovery coordination, and special test assistance.

1.2.6.5. NTTR Current Operations is responsible for the day-to-day management of the NTTR Range Operations Center (ROC). Specifically, through the NTTR Joint Data Network Cell (JDNC), it is responsible for all data link operations in support of NTTR test, tactic development, and advanced training events. NTTR Current Operations also submits Satellite Access Requests (SARs), when requested.

1.2.7. The NTTR hosts a Joint Staff/J7 provided JIOR Service Delivery Point (SDP) within the NTTR ROC. The JIOR SDP provides support to cyberspace and multi-domain testing, training, and tactics development.

**1.3. 99th Air Base Wing.**

1.3.1. The commander, 99th Air Base Wing (99 ABW/CC) is designated the Installation Command Authority (ICA) for DoD lands withdrawn from public for use as the NTTR.

1.3.2. 99th Civil Engineer Squadron (99 CES) provides real property management support as the Base Civil Engineer (BCE), with NTTR/MS overseeing contracted civil engineer operations under the NTTR Range Support Services contract. Additionally, 99 CES provides range Explosive Ordnance Disposal (EOD) support to the range. Environmental coordination is supported by the 99 CES in collaboration with the Air Force Civil Engineering Center (AFCEC) and AF Installation and Mission Support Center (AFIMSC). Specific BCE responsibilities are further articulated in 99 ABW, AFCEC, and/or AFIMSC plans and publications.

**1.4. 57th Wing (57 WG).**

1.4.1. 57th Operations Group (57 OG) directs and executes Air, Air Defense, Space, and Information Aggressor tactics on the NTTR.

1.4.2. 57th Operations Support Squadron Weather Flight (57 OSS/OSW) is the focal point for Nellis AFB weather services and is supported, via memorandum of agreement, by 25th Operations Weather Squadron (25 OWS).

1.4.3. 57th Operations Support Squadron Airspace Management Office (57 OSS/OSP) is responsible, in conjunction with the FAA, for investigating deviations on the NTTR that involve the National Airspace System, as well as other airspace management functions as described in AFI 13-201, Airspace Management.

1.4.4. 57th Operations Support Squadron Airfield Operations (57 OSS/OSA) manages Nellis AFB air traffic control through the Nellis Tower and the Nellis Air Traffic Control Facility (NATCF), which encompasses both Nellis Approach and Nellis Control.

1.4.5. 57th Operations Support Squadron Scheduling (57 OSS/OSOS). 57 OSS/OSOS oversees the Nellis AFB operations scheduling process and is the focal point for operations scheduling related issues. Additionally, the 57OSS/OSOS is responsible for operating and administering Patriot Excalibur (PEX) scheduling software.

**1.5. 432nd Wing (432 WG).**

1.5.1. 432nd Mission Support Group (432 MSG) provides base support for Creech AFB, including controlling and managing ground access to, and providing integrated defense for, Creech AFB.

1.5.2. 432nd Security Forces Squadron (432 SFS) provides physical security assistant support as required to the NTTR. The Southern Range Base Defense Operations Center (BDOC-Contract Security) facilitates access to the southern ranges via the By-pass Gate (providing transit between by-pass road and Mile Range Road) and the Box Canyon Road.

1.5.3. 432d Operations Support Squadron Weather Flight (432 OSS/OSW) is the focal point for Creech AFB weather services.

1.5.4. 432d Operations Support Squadron Airfield Operations (432 OSS/OSA) manages Creech AFB air traffic control through the Creech Tower.

1.5.5. 432d Operations Support Squadron Current Operations (432 OSS/OSO). 432 OSS/OSO oversees Creech AFB operations scheduling along with the 25th ATKW Operations Support Squadron Operating Location Charlie (25 OSS/OLC). Both units are the focal points for operations scheduling related issues. Additionally, both units are responsible for operating and administering Patriot Excalibur (PEX) scheduling software.

**1.6. Tonopah Test Range Operations Office (TTRO).** TTRO acts as a liaison between the Air Force Test Center (AFTC) and HQ NTTR. TTRO manages the Tonopah Test Range (TTR) installation and airfield services, as well as providing physical security support to the northwestern portion of the NTTR.

**1.7. NNSS.** NNSS is responsible for safety, security, and scheduling of activities in R4808 at all altitudes, and Pahute Alpha and Bravo at or below 500’ Above Ground Level (AGL).

1.7.1. NNSS Air and Ground Operations Support (AGOS) callsign “Birddog” is responsible for real time range scheduling/deconfliction for the NNSS.

**1.8. Sponsoring Agencies.** In addition to HQ NTTR, DOE, NNSS/NFO, and NNSA/SFO act as key sponsoring agencies for range activities.

**1.9. Range Users.** The range may be used by DoD (both local and off-station) units, other US government agencies, state and local governments, allied foreign governments, and commercial users. All users require expressed HQ NTTR approval to operate on range.

1.9.1. Local units. 57 WG, 432 WG, and Nellis/Creech AFB-assigned tenant units are considered local units.

1.9.2. Off-station units. IAW NAFBI 11-250 and CAFBI 11-250, a local host unit must be assigned and identified to facilitate range support for the off-station organization requesting range activity and launching from or recovering to Nellis/Creech AFB. Off-station requestors who will not launch from or recover to Nellis/Creech AFB are not required to have a host unit; however, their request must be managed through the NTTR BMP.

1.9.2.1. Off-station ground parties are required to have a local area briefing from a local host unit covering NTTR procedures prior to range activity. Deployed unit commanders will ensure their host unit is informed of any additional events the deployed unit will accomplish outside of their scheduled, approved activity. Nellis host unit will provide an entry access list (EAL) identifying all Foreign National participants in ground activities on the NTTR to NTTR/DO no later than 7 days prior to range activity. All off-station ground party range activity must be managed through the NTTR BMP.

1.9.3. Use of the range by other US government agencies, state and local governments, allied foreign governments, and commercial users is carefully managed through the NTTR BMP.

1.9.3.1. Detailed organizational, programmatic, and operational information must be obtained from any prospective commercial user in order to assess any conflicts with NTTR core activities. Developmental commercial activity, in particular, will be subjected to rigorous review through the NTTR BMP.

1.9.3.2. HQ NTTR will accomplish commercial organizational vetting, review proximity to sensitive activities, and assess possible Electromagnetic Interference (EMI)/Radio Frequency Interference (RFI) encroachment through the NTTR BMP.

1.9.4. Prior to operating on the range, all users must obtain the NTTR Orientation Briefing from NTTR/SD and conducted in the HQ NTTR building. Local host units are responsible for ensuring all off-station, hosted units receive the briefing. Off-station units may obtain the briefing from the video link on the NTTR Mission Planning website.

**1.10. Scheduling Policy.** Competition for limited range resources is an ever-present fact on the NTTR. In addition to testing, training, and tactics development uses, range infrastructure maintenance, target maintenance, and EOD operations are fundamental range requirements that must be accommodated. These fundamental activities will be accomplished IAW applicable directives and as required to provide a safe and secure range environment. These fundamental requirements are carefully balanced with user requirements to ensure effective and efficient availability for all range users.

1.10.1. Scheduling Priorities. The following prioritized list, while authoritative, is not directive and requires judgment in application. Nevertheless, it provides range users and range decision makers a standard frame of reference with which to balance competing requirements. Scheduling will be accomplished through the NTTR Scheduling Process and scheduling conflicts will be resolved through NTTR Conflict Adjudication Process. Use of the range is prioritized in accordance with table 1.1.

**Table 1.1. NTTR Range Priorities.**

| |
|---|
| Priority 1. DOE. |
| Priority 2. NNSS/NFO. |
| Priority 3. NNSA/SFO. |
| Priority 4. DoD higher headquarters-directed, priority uses including USAF operational testing of Office of the Secretary of Defense (OSD) Test and Evaluation (T&E) Oversight List systems; Red Flag; USAF Weapons School Integration Phase (WSINT); Black Flag; Marine Aviation Weapons and Tactics Squadron One (MAWTS-1) Strike Three; Neptune Series Exercises, and other LFEs as determined by the NTTR/DO. |
| Priority 5. Range Maintenance. |
| Priority 6. ACC Test Priority Order (TPO), exclusive of operational testing of OSD T&E Oversight List systems. |
| Priority 7. ACC formal training syllabus (exclusive of WSINT). |
| Priority 8. Other Nellis AFB and Creech AFB testing and operational training. |
| Priority 9. Other DoD uses. |
| Priority 10. Other US Government agency, state, and local government uses. |
| Priority 11. Allied military uses. |
| Priority 12. Commercial uses. |

1.10.2. In addition to the above prioritized list, statutorily “mandated” driven uses (such as the Stonewall Mountain and Desert National Wildlife Refuge (DNWR) Bighorn Sheep hunts, BLM Horse census and gathers, and USFWS bighorn sheep census and guzzler maintenance flights) will be supported and determined Priority 1 when deemed necessary by the ROA with full collaboration with appropriate counterparts.

1.10.3. 432 SFS normally has first priority among DoD users for firing range 1-2 use on R-67A.

1.10.4. Late emerging requirements (i.e., post-NTTR Range War meeting) will be entertained to the extent feasible; however, the priority that would otherwise be attendant cannot be guaranteed. Late emerging requirements will be considered through the NTTR Scheduling Adjudication Process.

## Chapter 2

## DESCRIPTION OF RANGE AND MILITARY OPERATING AREAS

**2.1. Basic Boundaries.** The NTTR is comprised of Reveille North, Reveille South, and the Desert Military Operations Areas (MOA), Air Traffic Control Assigned Airspace (ATCAA), and restricted areas R-4806, R-4807, and R-4809. It provides Class B and Class C ranges that include Air-to-Air, Air-to-Surface, Surface-to-Air, and Electronic Combat with Air Combat Training System instrumentation. Access to each area is allocated IAW the daily range schedule. Since R-4808 overlies the NNSS, NTTR users who wish to use R-4808 airspace will coordinate their desires through the NTTR Scheduling Process to obtain NNSS/NFO and DOE approval.

**2.2. Ground Boundaries.** The NTTR's ground boundaries do not match airspace boundaries. The NTTR's perimeter only aligns with its airspace in a few locations. Consult the NTTR Mission Planning webpage for specific information.

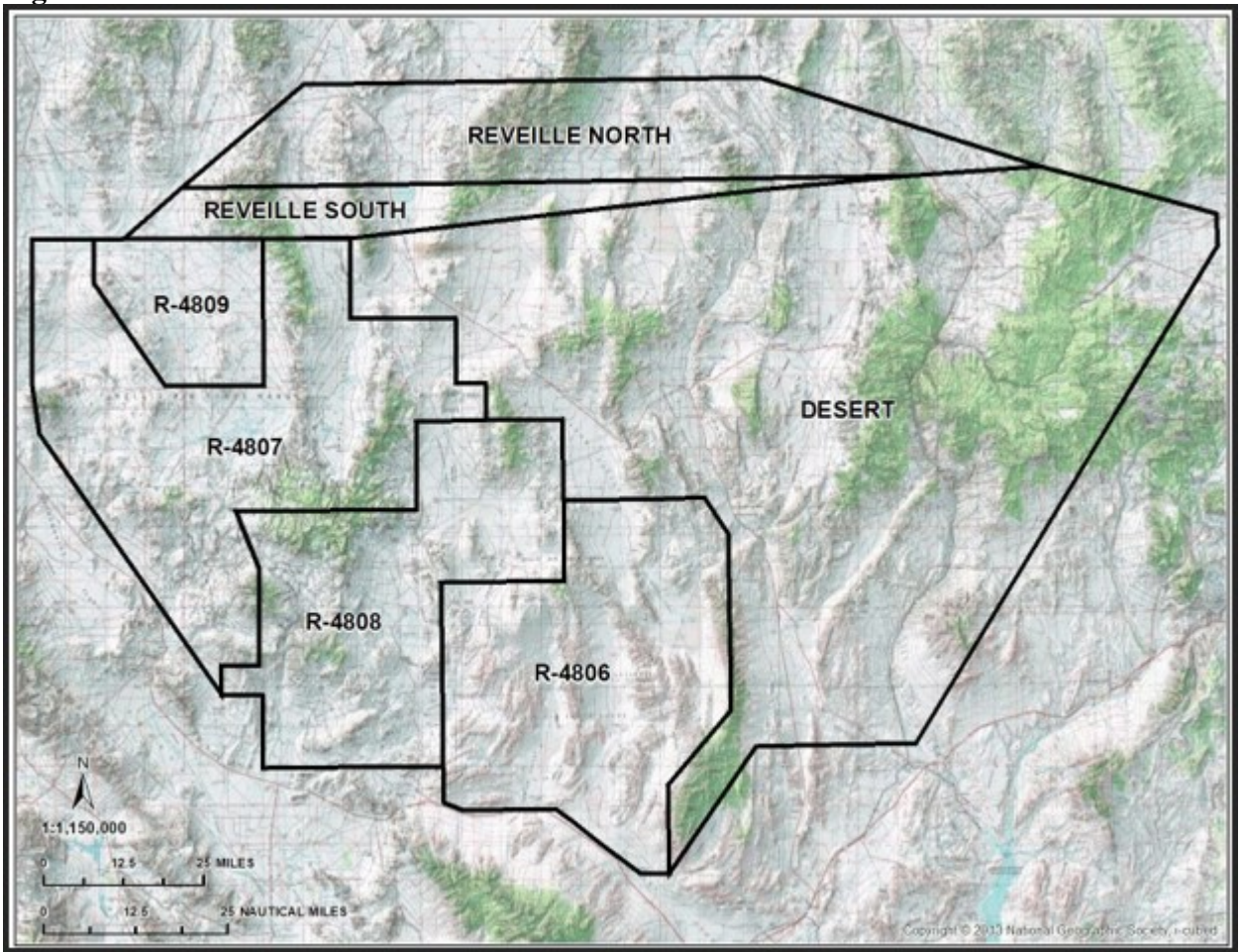
**2.3. Management Agreements.** The NTTR is managed under multiple cooperative agreements comprising the USAF, the Bureau of Land Management (BLM), US Fish and Wildlife Service (USFWS), and the Department of Energy (DOE). The jointly managed lands have some unique rules and regulations for each area. The 4808s are administered by the Nevada National Security Site (NNSS) owned by DOE. In the north Ranges (commonly refereed to as the 70s) BLM has ownership, while in the South Ranges (commonly referred to as the 60s) the USFWS has ownership and primary jurisdiction. The South Ranges have additional restrictions due to their administration by USFWS. Some of the Range's notable ground restrictions are as follows:

2.3.1. The southern ranges are owned by USFWS and co-managed with DAF under a MOU. The southern ranges are also the Desert National Wildlife Refuge (DNWR) and legislative proposed wilderness in the joint use area managed by USFWS. Although the land is withdrawn for USAF use, 730k of the 842k acres is generally off limits to military ground activities. USAF can normally utilize the 112k acres in the South Range also known as the Impact Areas. Any ground activity outside those 112k acres is significantly restricted and must be in accordance with the MOU between DAF and USFWS. For information regarding the MOU and approved uses outside the 112k impact areas contact NTTR/XP.

2.3.2. All ground activity within NNSS's boundaries (R4808) is tightly controlled and must be coordinated through NTTR/XP who will coordinate with NNSS BirdDog Ops for access.

2.3.2.1. NNSA/NFO is responsible for safety and security associated with R4808 and Pahute A/B ground space.

**Figure 2.1. Basic Boundaries.**




NOTE: R-2508 and the Leach Lake Training Range which lies to the southwest of the NTTR, is not part of the NTTR complex and is not scheduled through the NTTR Scheduling Process.

**2.4. NTTR Subranges**. To ease scheduling and deconfliction, the basic range boundaries are further subdivided into a series of more-easily managed subranges.

**Figure 2.2. NTTR Subrange Boundaries.**

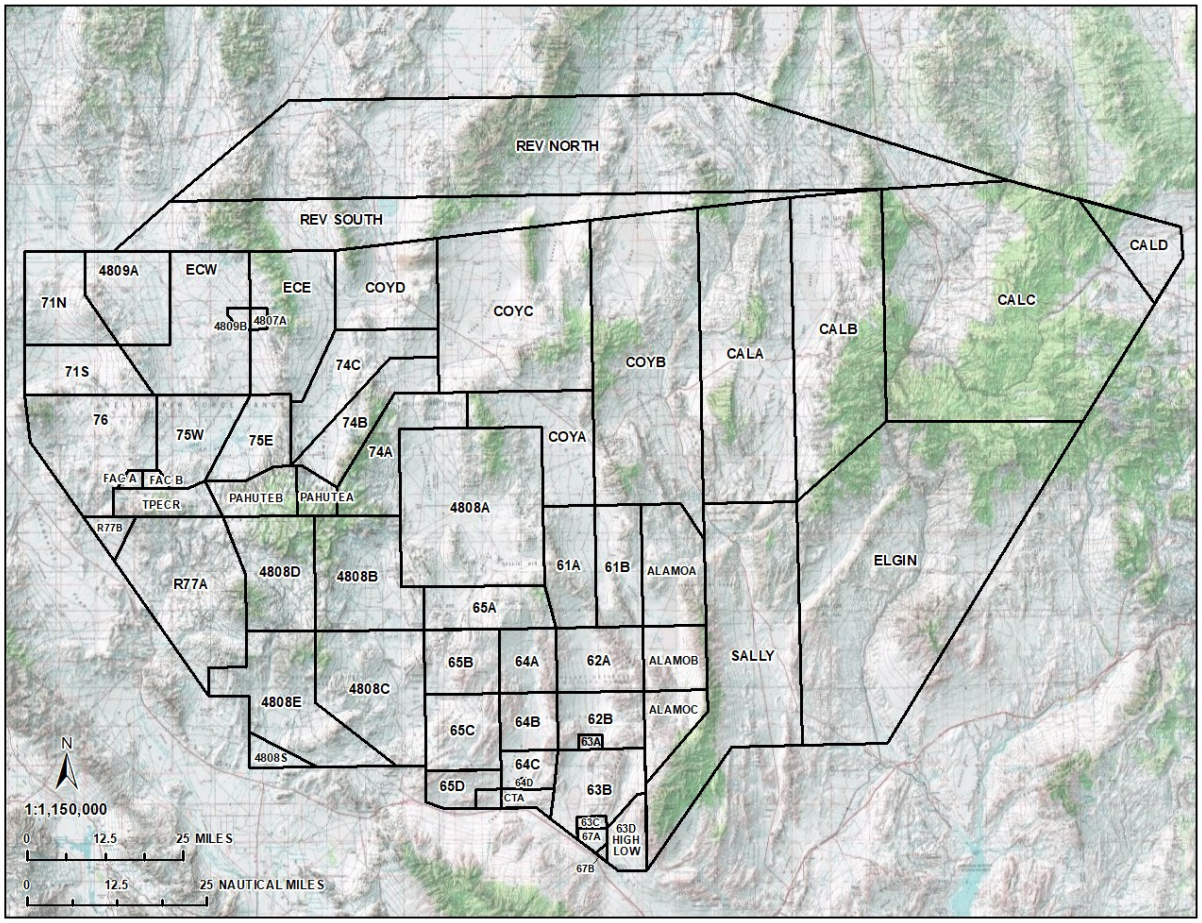


**Table 2.1. FAA Airspace Designations and Associated Subranges.**

| FAA Airspace Designation | FAA Sub-Airspace Designation | NTTR Subrange Designations |
|---|---|---|
| R-4806 | R-4806E | Alamo A/B/C[3] |
| | R-4806W[2,3] | R-61A/B |
| | | R-62A/B |
| | | R-63A/B/C/D |
| | | R-64A/B/C/D |
| | | R-65A/B/C/D |
| | | R-67A/B |
| R-4807 | R-4807A | R-71N/S[4] |
| | | R-74A/B/C |
| | | R-75E/W |
| R-4807 | | R-76[4] |
| | | R77A/B |
| | | R-4807A (Subset) |
| | | EC East[5] |
| | | Tolicha Peak Electronic Combat Range (TPECR)[6] |
| | R-4807B | Pahute Mesa A/B |
| R-4808 | R-4808N | R-4808A[7]/B/C/D[8]/E |
| | R-4808S | N/A |
| R-4809 | EC West | EC West |
| | R-4809A | R-4809A[5,9] |
| | R-4809B | R-4809B |
| Reveille North MOA | Reveille North MOA and ATCAA | Reveille North |
| Reveille South MOA | Reveille North MOA and ATCAA | Reveille South[10] |
| Desert MOA[1] | Desert MOA and ATCAA | Coyote A/B/C/D |
| | | Caliente A/B/C/D |
| | | Elgin |
| | | Sally Corridor |

Note 1: Although not part of the Desert MOA, an area to the southeast section of the MOA is identified as the "X-Ray Extension."
Note 2: Although not specified subranges, Creech Tower Airspace (CTA) lies within R-4806W as well as two USAF Aerial Demonstration Squadron (Thunderbirds) training areas.
Note 3: Although not a specified subrange, a corridor within Alamo A/B/C (FL190 to FL210) is locally identified as "Alamo Corridor."
Note 4: A No Impact Zone (NIZ) area lays within R-71S and R-76 and may be manned.
Note 5: Although not specified subranges, three NNSA/SFO supersonic target entry cones lie within EC East and R-4809A.
Note 6: For scheduling and supersonic operations purposes, TPECR includes FAC A/B ground/airspace.
Note 7: R4808A is not a schedulable sub-range.
Note 8: Although not a specified subrange, an area in the north of R-4808D (above 2,500ft AGL) is locally identified as the "Gomer Triangle."
Note 9: Although not specified subranges, Silverbow Tower Airspace (STA) lies within R-4809A as well as an area south of N37 42.90 (above 500ft AGL) which is locally identified as the "Shelf."
Note 10: Although not a specified subrange, an area in the west of Reveille South is locally identified as the "Dorito."

**2.5. Military Assumes Responsibility for Separation of Aircraft (MARSA).** Military aircraft operating on the NTTR normally operate under MARSA rules and will use the Nellis AFB local altimeter setting. MARSA applies only between participating military aircraft and does not apply for separation from civil aircraft (except contractor aircraft participating in a DoD event). NATCF will not clear civil and military aircraft operating under IFR into a MOA and ATCAA unless they can provide IFR separation from participating aircraft and have NTTR Range Monitoring coordination. Advisory control to aircraft operating in NTTR-scheduled airspace is provided by Ground Control Intercept (GCI) controllers.

**Table 2.2. Routine Civil Air Presence.**

| FAA Airspace Designation | Civil Presence |
|---|---|
| R-4806E | Designated as Joint-Use through agreement with the FAA; available for non-military aircraft transition under NATCF control when not activated for NTTR-scheduled use. |
| R-4806W | Designated as Non Joint-Use through agreement with the FAA; not available for civil aircraft. |
| R-4807 | Designated as Joint-Use through agreement with the FAA; available for non-military aircraft transition under NATCF control when not activated for NTTR-scheduled use. |
| R-4808 | Designated as Non Joint-Use through agreement with the FAA; not available for civil aircraft. |
| R-4809 | Designated as Non Joint-Use through agreement with the FAA; not available for civil aircraft. |
| Reveille North MOA | IAW FAA regulations, civil air traffic may be present. |
| Reveille South MOA | IAW FAA regulations, civil air traffic may be present. |
| Desert MOA | IAW FAA regulations, civil air traffic may be present. |

**2.6. Air Traffic Control (ATC).** Controlling Agency responsibilities for the NTTR are designated IAW FAA Order JO 7400.8Y. Key ATC agencies include Los Angeles Air Route Traffic Control Center (LA ARTCC) and Salt Lake City ARTCC (SLC ARTCC), Dreamland Approach Control (DML), and NATCF.

2.6.1. Specific NTTR ATC responsibilities are further delegated and designated through Letters of Agreement (LOAs) and Certificates of Authorization (COAs).

2.6.2. In addition to NTTR Range Monitoring, NNSS/NFO accomplishes independent R-4808 range monitoring through Birddog Operations (callsign BIRDDOG). Birddog Operations may be contacted at (702) 295-2467.

**Table 2.3. ATC Responsibilities.**

| FAA Airspace Designation | FAA Sub-Airspace Designation | NTTR Subrange Designations | Controlling Agency[1] |
|---|---|---|---|
| R-4806 | R-4806E | Alamo A/B/C | NATCF |
| | R-4806W | R-61A/B | NATCF |
| | | R-62A/B | NATCF |
| | | R-63A/B/C/D | NATCF |
| | | R-64A/B/C/D | NATCF |
| | | R-65A/B/C/D | NATCF |
| | | R-67A/B | NATCF |
| R-4807 | R-4807A | R-71N/S | NATCF |
| | | R-74A/B/C | NATCF |
| | | R-75E/W | NATCF |
| | | R-76 | NATCF |
| | | EC East | NATCF |
| | | R-4807A (Subset) | NATCF |
| | | R77A/B | NATCF |
| | | TPECR | NATCF |
| | R-4807B | Pahute Mesa A/B | NATCF |
| R-4808 | R-4808N | R-4808A/B/C/D/E | DML |
| | R-4808S | R-4808S | LA ARTCC: FL280 and above NATCF: 11,000ft MSL to FL270<br>DML: Surface to 10,000ft MSL |
| R-4809 | EC West | EC West | NATCF |
| | R-4809A | R-4809A | NATCF |
| | R-4809B | R-4809B | NATCF |
| Reveille North MOA | Reveille North MOA and ATCAA | Reveille North | SLC ARTCC: 100ft AGL to FL600[2] |
| Reveille South MOA | Reveille South MOA and ATCAA | Reveille South | SLC ARTCC: 100ft AGL to FL600[2] |
| Desert MOA | Desert MOA and ATCAA | Coyote A/B/C/D | NATCF |
| | | Caliente A/B/C/D | NATCF |
| | | Elgin | NATCF |
| | | Sally Corridor | NATCF |

Note 1: DML serves as the controlling agency for aircraft operating in DOE-scheduled airspace that is otherwise NATCF controlled. NATCF, by LOA, provides ATC services for NNSS/NFO when DML is unavailable.

Note 2: NATCF serves as the controlling agency for aircraft operating in NTTR-scheduled Reveille North and Reveille South airspace that is otherwise SLC ARTCC controlled.

**2.7. Concurrent-Use Operations.** Defined as two or more military air operations performed at the same time and scheduled in the same NTTR range space. Participating flight leads assume equal responsibility for aircraft separation and weapons delivery deconfliction amongst all participating aircraft and ground parties.

2.7.1. To provide close air support training, ground parties are placed in close proximity to impact areas at pre-surveyed Observation Posts (OPs). To prevent weapons deliveries on manned sites, aircrew working in the vicinity of an OP must be familiar with all safety restrictions/ considerations and adhere to planned weapons delivery parameters.

**Table 2.4. Pre-Surveyed Ops (DD MM.MM).**

| OP | Coordinates | | | OP | Coordinates | |
|---|---|---|---|---|---|---|
| Alpha | N37 21.83 | W116 47.83 | | Seahawk | N37 21.95 | W116 52.28 |
| Bravo | N37 21.42 | W116 42.33 | | Sierra | N36 54.32 | W115 35.79 |
| Charlie | N37 33.09 | W116 16.87 | | Steeler | N37 28.22 | W116 22.45 |
| C3 | N36 51.31 | W115 28.50 | | Viking | N37 26.57 | W116 49.60 |
| Delta | N37 25.41 | W116 34.21 | | Viv | N36 53.50 | W115 37.52 |
| Hill | N36 39.81 | W115 46.88 | | Whisky | N36 46.59 | W115 25.82 |
| Jack | N36 45.12 | W115 26.22 | | X-Ray | N36 46.84 | W115 26.42 |
| Jill | N36 37.30 | W115 31.53 | | Zulu | N37 23.07 | W116 50.11 |
| Romeo | N36 49.63 | W115 28.28 | | | | |

2.7.2. Participating Joint Terminal Attack Controllers (JTAC), when present, will assume overall responsibility of the operation.

**2.8. Obstructions.** Vertical obstructions (towers, antenna, etc.) exist on the NTTR and must be considered throughout mission planning and execution. Chart Update Manual (CHUM) data used to construct navigational charts does not address obstructions below 200ft AGL. Aircrews will use all applicable Area Planning, Military Training Routes (North and South America) (AP/1B) guidance and Mission Design Series (MDS) directives while operating on the NTTR.

2.8.1. Aircrews should assume obstructions extending from the surface to 50ft AGL throughout the NTTR and employ appropriate “see and avoid” techniques. Obstructions of greater than 50ft AGL up to 200ft AGL are reflected on daily range schedule Graphical Range Restrictions (GRR) regarding the restricted areas, but not the MOAs. Aircrews should refer to the National Geospatial-Intelligence Agency (NGA) website (https://aerodata.nga.mil/AeroBrowser) to obtain all FAA reported obstructions that extend to 200ft AGL or higher.

**Table 2.5. Vertical Obstruction Summary.**

| FAA Airspace Designation | ≤ 50ft | 50-200ft | > 200ft |
|---|---|---|---|
| R-4806 | See and Avoid | Reflected on GRR | NGA website |
| R-4807 | See and Avoid | Reflected on GRR | NGA website |
| R-4808 | See and Avoid | Reflected on GRR | NGA website |
| R-4809 | See and Avoid | Reflected on GRR | NGA website |
| Reveille North MOA | See and Avoid | See and Avoid | NGA website |
| Reveille South MOA | See and Avoid | See and Avoid | NGA website |
| Desert MOA | See and Avoid | See and Avoid | NGA website |

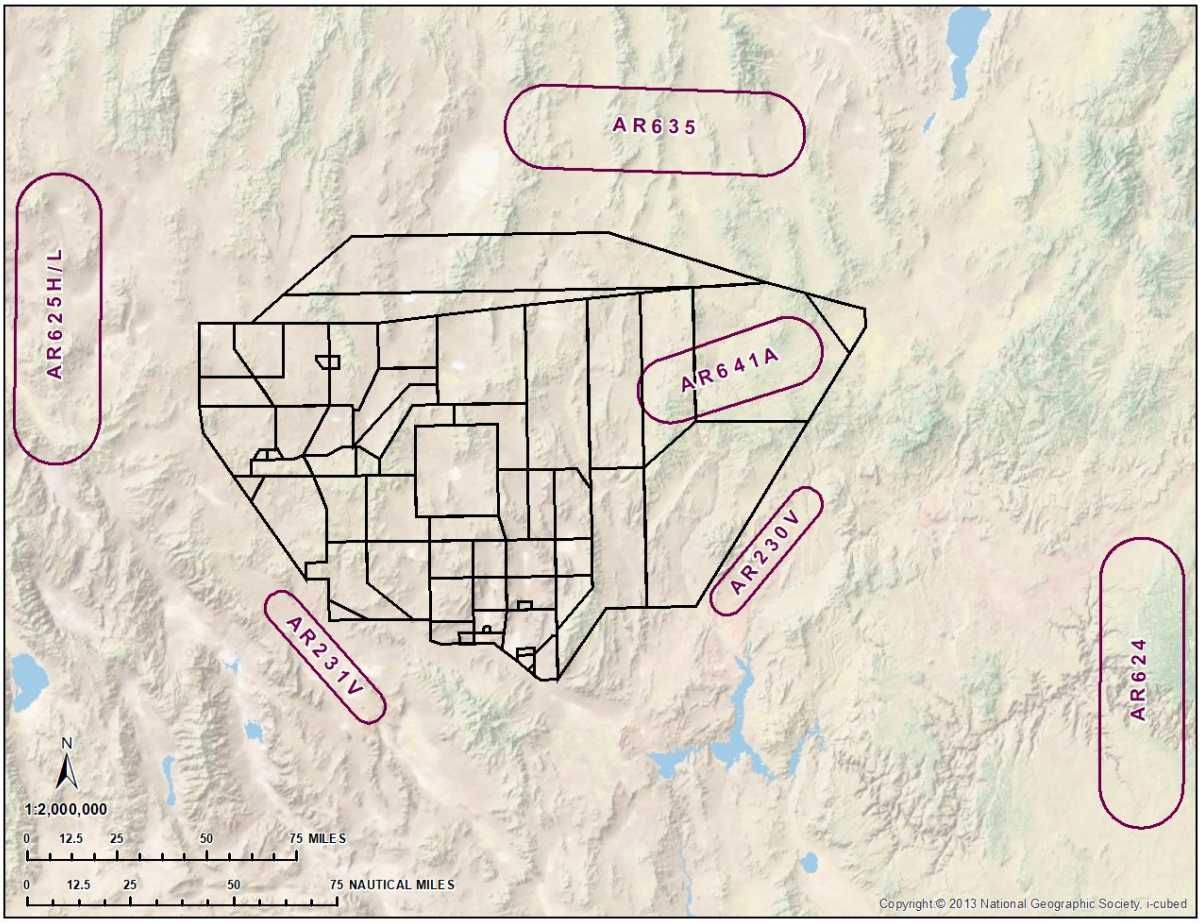
**2.9. Military Training Routes.** VFR/IFR Military Training Routes (VR/IR). Refer to FLIP AP-1B and NAFBI 11-250 (as applicable) for more information.

**2.10. RPA Operating Areas.** The NTTR includes multiple transit corridors and transit routes designated for RPA use.

**2.11. NTTR/Utah Test and Training Range (UTTR) Transition Corridor**. A transition corridor has been established for the purpose of allowing military aircraft to transition unrestricted and without delay, to/from the NTTR and UTTR. Transition corridor is normally only available for use during scheduled Large Force Exercises.

**2.12. Air Refueling (AR) Tracks.** Access to AR-641A is obtained by scheduling the Caliente MOA through the NTTR Scheduling Process. Access to AR-624, AR-625, and AR-635 is obtained through the 57 OSS/OSOS scheduling process. Helicopter air refueling may be conducted on VFR helicopter air refueling tracks AR-230V and AR-231V or on “ad hoc” on-range refueling tracks coordinated between the tanker and receivers.

**Figure 2.3. AR Tracks.**




**2.13. Drop Zones (DZs) and Landing Zones (LZs).** DZs, LZs, and Helicopter Landing Zones (HLZs) are available on range. Use of these zones will be IAW DAFMAN 13-217, Drop Zone, Landing Zone, And Helicopter Landing Zone Operations, and coordinated through the NTTR/DO. Refer to the NTTR Mission Planning website or HQ Air Mobility Command (AMC) Zone Availability Report (ZAR) for a current/approved list of zones and associated restrictions. Users will confirm coordinates and check for any zone changes/updates or local restrictions prior to submitting requests to NTTR/DO.

2.13.1. There are several HLZs available on the NTTR. Helicopter landing operations within the NTTR will be accomplished IAW all published wing and squadron guidance as well as applicable 11-2MDS publications and IFGs. All landing operations are conducted at the aircrew's own risk. All landing areas should be visually surveyed before use. Landings are not permitted in certain areas known to contain unexploded ordnance hazards. For current maps with all associated hazard areas, reference the NTTR targets webpage.

2.13.2. Public Facility/Land Use. BLM requires 60 days advance notice of the military's intended use of public facilities/lands in order to notify other users. Therefore, range users will contact NTTR/DO at least 60 days in advance of use of public or private lands within the NTTR boundary.

2.13.3. Range users desiring use of the following civilian airstrips/airports will contact NTTR/DO at least 60 days in advance:

2.13.3.1. Alamo Landing Field, NV (L92)

2.13.3.2. Beatty Airport, NV (BTY)

2.13.3.3. Desert Rock Airport (DRA), NV (NV65) Use of DRA should be coordinated through NNSS BirdDog Ops.

2.13.3.4. Lincoln County Airport, NV (1L1)

2.13.3.5. Tonopah Airport, NV (TPH)

**2.14. Forward Area Refueling and Arming Points.** Forward Area Refueling Point (FARP) operations, as well as Forward Area Arming and Refueling Point (FAARP) operations are available at various locations across the NTTR. These operations must abide by the requirements listed in DAFMAN 13-217, Drop Zone, Landing Zone, And Helicopter Landing Zone Operations, and must be coordinated through the NTTR/DO. For a list of approved FARP and FAARP locations, refer to the documents section on the NTTR Mission Planning website.

**2.15. Range Capabilities.** The range includes a variety of capabilities that may be used to present a high-fidelity, integrated battlespace. Range capabilities include targets, threats, feedback systems, and the NTTR ROC supporting Blue and Red command and control (C2).

2.15.1. Targets. The range presents over 260 target arrays and more than 3,500 Desired Mean Points of Impact (DMPIs). Target sets include the Urban Operations Complex (UOC), armored convoys, operational and simulated airfields, moving vehicles, and various tactical complexes.

2.15.2. Targets are categorized for live ordnance, inert-only, and no-drop targeting. Refer to the NTTR Mission Planning website for the latest mission planning information and weapons employment restrictions. Access to space and cyberspace targets may be provided through the 25 SRS space test and training range and/or the JIOR.

2.15.2.1. The range does not have targets that are artificially illuminated at night.

2.15.2.2. IR Enhanced Targets. Numerous targets are enhanced to provide a realistic IR signature. Refer to the NTTR Mission Planning website. Unit schedulers should request these targets IAW the NTTR Scheduling Process.

2.15.2.3. High-Speed Moving Target (HSMT). Target Complex 6208, also known as the High- Speed Moving Target complex or the “Fast Track,” is a 250ft wide by 2.6NM long dirt "highway" cut into the Dog Bone dry lake bed in R-62A and R-62B. The HSMT is a day or night target capable of speeds up to 70 mph and the only moving target on the NTTR capable of supporting live/inert ordnance employment in a moving target scenario. Refer to the NTTR Mission Planning website for additional details on how to coordinate moving target operations.

2.15.3. Threats. Threat emitters provide an electronic warfare (EW) environment designed to simulate enemy threats in a realistic combat scenario. The range includes fixed manned and unmanned threat emitters and mobile threat emitters. Refer to the NTTR Mission Planning websites for further information.

2.15.3.1. Simulators. NTTR/RS manages a variety of threat simulators. They include manned and unmanned threats that simulate SA-2/3/6/8/13/15, Air Defense Artillery (ADA) and Early Warning/Acquisition (EW/ACQ) threats, as well as mobile ROLAND threats. Both transportable (trailer) and movable (truck mounted) Unmanned Threat Emitters (UMTE) systems are located on range, each consisting of remote emitter units (REU) that provide an electronic threat environment while normally being controlled remotely from Point Bravo. Up to three UMTEs can be used in the UMTE-M configuration mounted on a truck and operated remotely or manually, depending on customer requests and the limits of site configurations. Additionally, high-power, transportable, remotely-operated Joint Threat Emitters (JTE) are located on the range capable of emulating SA-2, SA-3, SA-6, SA-13 (ACQ only), and ADA threats. RADAR Signal Emulator (RSE) high and low power phased array systems are available and capable of simulating a variety of real-world threats. The Radio Electronic Combat Systems (RECS) provides communications jamming and spoofing capabilities. Mobile Radio Electronic Combat System (M-RECS) is a one, two or three man locally or remotely operated system. M-RECS is a standalone Electronic Warfare (EW) unit comprised of direction finding hardware, communication disruption, and electronic countermeasure/ targeting simulation (ECM/TS) equipment. The High Power Noise Jammer (HPNJ) simulator is an electronic (actively-scanning) array noise jammer. Digital Radio Frequency Memory (DRFM) is a trailer- mounted, mobile radar jamming and spoofing threat (two systems, one transportable the other fixed). Synthetic Aperture Radar (SAR) Jammer is a Man Portable, Vector Modulator-based SAR Jamming System.

2.15.3.2. Battlefield Communication Simulator System (BCSS). The range maintains two BCSS vehicles, both equipped with an advanced suite of communications software providing advanced signal generation. The BCSS is an exploitable and attackable opposing force command and control simulation system. The system provides automated communications simulations to support DoD training exercises as well as a variety of analog and complex digital communications signals. The system is designed to develop battlefield scenarios. Refer to the NTTR Mission Planning websites for further information.

2.15.3.3. Feedback systems. The range offers three primary feedback systems: The NTTR P5 Combat Training System (P5CTS) paired with the Individual Combat Aircrew Display Station (ICADS) and the Television Ordnance Scoring System (TOSS). Additionally, NTTR/DO can facilitate access to the ACC-sponsored untethered Integrated Tactics Assessment Software (ITAS). Vehicle Tracking System (VTS) is a radio network that provides the means to track ground vehicles on the NTTR Southern (60 series) and Northern (70 series) ranges. Requests for feedback systems are made through the NTTR BMP. Refer to the NTTR Mission Planning websites for further information.

2.15.4. NTTR ROC. The ROC provides a venue for Blue force weapons control, Link-16, and Situational Awareness Data Link (SADL) use. Red forces utilize the ROC to simulate aspects of an Integrated Air Defense System (IADS). The NTTR ROC provides multiple secure work areas, to include space and cyber vaults.

2.15.4.1. ROC-controlled systems include radar simulators with opposing force early warning (EW) and GCI radars, SAM acquisition, fire control, missile guidance radars, and ADA fire control radars. Realistic training is further augmented through echelon command and control, communications jamming, interception/exploitation, deception, intrusion, and radar jamming. Tailored data collection and mission debriefing products and services are available to include tracking, detection/launch or fire data, aircrew tactics commentary, and video replay of radar and optical engagements.

2.15.5. Surrogates provide cognitive skills training to aircrews. They replicate the physical threat representative multi-spectral signatures (Electro-optical, IR and Radar Cross Section [RCS]) located in close proximity to a radar signal emitter. Surrogates are transportable and can be located on or off the range.

2.15.6. AAA Visual Simulator provides four independent, standalone systems that provide visual simulation of real-world AAA threats. The AAA Visual Simulator employs a series of strobe lights, spaced in a rectangular formation, which sequentially flash to imitate the firing of an anti-aircraft artillery weapon. Since the AAA Visual Simulator only emits visible light, it will be difficult to detect. Strobe rates of ZPU-1, ZPU-2, ZPU-4, and ZSU-23-4.

# Chapter 3

# OPERATIONS/WEAPONS DELIVERY PROCEDURES

**3.1. Operations.** The following provides general procedures for range operations. Detailed procedures are provided as attachments to this publication. Users should refer to the NTTR Mission Planning website, daily range schedule, and GRR for current range restrictions and status. NTTR Range Monitoring will relay GRR updates upon check-in. Aircrew must comply with MDS-specific instructions and technical orders, NAFBI 11-250 and CAFBI 11-250 (as applicable), and applicable In-flight Guides (IFG).

3.1.1. Range Procedures.

3.1.1.1. Aircraft Range Entry/Exit/Holding. To request range entry, users will contact NATCF and NTTR Range Monitoring (in order). To request NTTR exit, aircrew will contact NTTR Range Monitoring and NATCF (in order). Refer to the NTTR Mission Planning websites for assigned frequency clearances. If holding is required, aircrew will hold in adjacent ranges if scheduled, hold VFR off-range, or hold IFR off-range under ATC control. Departures should be planned as to enable on-time range arrival at scheduled range start time.

3.1.1.2. Ground Party Range Entry/Exit and transition between subranges. The NTTR ground environment includes areas of relative safety though which authorized users may move with limited oversight and other areas that include a variety of hazards (including housing strikeable targets) that require specific scheduling and access monitoring. To request scheduled range entry, users will stop at the range boundary entry point and contact NTTR Range Monitoring. To notify NTTR Range Monitoring of range exit, users will stop at the range boundary entry point and contact NTTR Range Monitoring.

3.1.1.3. Ground Party Range Entry/Exit To/From Unmanned Systems. The NTTR ground environment includes unmanned threat system site locations that have the potential for high power radio frequency (RF) radiation hazard and movement hazard. To request entry to unmanned sites, users will hold at the site entrance sign and contact NTTR Range Monitoring for instructions. Unless otherwise informed, if active system lights (solid or flashing) or any system movement (platform spinning, vertical or horizontal system movement, etc.) are observed when approaching an unmanned system, return immediately to the site entrance sign and contact NTTR Range Monitoring for instructions. To notify NTTR Range Monitoring of site exit, users will contact NTTR Range Monitoring when past the site entrance sign.

3.1.1.4. Incidents wherein aircraft intrude into airspace for which they are not scheduled (i.e., deviate from the published schedule) are reviewed by the ROO for the purpose of gathering relevant facts to help ensure safety and security. NNSS/NFO and DOE will be notified of airspace deviations in R-4808. The deviating/host unit plays a key role in gathering deviation information and assigning appropriate corrective actions, and may be called upon to brief senior USAFWC and interagency leaders on the facts of the incident. Deviation information is shared with all affected range entities and may result in restrictions being placed on individuals and organizations regarding their use of the NTTR.

3.1.1.5. It is the responsibility of the Pilot in Command (PIC) to maintain awareness of and avoid high-risk airspace borders in the NTTR. Control agencies will strive to enhance situational awareness and provide PICs with intervening comm for avoidance of high-risk borders when possible. Table 3.1 identifies NTTR airspace borders that are designated high-risk:

**Table 3.1. NTTR High Risk Airspace Borders.**

| Airspace | High Risk Boundary |
|---|---|
| 4809A | Vertical and Lateral |
| 4808A/B/C/D/E | Vertical and Lateral |
| Caliente C and Elgin | Eastern Lateral Border |
| Reveille North | Vertical Border |
| Airspace Scheduled by NNSS/NFO and DOE | All |

3.1.1.6. Any aircrews infringing on any high-risk airspace borders specified in Table 3.1 must execute the following steps:

3.1.1.6.1. Immediately cease tactical maneuvering, maneuver the aircraft in the shortest direction to return to approved airspace, and initiate a kill removal/return to SAFE area or FRA. The intent is to establish a tactical pause where aircrew can assess ORM factors leading to the spillout. Mission commanders/air b3.1.9osses/test directors retain the option of directing RTBs for egregious or repeated violations.

3.1.1.6.2. Debrief the airspace spillout with the owning Squadron CC (delegated no lower than Squadron DO) immediately after the mission and establish a written record including at a minimum: date and time, flying callsign, aircraft type, owning squadron, border violated, extent/length of the violation, ORM factors leading to the infringement, and lessons learned.

3.1.1.6.3. Squadron CCs will route the written debrief of airspace violations to owning Group/CCs, NTTR/DOX, 57 OSS/CC, and 57 OSS/DO. Remedial actions (such as grounding) are at the discretion of the owning Squadron CC.

3.1.1.6.4. 57 OSS will interface with FAA (ATREP/FSDO/ARTCC) for follow-on actions in instances where airspace spillouts resulted in civilian TCAS resolution advisories, or otherwise involved civilian traffic. NTTR ROO/DOX will interface with NNSS/NFO and DOE in spillouts involving R-4808 and R-4809. NTTR and 57 OSS may choose to elevate the spill-out data to 57 WG/CC where applicable.

3.1.2. All aircraft are required to monitor Guard (243.0 MHz or 121.5 MHz) while operating within NTTR boundaries.

3.1.3. Transponder Identification. Unless specifically authorized, all aircraft will have an operational identification friend or foe (IFF) system and will squawk Mode C and a tactical Mode 3 (non-1200 squawk) on all missions on range.

3.1.4. Reference Attachment 17 of this Addendum for Lights-Out flying operations.

3.1.5. Airspace Recall. Emergency situations, including extreme weather or operational requirements, may demand the recall of previously scheduled airspace.

3.1.6. Supersonic Operating Areas. Supersonic flight is approved within designated areas based on the requirement for realistic testing and training. Aircrew will conduct supersonic flight only when necessary to accomplish the mission. In order to avoid public impact, and to the maximum extent possible, aircrew should avoid supersonic flight over populated portions of the range.

3.1.7. Altitude Restrictions. Aircrews must comply with MDS-specific instructions and technical orders, NAFBI and CAFBI 11-250 (as applicable), and the daily range schedule. For vertical/short takeoff and landing (V/STOL) aircraft, fixed-wing altitude restrictions apply when in forward flight, helicopter altitude restrictions apply when in vertical flight. Low-level helicopter operations are defined as flight below 1,000ft AGL with operating altitudes as low as 10ft AGL during hover, observation, and weapons employment.

3.1.8. Manned Site Overflight Restrictions. All aircraft must avoid all manned sites and ground parties by at least 500ft, vertically and laterally, unless further restricted within Attachment 4.

3.1.9. Targets and target material. Targets and Target Material may be procured from local sources or the Defense Logistics Agency (DLA) IAW DoDM 4160.21 Volume 1, Defense Materiel Disposition: Disposal Guidance and Procedures; DoDM 4160.21V2, Defense Materiel Disposition: Property Disposal and Reclamation; DoDM 4160.21 Volume 3, Defense Materiel Disposition: Reutilization, Transfer, and Sale of Property; and DoDM 4160.21 Volume 4, Defense Materiel Dispostion: Instructions for Hazardous Property and Other Special Processing Materiel. The range operating authority will ensure acquired items repurposed for the transport of an operator or passengers comply with DoDI 6055.04, DoD Motor Vehicle and Traffic Safety; AFI 24-302, Vehicle Management; DAFI 91-207, The Traffic Safety Program; AFI 24-302, Vehicle Management; DAFI 24-301, Ground Transportation; DAFI 91-207, The Traffic Safety Program; and DoDI 6055.04, DoD Motor Vehicle and Traffic Safety. Procured items shall not be used to support any off-range functions.

3.1.10. Target Maintenance and/or EOD Ground Party Overflight Restrictions. Range cleanup periodically requires range closures and altitude restrictions for the mutual protection of aircraft, aircrew, and ground parties. During range cleanup, EOD personnel may detonate ordnance in place. The daily range schedule will reflect the range closures and altitude restrictions.

3.1.10.1. Noise associated with overflight can create serious hazards to personnel working in an Unexploded Ordnance (UXO) environment. To preclude overflight hazards to ground parties, aircrews must remain aware of all range restrictions associated with ground party activity. EOD overflight restrictions are published in the daily range schedule. NTTR Range Monitoring will advise aircrew of range restrictions, as required.

3.1.10.2. When ground parties are on range conducting Target Maintenance and/or EOD operations, and demolition operations are not taking place, aircraft shall fly no lower than 3,000ft AGL at subsonic speeds, and no lower than 5,000ft AGL at supersonic speeds over the target area currently in maintenance.

3.1.10.3. When demolition operations are taking place, aircraft shall fly no lower than 10,000ft AGL over that range, unless more specific guidance is provided by EOD and relayed through Range Monitoring.

3.1.11. Small Unmanned Aerial Systems (sUAS) activity will be annotated on the GRR for affected airspace. sUAS requests must be submitted through the NTTR Business Management Process (BMP) with the NTTR Projects Office. 57th OSS/OSOP Airspace Management Office is the OPR for approving all sUAS flights in Desert MOA airspace.

3.1.12. Aircraft will not fly within 1,250 feet of the munitions storage area (MSA) nor the explosive material storage facilities located at N37 50.77 W116 42.62 in R-4809A.

3.1.13. Use of any laser or directed energy weapon (DEW), regardless of the mode of operation, may be potentially hazardous to air and ground personnel utilizing the NTTR. Use of lasers in combat mode is expressly forbidden unless reflected on the daily range schedule.

3.1.14. Range users will adhere to NAFBI 11-250 and CAFBI 11-250 (as applicable), regarding noise-sensitive range activities.

3.1.15. Night Vision Devices (NVDs). Use of NVDs on range will be IAW MDS-specific instructions, technical orders, NAFBI 11-250 and CAFBI 11-250 (as applicable). Ground parties using NVDs will adhere to NTTR/SD security procedures for sensitive equipment on range.

3.1.16. NTTR specific safety and security restrictions may be annotated in CSE under “Reports,” “Block Restrictions” and must be strictly adhered to:

**Table 3.2. Specified CSE Safety and Security Restrictions.**

| Restriction | Description |
|---|---|
| No Ground Party | Ground personnel are not authorized on the specified NTTR ranges and sub-ranges. |
| Sanitization | Powered on maintenance activities of site equipment is strictly prohibited on the specified NTTR ranges and sub-ranges. Ground personnel may be present at an emanating, collection, or optical support site, system, or sub-system but are not authorized to enter into or apply power to the equipment.<br>System maintenance activities are prohibited. Personnel are allowed to perform external ground maintenance activities that do not require the system to be powered on or the system to be entered.<br>(Examples: corrosion prevention, weed abatement, picking up discharged Smokey SAMs, maintaining grounds and painting.) |
| Down and Quiet | Maintenance and/or startup activities that require transmission or radiation are prohibited. Ground personnel may be present at and enter the system to perform system startup, diagnostic or ground maintenance activities; dummy load and/or open air transmissions are prohibited. All system optics and/or cameras shall be turned off or disabled. Systems will not be started beyond a standby state.<br>Note: Exceptions to the Down and Quiet restriction shall be coordinated with the organization that scheduled the restriction and applicable O&M Section.<br>Ground communications, such as LMR communications with NTTR Range Monitoring, are excluded from this restriction. |
| Evacuate | Ground personnel are not authorized on R-4807, EC West, R-4809B, or the Forward Air Control (FAC) A/B portions of TPECR. Cactus security will move to the TPECR gate. |
| TPECR Containment | Any ground personnel within the confines on TPECR must remain indoors while the restriction is active. Ground personnel are not authorized on FAC A/B portions of TPECR. The TPECR access road will be closed during TPECR containment. |
| Modified TPECR Containment | Any ground personnel within the confines on TPECR shall remain indoors and are not authorized on FAC A/B portions of TPECR. Personnel are authorized to be outdoors for brief periods of time, as required.<br>(Examples: travel between buildings, restroom breaks, smoke breaks, etc.) |

| | |
|---|---|
| No Optics | Visual enhancement devices used by ground personnel or systems are prohibited on the specified NTTR ranges and sub-ranges. |
| No Cameras | Photography is prohibited by ground personnel or systems while on the specified NTTR ranges and sub-ranges. |

3.1.17. May through October are typically seasonally dry months in Nevada and pose the greatest fire hazard on range.

3.1.17.1. Ordnance delivery and range operations may be negatively impacted by fire code restrictions. Fire codes will be depicted on the NTTR Mission Planning websites and the GRR. Aircrew must report all fires on the range to NTTR Range Monitoring as soon as possible.

3.1.17.2. Flares and chaff may be utilized only on specified areas of the range IAW specific safety procedures.

3.1.18. Preservation of Wildlife. Range users, both aircrew and ground parties, will not harass any wildlife including desert tortoises, wild horses, or burros and will avoid animal concentrations and watering holes.

3.1.18.1. The 99 ABW publishes an Integrated Natural Resource Management Plan (Plan 126-4) that details wildlife management on the NTTR. Environmental management activities also occur within R-4808 and may require coordination between multiple agencies.

3.1.18.2. Portions of the NTTR (Alamo A/B/C, R-61A/B, R-62A/B, R-63A/B/C, R-64A/B/CG, R-65A/B/C/D and R-67A/B) overlie the DNWR. Aircraft will remain above 2,000ft AGL in these areas unless a lower altitude is required for tactical training and IAW the daily range schedule.

**3.2. Weapons Delivery Procedures.** The range can support a wide variety of live, inert, and training ordnance for air-to-ground and ground-to-ground operations. Refer to the NTTR Mission Planning website for the authorized ordnance delivery profiles for specific targets. In all cases, aircrew should avoid overflight of populated areas when carrying ordnance and must positively identify the target prior to expending any ordnance. Transit of manned ranges with live ordnance is permitted if armament switches are in a SAFE/SIM/TRAINING/OFF position.

3.2.1. Transit with hung, live ordnance should be avoided whenever feasible. If transit with hung, live ordnance is required, an in-flight emergency (IFE) will be declared and armament switches will remain in the SAFE/SIM/TRAINING/OFF position. Aircrew carrying internally loaded ordnance (live or inert) will not open aircraft bomb bay doors while transiting manned ranges (except in support of NNSA/SFO-sponsored missions). Transit of R-4808B/C/D/E with hung, live ordnance is forbidden.

3.2.2. Minimums, fouls, and attack axes are IAW applicable publications and AFMAN 11-214, Air Operations Rules and Procedures.

3.2.3. Before releasing ordnance on range, each aircrew member will accomplish a dry/familiarization pass before weapons deliveries that require visual acquisition of the target (e.g. visual delivery, TGP, FLIR delivery) to be familiar with range boundaries, location of manned sites (e.g., threat systems, work facilities, etc.), and assigned target area. Aircrew members that have flown on range within the past 12 months will be exempt from this requirement.

3.2.4. WDZ/SDZ. The WDZ/SDZ Tool is used for each weapon safety footprint on each target on range. When weapon safety footprints are not supported by WDZ/SDZ, an RSA is performed using System Program Office (SPO) or test data in which severity and probability of risk is analyzed.

3.2.4.1. Approval levels are tied to the assessed risk:

3.2.4.1.1. Unlikely: No lower than NTTR/SE approval.

3.2.4.1.2. Low: No lower than ROO approval.

3.2.4.1.3. Medium: No lower than ROA approval.

3.2.4.1.4. High: No lower than USAFWC/CC approval.

3.2.4.2. Normally, WDZ profiles must be contained within the boundaries of the scheduled subrange(s) in order to employ live/inert weapons. Extreme caution must be exercised when utilizing segmented hazard areas (i.e., when a desired profile extends into a subrange that contains scheduled ground parties). Detailed coordination with, and approval of, the segmented operations must be provided by the ground party host unit. Moreover, specific NTTR/DO approval for segmented operations must be obtained. Segmented operations will be annotated on the daily range schedule and GRR.

3.2.5. Aircrews are prohibited from expending ordnance, except chaff and flares, on any electronic combat (EC) range. There are no DoD strikeable targets within the EC ranges (EC West, EC East, and TPECR) or in the Alamo and Pahute ranges. Transit of EC ranges, Alamo or Pahute ranges with expendable (except chaff and flare) ordnance is permitted if armament switches are in a SAFE/SIM/TRAINING/OFF position. Transit with hung live ordnance should be avoided whenever feasible. If transit with hung live ordnance is required, an in-flight emergency will be declared and armament switches will remain in the SAFE/SIM/TRAINING/OFF position. Aircrew carrying internally loaded ordnance (live or inert) will not open aircraft bomb bay doors while transiting EC ranges (except in support of NNSA/SFO-sponsored missions).

3.2.6. There are no DoD strikeable targets within R-4808B/C/D/E. Transit of these ranges with live ordnance is permitted if armament switches are in a SAFE/SIM/TRAINING/OFF position. Transit of R-4808B/C/D/E with hung, live ordnance is forbidden.

3.2.7. Cluster Bomb Units (CBU). CBU-87/103 deliveries are restricted to targets 62-01 and 75-46. Random delay fuse settings are not authorized. A 200 CBU canister limit is imposed on CBU targets between EOD range clearances. To control CBU expenditures, a CBU authorization number is required to authorize employment. Contact NTTR Target Management Office (nttr.dos.targets@us.af.mil) to obtain a CBU authorization number. CBU-105 deliveries are restricted to target 63-12 and must be coordinated through NTTR Target Management Office.

3.2.8. DU ammunition is only authorized for use on Target 63-10. Use of DU ammunition must be conducted IAW the Radioactive Material Permit and the annual expenditure of DU rounds is controlled through NTTR/SE. A DU authorization number must be obtained through NTTR Target Management Office prior to employment.

3.2.9. Support vehicles, vehicle shelters and threat simulators on range shall be painted white to distinguish them from targets. Exceptions to this policy must be approved by NTTR/DO. To further aid in identification, manned sites are marked by flashing white strobes (daytime) and steady white lights and/or infrared strobes (night). Aircrew will not deliver ordnance or make practice attacks with releasable ordnance on any structure or location painted white or marked with steady or flashing lights. Captive carry air-to-air and air-to-ground missiles, internally loaded weapons with bomb doors closed and mechanically-safed guns are not considered releasable ordnance.

3.2.10. Two Unmanned Threat Emitters (UMTE) systems are normally located on R-75 provide an electronic threat environment collocated with strikeable targets, to provide as realistic a combat environment as possible. Additionally, there is an RSE site (pads ECS05A and 2-T1) located approximately 2 NM southwest of target 77-02 in R77A. Each UMTE/RSE site is surrounded by white concrete bunkers or white sea-land containers. Aircrew should exercise extreme caution not to expend ordnance on UMTEs/RSEs. Refer to NTTR Mission Planning website for further information.

3.2.11. TOSS is currently active on several strikeable ranges. Aircrew will avoid expending ordnance on TOSS towers. Refer to the NTTR Mission Planning website for further information on TOSS capabilities and locations.

3.2.12. Armament switches will be in the SAFE/SIM/TRAINING/OFF position when aircraft are in a numbered range that has a “No Ordnance” Restriction per the GRR unless within an authorized WDZ or aircraft are not on the numbered ranges. Para 3.2.5 still applies. For a combined load of captive, training, inert or live ordnance, aircrew will not select armament switches that could possibly induce the inadvertent release of any ordnance. Captive training ordnance may be selected at any time.

3.2.13. An armament safety check will be completed after ordnance delivery and all armament switches are returned to the SAFE/SIM/TRAINING/OFF position. The aircrew or wingman must also confirm the release of all ordnance by visually scanning the aircraft. It is not necessary to verify with NTTR Range Monitoring that this procedure has been accomplished.

3.2.14. Aircraft with externally loaded ordnance will accomplish an ordnance check before exiting the range. Upon recovery, aircrew will fly a hung ordnance pattern as defined in NAFBI 11-250 and CAFBI 11-250 (as applicable) if ordnance status cannot be positively verified.

3.2.15. UXO avoidance. UXO awareness is required on range to protect aircraft from inadvertent damage when employing ordnance. Aircrew will employ safe escape parameters assuming a 2,000 lbs. UXO detonation.

**3.3. Target Identification.** To ensure correct targets are identified prior to expending any munitions, all air and ground range users intending to employ munitions on the NTTR are required to obtain imagery and maps for every intended target (including secondary targets) to ensure correct targets are identified prior to expending any munitions.

3.3.1. Aircrew will not drop ordnance on ranges occupied by ground parties unless specifically authorized by NTTR/DO, or operating under the control of a JTAC in accordance with attachment 26.

3.3.2. Aircrew will report operations resulting in unauthorized ordnance deliveries over manned sites or closed ranges. If this occurs, NTTR/SE will conduct a safety investigation and will determine the reportable mishap category. Local organizations may conduct supplemental investigations as necessary to support the primary investigation.

3.3.3. Aircrew must be aware that ground parties may be located on bomb impact ranges adjacent to a range scheduled for air to ground operations. Flight Leads assume flight lead control responsibilities IAW AFMAN 11-214, and Joint Pub 3-09.3. Flight Leads must know and brief positions of ground parties to their flight members. Ground parties may be mistaken for range targets, therefore caution should be exercised at all times. Aircrew will advise NTTR Range Monitoring of their intentions to deliver live, inert, or training ordnance prior to takeoff. This coordination should include weapon/target pairings and airspace deconfliction requirements, and should primarily take place via phone call.

3.3.4. For Large Force Exercises, the Air Boss will inform NTTR Range Monitoring of the intent to deliver live, inert, or training ordnance prior to mission launch. This includes weapon/target pairings and any airspace deconfliction requirements.

3.3.5. Aircrew will not perform simulated or mock attack passes on ground parties or manned sites with releasable ordnance. Captive carry air-to-air missiles, air-to-ground missiles, and mechanically-safed guns are not considered releasable ordnance.

**3.4. High Speed Moving Target (HSMT).** The High Speed Moving Target is the name for Target Complex 62-08. The HSMT spans Range 62A & 62B, on Dog Bone Lake and some of the surrounding area.

3.4.1. The HSMT complex contains no fixed targets. Weapon delivery profiles and restrictions are listed on the mission planning website and CSE under target 62-08.

3.4.2. Target vehicles are maintained by contractor personnel and may be stored at a storage facility at the south end of the track at any time. The storage facility is painted white and is not a valid target. When the HSMT is activated, one or more vehicles may be staged on or near the track. The vehicles may be controlled manually by personnel during the activities to activate and de-activate the HSMT. The vehicles are controlled remotely from the control station in 63A while the range is “hot”. Prior to the mission, the plan for the number of vehicles to be used, their staging area, the sequence for operating them, and any other information necessary to the successful completion of the mission, will be briefed to all personnel involved.

3.4.3. The HSMT vehicles are despecularized to the same standard as any other piece of target material on NTTR. Any laser system, air- or ground-based, that is approved for use on NTTR may be used with the HSMT. Magnifying optics will not be permitted when 2M, Class 3, Class 3B or Class 4 laser systems are in use, unless a separate safety assessment is performed and approved by the appropriate authority. This restriction applies to personnel that may be present in Range 62 AND at the control station in Range 63A. Laser Eye Protection will not be required for personnel who occupy the control station area in Range 63A.

**3.5. Emergency Procedures (EPs).** EPs will be handled IAW with MDS-specific instructions and technical orders, NAFBI and CAFBI 11-250 (as applicable), and applicable IFGs. The reporting of NTTR range incidents is mandatory. Personnel operating on the NTTR (aircraft or ground parties) will notify NTTR Range Monitoring following major accidents/incidents such as crashes, munitions that impact off-range or outside the target complex, inadvertent releases/dropped objects, fires or personal injuries and provide information requested.

3.5.1. In addition to Nellis AFB and Creech AFB, TTR Airfield (TTRA) and Desert Rock Airport (no services available) are readily available emergency airfields. Additional information is available in Flight Information Publications (FLIP) and NAFBI 11-250 and CAFBI 11-250.

3.5.2. Dropped Object and Inadvertent Release. For any accidental weapons release or dropped object on range, the following applies:

3.5.2.1. UNINTENTIONAL. Aircrew induced accidental release on range is not considered a malfunction. Continue at FAC/flight lead discretion.

3.5.2.2. INADVERTENT. System induced accidental release on range is considered a malfunction.

3.5.2.3. Off-Range (Unintentional or Inadvertent).

3.5.2.3.1. Safe armament switches.

3.5.2.3.2. Return to base immediately. If the dropped object resulted from an inadvertent release, declare an IFE, treat remaining stores as hung ordnance and fly hung ordnance procedures IAW the applicable IFG. If practical, the incident aircraft will be escorted to the base. No further releases will be attempted.

3.5.2.3.3. Record switch positions at the time of release.

3.5.2.3.4. Attempt to determine if the dropped object caused any injury or damage.

3.5.2.3.5. Contact NTTR Range Monitoring or NATCF as appropriate and provide the following information: Aircraft call sign, number, and type, time and location of incident, description of dropped object, location of impact, all known circumstances

3.5.2.3.6. Contact the Nellis Command Post as soon as possible after landing and give the information above. Contact aircrew unit leadership and 57 WG/SE ASAP after landing.

3.5.2.3.7. Aircrew will notify NTTR Range Monitoring of any weapons release that does not impact in the vicinity of the target. Provide the best estimated location of the impact point.

3.5.2.3.8. For weapon profiles in CSE/Mission Planning Website that state to contact NTTR/SE in addition to the NTTR/Duty Officer if the munition impacts off the designated target area immediately notify NTTR/SE. There are potential radiological concerns with these specific profiles.

3.5.3. Hung Ordnance and Unsafe Gun Procedures IAW NAFBI 11-250 and CAFBI 11-250 (as applicable).

3.5.4. Jettison Procedures. Inert ordnance, training ordnance, and non-ordnance stores may be jettisoned at the pilot's discretion on any approved range or target provided clearance from Blackjack has been received.

3.5.4.1. Live ordnance will be jettisoned live on scheduled targets if possible. Minimum altitude will be 2,000 feet AGL or the minimum frag clearance altitude, whichever is higher. If jettison cannot be made on scheduled targets, contact Blackjack for a target. If unable, inform Nellis Control and expect to use Jettison Hill procedures.

3.5.5. Night or IFR Jettison. At night or IMC conditions aircrew will use all available means to ensure ordnance or store impacts the desired area. If weather precludes jettison of ordnance, NATCF will provide radar vectors to Range 62 Alpha, target 62-2 (airfield) or Jettison Hill. Target 62-2 is located N36-55.015, W115-25.887 with elevation 3,463 feet MSL. The minimum vectoring altitude (MVA) for Range 62 Alpha is 10,400 feet MSL and the MVA for Jettison Hill is 7,000 feet MSL. The controlling agency will inform the aircrew when 1 NM short of designated impact point. Heading and altitude restrictions will be followed to the maximum extent possible. In all cases the aircrew is ultimately responsible for proper impact location of jettisoned ordnance or stores.

## Chapter 4

## ELECTRONIC WARFARE RANGES

**4.1. Electronic Warfare Ranges.** The NTTR electronic warfare ranges consist of EC West, EC East, and TPECR. Systems include radar simulators with opposing force early warning and GCI radars, SAM acquisition, direct control, missile guidance radars, and ADA fire control radars. EC ranges and resources are scheduled through the NTTR Scheduling Process. Refer to the NTTR Mission Planning websites for current resource availability.

4.1.1. The EC ranges have manned sites that may consist of shelters and vehicles, and associated threat systems. Manned sites may have Smokey Surface-to-Air Missile (SAM) launchers collocated at the site to provide visual launch cues to aircrew.

4.1.2. Adversary tactics are simulated through input from the 507th Air Defense Aggressor Squadron (507 ADAS). EC simulations can provide scenarios ranging from individual signals under user control to the full combat array of opposing air defense forces. Simulators are sited and positioned IAW current opposing force deployment doctrine, topography, range safety, and fiscal constraints. For questions regarding adversary tactics simulation, contact 507 ADAS.

4.1.3. Feedback from threat emitters is accomplished through either the 507 ADAS or Nellis AFB debriefing systems. 507 ADAS provides threat assessment for event participants. NACTS/ICADS or ITAS offer a threat debriefing capability for individual flights.

4.1.4. For a complete list of available threat systems, contact the NTTR RSO office at nttrrsoallpersonnel@us.af.mil.

JEFFREY D. SIMMONS, Colonel, USAF
Commander

## Attachment 1

## GLOSSARY OF REFERENCES & SUPPORTING INFORMATION

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NTTR OI 31-101, Security Controlled Area Operating Procedures, 12 Mar 2019
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Product and Service Catalog

***Adopted Forms***

DD Form 448, Military Interdepartmental Purchase Request (MIPR)
AF Form 847, Recommendation for Change of Publication
NAFB IMT 22, Range Officers Report
NAFB IMT 74, Nevada Test and Training Range Visit Request/Authorization

***Abbreviations and Acronyms***

**AAR**—After Action Report
**ABW**—Air Base Wing
**ACC**—Air Combat Command
**ACM**—Air Combat Maneuvering
**ACQ**—Acquisition
**ADA**—Air Defense Artillery
**ADAS**—Air Defense Aggressor Squadron
**AFB**—Air Force Base
**AFCEC**—Air Force Civil Engineering Center
**AFI**—Air Force Instruction
**AFIMSC**—Air Force Installation and Mission Support Center
**AFMAN**—Air Force Manual
**AFRIMS**—Air Force Records Information Management System
**AFTC**—Air Force Test Center
**AGL**—Above Ground Level
**AGRCO**—Aerial Gunnery Range Control Officer
**AGTS**—Aerial Gunnery Target Sets
**AI**—Airborne Instrumentation
**AMC**—Air Mobility Command
**AR**—Air Refueling
**ARTCC**—Air Route Traffic Control Center
**ATC**—Air Traffic Control
**ATCAA**—Air Traffic Control Assigned Airspace
**ATM**—Asynchronous Transfer Mode
**BCSS**— Battlefield Communication Simulation System
**BCE**— Base Civil Engineer
**BDOC**—Base Defense Operations Center
**BLM**—Bureau of Land Management
**BMP**—Business Management Process
**C2**—Command and Control
**CAFB**—Creech Air Force Base
**CAFBI**—Creech Air Force Base Instruction
**CARC**—Combined Arms Range Complex
**CBU**—Cluster Bomb Unit
**CES**—Civil Engineering Squadron

**CFA**—Controlled Firing Area
**CFR**—Code of Federal Regulations
**CHUM**—Chart Updating Manual
**CIS**—Communications Instrumentation Subsystem
**COA**—Certificate of Authorization
**CONOP**—Concept of Operations
**COP**—Common Operational Picture
**CS**—Computation Subsystem
**CSE**—Center Scheduling Enterprise
**CSS**—Customer Satisfaction Survey
**CT**—Continuation Training
**CTA**—Creech Tower Airspace
**CTS**—Combat Training System
**DAF**—Department of the Air Force or Device Assembly Facility
**DBA**—Direct Budget Authority
**DEW**—Directed Energy Weapon
**DISA**—Defense Information Systems Agency
**DLA**—Defense Logistics Agency
**DML**—Dreamland Approach Control
**DNWR**—Desert National Wildlife Refuge
**DoD**—Department of Defense
**DOE**—Department of Energy
**DU**—Depleted Uranium
**DZ**—Drop Zone
**EA**—Electronic Attack
**EC**—Electronic Combat
**ECE**—Electronic Combat East
**ECW**—Electronic Combat West
**ELINT**—Electronic Intelligence
**EMI**—Electromagnetic Interference
**EO—**Electro Optical
**EOD**—Explosive Ordnance Disposal
**EW**—Electronic Warfare; Early Warning
**FAA**—Federal Aviation Administration
**FAC** —Forward Air Control
**FAR**—Federal Aviation Regulation
**FDO**—Foreign Disclosure Officer
**FEO**—Foreign Exchange Officer
**FLIP**—Flight Information Publication
**FLPC**—Flight Limitation Profile Curve
**FM**—Frequency Modulation
**FMV**—Full-Motion Video
**FORNAT**—Foreign National
**FP**—Firing Point
**FVR**—Foreign Visit Request
**FVS**—Foreign Visit System

**GCI**—Ground Control Intercept
**GCS**—Ground Control Stations
**GIANT**—GPS Interference and Navigation Tool
**GIS**—Geographic Information Systems
**GPS**—Global Positioning System
**GRR**—Graphical Range Restrictions
**GTL**—Gun-to-Target Lines
**HDMI**—High-Definition Media Interface
**HE**—High Explosive
**HLZ**—Helicopter Landing Zone
**HQ**—Headquarters
**HSMT**—High-Speed Moving Target
**IADS**—Integrated Air Defense System
**IAW**—In Accordance With
**IC**—Interim Change
**ICA**—Installation Command Authority
**IDL**—Initialization Data Load
**IER**—Information Exchange Requirements
**IFE**—In-flight Emergency
**IFF**—Identification Friend or Foe
**IFG**—In-flight Guide
**IFR**—Instrument Flight Rules
**IO**—Information Operations
**IR**—Infra-Red
**ITAS**— Integrated Tactics Assessment Software
**JDN**—Joint Data Network
**JDNC**—Joint Data Network Cell
**JICO**—Joint Interface Control Officer
**JIOR**—Joint Information Operations Range
**JOCAS**—Job Order Cost Accounting System
**JON**—Job Order Number
**JTAC**—Joint Terminal Attack Controller
**JTE**—Joint Threat Emitter
**JTECH**—Joint Range Technical Services Contract
**LA ARTCC**—Los Angeles Air Route Traffic Control Center
**LEP**—Laser Eye Protection
**LFE**—Large Force Exercise
**LMR**—Land Mobile Radio
**LO**—Low Observable
**LOA**—Letter of Agreement
**LMS**—Live Monitor System
**LZ**—Landing Zone
**MAJCOM**—Major Command
**MARSA**—Military Assumes Responsibility for Separation of Aircraft
**MDS**—Mission Design Series
**MIPR**—Military Interdepartmental Purchase Request

**MOA**—Military Operations Area
**MOU**—Memorandum of Understanding
**MRTFB**—Major Range and Test Facility Base
**MSA**—Munitions Storage Area
**MSL**—Mean Sea Level
**MTN**— Multi-Tactical Data Link Network
**NACTS**—NTTR Air Combat Training System
**NAFB**—Nellis Air Force Base
**NAFBI**—Nellis Air Force Base Instruction
**NATCF**—Nellis Air Traffic Control Facility
**NDF**—Network Design Facility
**NAFC**—Nevada Area Frequency Coordinator
**NGA**—National Geospatial-Intelligence Agency
**NIZ**—No Impact Zone
**NLT**—Not Later Than
**NNSS**—Nevada Nuclear Security Site
**NNSA/SFO**—National Nuclear Security Administration/Sandia Field Office
**NNSS/NFO**—Nevada Nuclear Security Site/Nevada Field Office
**NOTAM**—Notice to Airman
**NPTEC**—Nonproliferation Test Evaluation Complex
**NTTR**—Nevada Test and Training Range
**NVD**—Night Vision Device
**OCR**—Office of Collateral Responsibility
**OD**—Optical Density
**OG**—Operations Group
**OI**—Operating Instruction
**O&M**—Operations and Maintenance
**OPR**—Office of Primary Responsibility
**OPTASKLINK**—Operational Tasking Link
**OSD**—Office of the Secretary of Defense
**OSS**—Operations Support Squadron
**OT&E**—Operational Test and Evaluation
**OWS**—Operations Weather Squadron
**OTF**—Over-the-Field
**PEX**—Patriot Excalibur
**PID**—Program Introduction Document
**POC**—Point of Contact
**POM**—Program Objective Memorandum
**POV**—Privately Owned Vehicle
**PSP**—Project Support Plan
**RA**—Resource Advisor
**RBA**—Reimbursement Budget Authority
**RCO**—Range Control Officer
**RDS**—Records Disposition Schedule
**RF**—Radio Frequency
**RFI**—Radio Frequency Interference

**ROA**—Range Operating Authority
**ROC**—Range Operations Center
**RON**—Remaining Overnight
**ROO**—Range Operations Officer
**RPA**—Remotely-Piloted Aircraft
**RPM**—Reliability Prediction Model
**RRU**—Remote Range Unit
**RS**—Range Support or Reconnaissance Squadron
**RSA**—Range Safety Assessment
**RSS**—Range Support Services
**RTB**—Return to Base
**RTO**—Range Training Officer
**RWMC**—Radioactive Waste Management Complex
**SAA**—Satellite Access Authorization
**SADL**—Situational Awareness Data Link
**SAM**—Surface-to-Air Missile
**SAR**—Satellite Access Request
**SDZ**—Surface Danger Zone
**SE**—Safety
**SFS**—Security Forces Squadron
**ShOC-N**—Shadow Operations Center-Nellis
**SIGINT**—Signals Intelligence
**SLC ARTCC**—Salt Lake City Air Route Traffic Control Center
**SMO**—Spectrum Management Office
**SFO**—Sandia Field Office
**SOC**—Statement of Capability
**SOF**—Supervisor of Flying
**SUAS**—Small Unmanned Aerial System
**SRB**—Safety Review Board
**SRS**—Space Range Squadron
**STA**—Silverbow Tower Airspace
**T&E**—Test and Evaluation
**TEG**—Test and Evaluation Group
**TMG**—Test Management Group
**TOSS**—Television Ordnance Scoring System
**TP**—Target Practice
**TPECR**—Tolicha Peak Electronic Combat Range
**TPL**—Test Priority List
**TPT**—Target Practice Tracer
**TSDF**—Time Slot Duty Factor
**TSPI**—Time, Space, and Position Information
**TTR**—Tonopah Test Range
**TTRA**—Tonopah Test Range Airfield
**TTRO**—Tonopah Test Range Operations Office
**UAS**—Unmanned Aerial System
**UAV**—Unmanned Aerial Vehicle

**UHF**—Ultra-High Frequency
**UMTE**—Unmanned Threat Emitter
**UOC**—Urban Operations Complex
**USAFWC**—United States Air Force Warfare Center
**USAFWS**—United States Air Force Weapons School
**USFWS—**United States Fish and Wildlife Service
**UTC**—Coordinated Universal Time
**UTM**—Universal Transverse Mercator
**UTTR**—Utah Test and Training Range
**UXO**—Unexploded Ordnance
**VDZ**—Vertical Danger Zone
**VFR**—Visual Flight Rules
**VHF**—Very-High Frequency
**VR**—VFR Route
**VTS**—Vehicle Tracking System
**V/STOL—**Vertical/Short Takeoff and Landing
**WD**—Weapons Director
**WDZ**—Weapon Danger Zone
**WG**—Wing
**WSINT**—Weapons School Integration Phase

***Terms***

**Air Traffic Control Assigned Airspace (ATCAA)** — Airspace with defined vertical/lateral limits, assigned by ATC, for the purpose of providing air traffic separation between the specified activities being conducted within the assigned airspace and other IFR air traffic. See FAA Order JO 7400.2N, Procedures for Handling Airspace Matters, for further information.

**Joint-Use** — Term used to describe a restricted area procedure that provides for the operation of nonparticipating IFR and/or VFR aircraft within the area. Flight within the restricted area is controlled by the using agency except when the area has been released to the controlling agency. During such periods, the controlling agency may permit nonparticipating aircraft operations in the restricted area. Implementation of joint-use in NTTR restricted areas has been established by assigning an ATC facility as the controlling agency and by executing a joint-use letter of procedure between the controlling and using agencies.

**Military Operations Area (MOA)** — Airspace designated outside of Class A/B/C airspace to separate certain non-hazardous military activities from instrument flight rules (IFR) traffic and to identify for visual flight rules (VFR) traffic where these activities are conducted. MOAs are designated to contain non-hazardous, military flight activities including, but not limited to, air combat maneuvers, air intercepts, low altitude tactics, etc. Special Use Airspace VFR aircraft are not restricted from transiting MOAs. See FAA Order JO 7610.4W, Special Operations, for further information.

**No-Drop Target** — A no-drop target is defined as a target upon which neither live nor inert ordnance may be employed. A no-drop target may have previously been designated a live (i.e., available for live/inert ordnance employment) target, but may be designated as no-drop by HQ NTTR, regardless of current condition. A no-drop target may be reactivated as a live/inert target by HQ NTTR when: 1) all applicable range clearance actions have been accomplished for the target or 2) declared safe for weapons employment by qualified EOD personnel. Range clearance on no-drop targets are accomplished no later than five years since last completed.

**Range Control Officer (RCO)** — The person responsible for range operations and safety. Except in situations where the RCO delegates weapons release clearance to a qualified flight lead, individual pilot or forward air controller, or other briefed person.

**Range Operating Authority (ROA)** — The agency designated to operate and maintain the range. ROA may delegate the daily scheduling, management, and maintenance of the range to any appropriate subordinate unit.

**Range Operations Officer (ROO)** — The individual responsible for day-to-day operating activities.

**Remotely-Piloted Aircraft (RPA)** — A powered, aerial vehicle that does not carry a human operator, uses aerodynamic forces to provide vehicle lift, can fly autonomously or be piloted remotely, can be expendable or recoverable, and can carry a lethal or nonlethal payload. Also referred to as Unmanned Aerial System (UAS) or Unmanned Aerial Vehicle (UAV). Ballistic or semi-ballistic vehicles, cruise missiles, and artillery projectiles are not considered RPAs.

**Range Classification Service:**

**Class A** — Range is manned, has a ground-based scoring capability, and has a RCO on the ground that controls aircraft using the range.

**Class B** — Range is either manned or unmanned, has a ground-based scoring capability, but does not have a RCO on the ground controlling aircraft. The flight lead, individual pilot, FAC, or other briefed person performs the RCO function.

**Class C** — Range is unmanned, with no scoring or aircraft control from the ground. The flight lead, individual pilot, FAC, or other briefed person performs the RCO function.

**Class D** — An instrumented air-to-air range, manned by a Range Training Officer (RTO) who maintains radio contact with aircraft on the range during air combat training, as required.

**Restricted Area** — A restricted area is airspace established under 14 CFR part 73, Special Use Airspace, provisions, within which the flight of civilian aircraft, while not prohibited, is subject to restriction. Restricted areas are established when necessary to confine or segregate activities considered hazardous to nonparticipating aircraft. Restricted areas can include: An area (land, sea or air) in which there are special restrictive measures employed to prevent or minimize interference between friendly forces or an area under military jurisdiction in which special security measures are employed to prevent unauthorized entry; airspace where the flight of aircraft, while not wholly prohibited, is subject to restriction. When not activated by the using agency, the controlling ATC facility may authorize IFR or VFR operations in the area. If joint-use is authorized, the name of the ATC controlling facility is annotated on the map; or an area that must contain all "Hazardous Activity" as defined by branch of service for specific type of aircraft using the range. See FAA Order JO 7610.4W, Special Operations, for further information.

**Site** — A manned or unmanned, fixed ground location associated with a threat emitter/system, communications equipment, or scoring system.

**Traditional Aircraft** — Refers to directly manned aircraft, as opposed to RPAs.

## Attachment 2

## NTTR BUSINESS MANAGEMENT PROCESS (BMP)

**A2.1. BMP.** The NTTR BMP provides a procedural framework for the planning of range support to range events that are managed by HQ NTTR. As the HQ NTTR lead for the BMP, the NTTR Projects Office provides the initial entry point for use of the NTTR. The NTTR Projects Office conducts a monthly synchronization meeting which includes project managers, Range Safety, Range and Resource Scheduling, Range Support Operations, NTTR/FM, DOE liaison, NTTR/PM, JTECH representative, and an RSS representative. Other agencies may participate based upon their project support. NTTR Projects Office provides routine range projects updates to the ROO.

A2.1.1. All users conducting activities (other than CT) will initiate their desired usage through the BMP. New CT customers will coordinate through the Projects office initially.

A2.1.2. All users coordinating CT sorties will initiate their desired usage through the NTTR Scheduling Process, rather than the BMP. New CT customers will coordinate through the Projects office initially.

**A2.2. BMP Phases.** The NTTR BMP is segmented into five phases, each of which is further segmented into a time-phased sequence of milestones.

A2.2.1. Phase I: Requirements Definition. This phase begins with the identification of a new requirement, continues through the processing of the Program Introduction Document (PID) and ends with requirements identification completion.

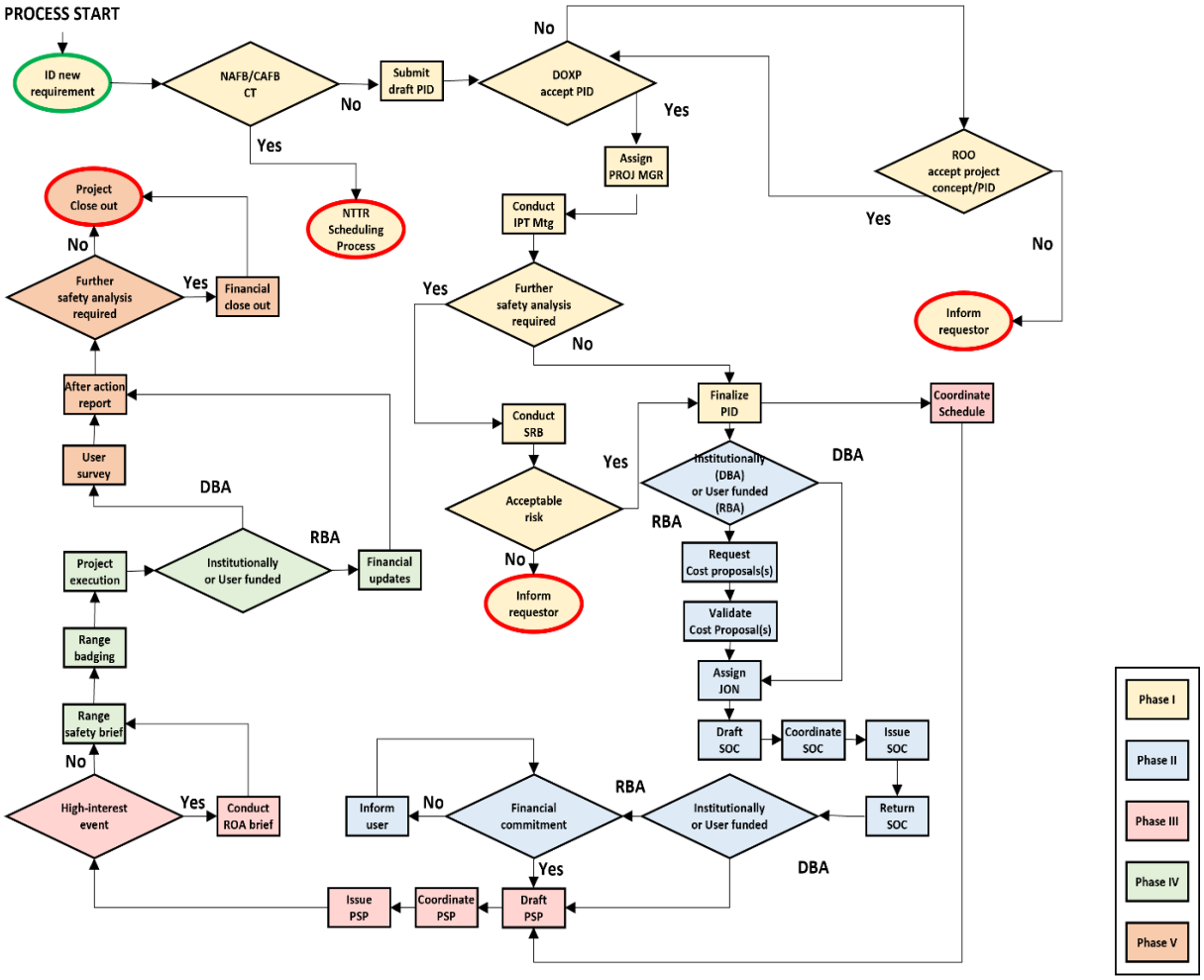
A2.2.2. Phase II: Financial Commitment. This phase begins with the determination of the event’s institutionally-funded or customer-funded status and ends with the return of the Statement of Capability (SOC) by the customer for institutionally-funded events or receipt of a certified funding document for customer-funded events.

A2.2.3. Phase III: Support Tasking. This phase begins with the start of schedule coordination and ends with publishing of the Project Support Plan (PSP) or conduct of ROA brief, when required.

A2.2.4. Phase IV: Execution. This phase begins with the delivery of the range safety briefing to the customer and ends with the completion of execution.

A2.2.5. Phase V: After Action. This phase begins with coordination of the Customer Satisfaction Survey and ends with project closeout.

**Figure A2.1. NTTR BMP Phases and Milestones.**




**A2.3. Late Submissions.** Late submission of required products or funding may limit the support available for a given activity and/or may increase costs to the requestor. PIDs received later than the BMP timeline suspense must be endorsed by the first O-6 (or equivalent) or higher in the chain-of-command of the requesting organization.

**A2.4. Phase I:** Requirements Definition. During this phase, the potential event is considered for acceptance as an NTTR event; the PID is drafted, submitted, and coordinated; the Integrated Project Team (IPT) meeting is held; the SRB, if required, is conducted; and the PID is finalized. The project manager is assigned during this phase and will act as the direct interface for all customer requirements. Local Nellis/Creech AFB users wishing to conduct CT sorties will be referred to the NTTR Scheduling Process.

A2.4.1. To be accepted onto the NTTR, at a minimum, events must fall within applicable safety and regulatory guidance; must not exceed NTTR's physical and logistical capabilities; and meet NTTR BMP timelines.

A2.4.2. The PID is a statement of project support requirements drafted by the prospective customer. If assistance is required contact the Chief of Projects. The PID formally documents the customer's requirements. The level of detail will depend on the scope and complexity of project support requested. The project manager will coordinate the PID with:

**Table A2.1. Typical PID Coordinating Agencies.**

| NTTR | External Agencies | Contractor Agencies |
|---|---|---|
| DO and associated Divisions | DOE | JTECH II Operations Manager |
| MS and associated Divisions | NNSS/NFO | RSS Operations Manager |
| FM, PM, RS, SE, XP, SD, AP | NNSA/SFO | |

A2.4.3. PID Response: Directors or designated representatives will staff the PID through their organizations and respond based on their staff's recommendations. Once directorates provide feedback on project support requirements, the project manager will inform the customer of any non-supportable requirements.

A2.4.4. The IPT meeting is a mandatory event chaired by the project manager. The IPT meeting is attended by the customer, representatives from all HQ NTTR directorates, and subject matter experts. The IPT meeting provides the user a forum to directly discuss project requirements with support agencies. The IPT meeting allows NTTR directorates to identify supportability requirements and issues and determine available methods to meet user needs. Directorate representatives at the IPT meeting must be prepared to discuss any activity within their area of responsibility and possess their directorate's decision-making authority. The project manager gathers additional information from the IPT meeting to include in the PID.

A2.4.5. The IPT will determine if the project requires a SRB. If so, NTTR/SE will convene a SRB to determine whether the project risk is acceptable. NTTR Projects Office relays the SRB decision, any RSA results, and associated reason(s) for restrictions or project rejection to the requestor.

**A2.5. Phase II:** Financial Commitment. During this phase, DBA (institutional funding) versus RBA (customer funding) source is determined; a proposal, if required, is developed; a JON is assigned; a SOC is developed and returned; and the user commits to providing any required funding.

A2.5.1. In support of customer-funded events, NTTR/PM will request a cost proposal from the RSS contractor articulating required materials and labor and identifies any long-lead time articles.

A2.5.2. A JON is assigned for all events. The JON is an NTTR/FM-provided accounting code that allows for accurate tracking of billable activities. A JON must be assigned and funds in place before accountable expenditures are made in support of an event.

A2.5.3. The SOC formally details HQ NTTR’s supportability response to the PID, including cost factors. It includes a list of those items deemed unsupportable. The project manager will coordinate the SOC with the same agencies who coordinated on the associated PID.

A2.5.3.1. The SOC is approved and signed by the NTTR ROO and the customer or agent authorized to commit project funding. Once the SOC is signed by these individuals, it becomes both a financial agreement to fund the project and the support contract to execute the project between the customer and the HQ NTTR. The customer's resource advisor (RA) and NTTR/FM will coordinate delivery of a certified funding document and funds transfer.

A2.5.3.2. Post-Approval Changes: The project manager will often need to address multiple project changes after the SOC is approved, up to the end of project execution. The project manager will coordinate customer needs with directorates for approval/ supportability and keep all parties informed on changes. The project manager will keep a record (printed e-mails, network folder, etc.) of all requested changes for documentation and associated cost recovery. This record will include the level of change by the requestor (i.e., DO) as well as detailed SOC changes. SOC amendments and revised cost estimates may be required to satisfy additional project requirements and cost recovery.

**A2.6. Phase III:** Support Tasking. During this phase, the PSP is developed; scheduling is accomplished; and ROA coordination is accomplished, when required.

A2.6.1. The PSP formally tasks NTTR-supporting agencies with support to the event. The PSP provides the framework of services required for contractors and government support agencies. The project manager will coordinate the PSP with the same agencies who coordinated on the associated SOC.

A2.6.2. Post-Approval Changes: The project manager will often need to address multiple project changes after the PSP has been approved, through the end of project execution. The project manager will coordinate customer needs with directorates for approval/ supportability and keep all parties informed on changes. The project manager will keep a record (printed e-mails, network folder, etc.) of all requested changes for documentation and associated cost recovery. This record will include the level of change by the requestor (i.e., DO) as well as detailed PSP changes. PSP amendments and revised cost estimates may be required to satisfy additional project requirements and cost recovery.

A2.6.3. Scheduling coordination responsibility through the NTTR Scheduling Process may fall to the project manager or the unit scheduler and is agreed upon at the IPT.

A2.6.4. Within the two-week period prior to execution start, the project manager will submit/coordinate customer-provided personnel and sensitive equipment information to NTTR/SD to obtain range safety briefings, restricted area temporary badges, and sensitive equipment access approval. NTTR Projects Office may assign additional project managers to provide on-site, bed down and coordination support.

**A2.7. Phase IV:** Execution. During this phase, HQ NTTR monitors event execution and adjusts IAW customer needs and organizational constraints.

A2.7.1. The project manager represents HQ NTTR for all aspects of project execution support during this phase. The project manager is responsible for resolving immediate execution support problems, providing broad, in-depth NTTR expertise and coordinating with directorates on support issues. Any change to the scope of effort that drives additional cost must be coordinated through project manager and NTTR/PM; NOT the contractor.

A2.7.2. Long-Duration Projects: For projects which exceed two months duration, the project manager will review NTTR/FM provided, JOCAS-based financial reports during the first week of each month to validate and update actual project costs and remaining funds on the project's account. The project manager will contact NTTR Range Scheduling and the customer as necessary, to reconcile project schedule/activity cost discrepancies. NTTR/FM will forward the financial reports to the user for their fiscal management planning. If the user requires a project's funds to be reallocated, the user's RA and NTTR/FM will coordinate funds transfer.

**A2.8. Phase V:** After Action. During this phase, HQ NTTR collects and analyzes feedback to help shape future activities as well as providing a final cost reconciliation to the user.

A2.8.1. The project manager will schedule a project-specific "hotwash" to discuss any significant issues identified during the project lifecycle. The hotwash participants will review identified issues, devise solutions, assign an OPR if necessary, and set a suspense date that is mutually agreed upon. Additionally, within two days upon completion of the project's activity, the project manager will send a Customer Satisfaction Survey (CSS) to the customer to obtain feedback on all aspects of the HQ NTTR's planning and execution support. The CSS is a crucial tool utilized in informing the decisions that shape NTTR processes and systems. Customers accept the obligation to complete the CSS.

A2.8.2. The project manager authors the NTTR AAR (if necessary). The AAR captures data via the project manager's observations while overseeing the project, hotwash, and CSS feedback. The AAR summarizes issues and lessons learned throughout the project's lifecycle, to include external inputs and internal action items. Once the AAR is complete, the project manager staffs the document to the NTTR directorates for review and comment, and obtains ROO approval of the report to close out the project. NTTR action items are referred to NTTR/XP for any required tracking and to support range capability requirements definition.

A2.8.3. The final step in the project is financial reconciliation. The customer, project manager, and NTTR/FM will collaboratively arrive upon final, reconciled project costs and NTTR Projects Office will distribute a project closeout letter to NTTR Range Scheduling, NTTR/FM, NTTR/PM, the JTECH II Operations Manager and the RSS Operations Manager. NTTR/FM subsequently will close out all financial commitments with the customer's RA and close the JON.

**A2.9. BMP Roles and Responsibilities.** The following tables lists key HQ NTTR BMP roles and responsibilities. It is not intended to be comprehensive, but rather it identifies those activities most commonly conducted as part of the NTTR BMP.

**Table A2.2. NTTR/DOXP (Projects Office) Key BMP Roles and Responsibilities.**

| |
|---|
| • Respond to initial contact by prospective customers<br>• Explain general BMP timeline requirements and flow<br>• Provide customers with PID template and assist in submitting necessary documents<br>• Gather customer project requirements and provide to NTTR directorates<br>• Coordinate the PID, SOC, PSP, and AAR with the NTTR directorates<br>• Schedule, conduct and chair the IPT and manage the development of the PID<br>• Provide the customer with the SOC and overall project cost estimate for approval<br>• Coordinate for the scheduling of air and ground space<br>• Provide project brief to ROO and ROA<br>• Review JOCAS-based monthly FM reports for project cost reconciliation<br>• Schedule, conduct and chair hotwash meeting<br>• Provide CSS to customer<br>• Author the AAR, validate and assign action items and suspense dates<br>• Provide execution oversight of the project<br>• Meet with NTTR/FM and customers to reconcile final project costs<br>• Issue project closeout letter<br>• Coordinate Special Use Airspace Letter approval and GPS testing approval, schedule/conduct meeting as required |

**Table A2.3. NTTR/DOXS (Range & Resource Scheduling) Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Schedule air and ground space and threat support<br>• Provide NTTR Projects Office range utilization reports, if required<br>• Review the AAR for assigned action items |

**Table A2.4. NTTR/DOO (Current Operations) Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Provide operational support for the project<br>• Review the AAR for assigned action items |

**Table A2.5. NTTR/DOS (Target Management) Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Provide operational support for the project<br>• Review the AAR for assigned action items |

**Table A2.6. NTTR/DO (ROO) Key BMP Roles and Responsibilities.**

| |
|---|
| • Review and approve the SOC, PSP and AAR<br>• When required, review project proposals for execution on NTTR |

**Table A2.7. NTTR/FM Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT (if required by the project manager)<br>• Issue a project JON to the project manager<br>• Coordinate customer funds transfer with unit RA<br>• Provide project manager JOCAS-based monthly project charge reports<br>• Review the AAR for assigned action items<br>• Provide the project manager with the final project cost report<br>• Provide the customer with the final project cost report<br>• Reconcile funding with unit RA |

**Table A2.8. NTTR/SD Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Provide support for the project IAW NTTR OI 31-101 |

**Table A2.9. NTTR/MS Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Perform technical analysis/validation of cost estimates as needed<br>• Monitor NTTR Projects Office project-associated 332s to completion<br>• Review the AAR for assigned action items |

**Table A2.10. NTTR/PM Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR; Ensure the SOC and PSP address PID-required RSS support<br>• Participate in the IPT<br>• Provide RSS contractor interface for the project<br>• Process proposal estimate requests and validate proposal estimate costs<br>• Review the AAR for assigned action items |

**Table A2.11. NTTR/RS Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Provide JTECH II contractor interface for project<br>• Process ROM estimate requests and validate proposal estimate costs<br>• Ensure the SOC and PSP address PID required JTECH II support<br>• Coordinate spectrum management clearances IAW the PID<br>• Provide technical support throughout project execution<br>• Review the AAR for assigned action items |

**Table A2.12. NTTR/SE Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Conduct SRBs<br>• Develop RSAs<br>• Ensure weapon profiles are provided<br>• Review the AAR for assigned action items |

**Table A2.13. NTTR/XP Key BMP Roles and Responsibilities.**

| |
|---|
| • Review the PID, SOC, PSP and AAR<br>• Participate in the IPT<br>• Consolidate, track and identify range capability shortfalls for POM input<br>• Review the AAR for assigned action items and HQ NTTR implications<br>• BLM land use and environmental coordination |

**A2.10. BMP Timeline.** The BMP 90 day timeline is used to support Nellis/Creech based requestors and non-Nellis/Creech based requestor LFEs. Complex support requirements may increase this timeline.

**Table A2.14. Nominal BMP Timelines.**

| Milestone | Project Timeline | | NTTR OPR/OCR |
|---|---|---|---|
| HQ NTTR receives PID[2] | ≥S - 90 | | Projects Office |
| Assign project manager | S - 90 | | Projects Office |
| Conduct IPT meeting | S - 85 | | Projects Office |
| Finalize PID | S - 85 | | Projects Office |
| Determine SRB requirement | S - 85 | | SE |
| Request JON | S - 69 | | FM |
| Request proposal *See Note 4 | S - 83 | | PM/RS |
| Validate proposal | S - 62 | | PM/RS |
| Begin SOC e-coordination | S - 63 | | Projects Office |
| Complete SOC e-coordination, provide to customer | S - 60 | | Projects Office |
| Complete SRB | S - 30 | | SE |
| Range customer return SOC | S - 30 | | Projects Office |
| Begin PSP e-coordination | S - 23 | | Projects Office |
| Complete PSP e-coordination | S - 20 | | Projects Office |
| Provide PSP to range support agencies | S - 20 | | Projects Office |
| Conduct senior leader brief (ROO) | S - 14 | | Projects Office |
| Receive certified funding document | S - 14 | | FM |
| Rescind JON (if funding not received) | S - 13 | | Projects Office |
| Issue task order | S - 03 | | PM |
| Conduct range safety brief | S - 01 | | SD/SE |
| Conduct senior leader brief (ROA) | S - 01 | | Projects Office |
| Project execution[3] | -------- | | --------- |

| | | | |
|---|---|---|---|
| Send CSS to range customer | E + 02 | | Projects Office |
| Hotwash | E + 07 | | Projects Office |
| Range customer returns CSS | E + 23 | | Projects Office |
| Begin AAR e-coordination | E + 30 | | Projects Office |
| Complete AAR e-coordination | E + 40 | | Projects Office |
| Finalize AAR and distribute | E + 45 | | Projects Office |
| Validate final costs | E + 90 | | FM/Projects Office |
| Reconcile final funding | E + 99 | | FM |
| Provide project closeout letter | E + 99 | | Projects Office |

Note 1: Expressed in Calendar Days, Plus/Minus Execution Days. “S” Day indicates the day of support start (which may not coincide with the start of event execution); “E” Day indicates the day of event execution end.
Note 2: In support of the unique planning timelines associated with Exercise RED FLAG, the PID is required NLT seven calendar days following the final planning event.
Note 3: NTTR Projects Office will provide financial updates to the customer, as required.
Note 4: Suspense for proposal is 21 days from complete requirements. Any requirement changes after a Request for Proposal is issued to the contractor may restart the 21-day count.

**A.2.11. Support to commercial activities.** All commercial activity managed by HQ NTTR and conducted on the NTTR and/or R-4808 must be specifically coordinated and approved through NTTR Projects Office. As part of the application of the NTTR BMP to commercial activities, special care must be given to ensure all key sponsoring agencies are provided the opportunity to consider the implications of the proposed activity.

A2.11.1. All commercial organization activity will be vetted through the Air Force Office of Special Investigation. It is crucial for the requesting agency to identify foreign-national ownership or participation in the requested activity no later than 45 days prior to execution. An EAL and Sensitive Equipment List (SEL) must be provided. Submissions after this date may not be approved.

A2.11.2. Operating restrictions on commercial activities may be imposed, and trained escorts may be required, based on proximity to sensitive range activities. Any data collection/distribution capabilities utilized as part of the proposed commercial activity must be identified and assessed with respect to risk to core NTTR sensitive range activities.

A2.11.3. Sensor payloads, flight test telemetry streams, command and control links, and other RF emanations, as well as any live, inert, and training ordnance associated with commercial activities must be fully reviewed for possible safety and security impacts to NTTR core missions.

**A2.12. Commercial Customer Questionnaires.** As part of the NTTR BMP, prospective commercial customers must submit “Commercial Customer Questionnaires” (Tables A2.15-A2.19) in conjunction with the PID IAW NTTR BMP timelines. Any questions that are not answered at the time of submittal may delay or preclude approval.

**Table A2.15. Commercial Customer Questionnaire: General Description.**

- Name of company requesting range access?
- Event name and desired dates and times (if known)?
- Event objective? For test events, provide test plan.
- Commercial and Government Entity (CAGE) Code?
- Company's primary address, telephone, and email address?
- Company point of contact/principal officer?
- Any foreign interest/investment involved in company, management, or proposed activity? If YES, explain.
- Any foreign nationals involved in the execution of the event? If YES, explain.
- Local host unit(s) (if any)?
- Has the requesting customer operated on the NTTR and/or R-4808 before? When? Why?
- Will this event require a pre-visit/site survey? If YES, when, where, air/ground/both?

**Table A2.16. Commercial Customer Questionnaire: General Equipment.**

- Individually list all mission equipment intended to be used. For each piece of equipment, provide name, type, specifications, manufacturer, and details of expected usage.
- Location of mission equipment on or near the NTTR supporting the event?
- Will it be necessary to erect towers/antennas?
- Can mission equipment be turned/powered on and off remotely?
- Is mission equipment required to be left overnight on the NTTR? If YES, describe desired safeguards.
- Will experimental equipment/software be used?
- What will be the disposition of mission equipment after the event is complete?
- Aircraft involved? If YES, type/model, call sign, and IFF Code.
- Will any type of unmanned system be involved? If YES, explain and provide type/model, call sign, and IFF Code (as applicable).
- Is space-based (commonly referred to as “overhead imagery”) and signals intelligence (SIGINT) collections intended as part of the event?

**Table A2.17. Commercial Customer Questionnaire: Sensors and Dissemination.**

- Are any imaging systems intended to be used? If yes, obtain sensitive equipment passes IAW Attachment 21 (Imagery Processes and Restrictions).
- Location of sensor equipment on or near the NTTR supporting the event?
- Can sensor equipment be turned/powered on and off remotely?
- Is sensor equipment required to be left overnight on the NTTR? If YES, describe desired safeguards.
- Will experimental sensor equipment/software be used?
- What will be the disposition of sensor equipment after the event is complete?
- Can the sensor(s) be turned on and off during the mission?
- Sensor(s) field of view?
- Sensor(s) field of regard?
- Sensor(s) resolution?
- Sensor(s) range?
- Where does sensor data flow after it is collected?
- Is it a live feed or will it be recorded/processed first?
- Is data downlinked to a ground station or retained on-board?
- If necessary, can the data be edited prior to distribution or storage?
- What is the final delivery and storage destination of the data?
- At what security level will the data be maintained?
- Will stored data be retained in it be encrypted format? If YES, at what encryption level?
- Will stored data be treated as “US Only?”
- Who will have access to collected data?
- How long will the data be retained after the event?
- Will FMV equipment to be used? If yes, submit FMV Questionnaire IAW Attachment 21, Imagery Processes and Restrictions).

**Table A2.18. Commercial Customer Questionnaire: EMI/RFI.**

- Will equipment transmit RF signals? If YES, what are the operating frequencies and power levels?
- Does equipment require special assistance from spectrum management?
- Provide "Cease Buzzer” information and locations (if applicable).

**Table A2.19. Commercial Customer Questionnaire: Ordnance.**

- What live, inert, and/or training ordnance is intended to be utilized?
- Has the intended ordnance been operated on the NTTR and/or R-4808 by the requesting user before? When? Why?
- Is experimental ordnance intended to be utilized?

## Attachment 3

## SCHEDULING AND CONFLICT ADJUDICATION

**A3.1. NTTR Subrange Scheduling.** NTTR Range Scheduling schedules the following activities in a manner that safely optimizes coordinated range use.

**Table A3.1. Scheduled Air and Ground Activities Upon NTTR Subranges.**

| FAA Airspace Designation | FAA Sub-Airspace Designation | NTTR Subrange Designations | Air | Ground |
|---|---|---|---|---|
| R-4806 | R-4806E | Alamo A/B/C | Yes | No |
| | R-4806W | R-61A | Yes | Yes |
| | | R-61B | Yes | Yes |
| | | R-62A/B | Yes | Yes |
| | | R-63A/B/C | Yes | Yes |
| | | R-64A/B/C | Yes | Yes |
| | | R-64D/E/ | Yes[2] | No |
| | | R-65B/C | Yes | Yes |
| | | R-65A/D | Yes | Yes |
| | | R-67A/B | Yes[1] | Yes |
| R-4807 | R-4807A | R-71N/S | Yes | Yes |
| | | R-74A/B/C | Yes | Yes |
| | | R-75E/W | Yes | Yes |
| | | R-76 | Yes | Yes |
| | | EC East | Yes[3] | No |
| | | R-4807A (Subset) | Yes[3] | No |
| | | R-77A/B | Yes | Yes[10] |
| | | TPECR | Yes[4] | No |
| | R-4807B | Pahute A/B | Yes | No |
| R-4808 | R-4808N | R-4808A | No[5] | No |
| | | R-4808B/C/D/E | Yes[6] | No |
| | R-4808S | R-4808S | No[5] | No |
| R-4809 | EC West | EC West | Yes[7] | No |
| | R-4809A | R-4809A | Yes | No |
| | R-4809B | R-4809B | Yes[7] | No |
| Reveille North MOA | Reveille North MOA and ATCAA | Reveille North | Yes[8] | No |
| Reveille South MOA | Reveille South MOA and ATCAA | Reveille South | Yes | No |
| Desert MOA | Desert MOA and ATCAA | Coyote A/B/C/D | Yes | No |
| | | Caliente A/B/C/D | Yes | No |
| | | Elgin | Yes | No |
| | | Sally Corridor | Yes[9] | No |

Note 3: For scheduling purposes, EC East includes R-4807A (Subset) ground/airspace.
Note 4: For scheduling and supersonic operations purposes, TPECR includes FAC A/B ground/airspace. FAC A/B includes the road leading to TPECR and are to be considered continually manned.

Note 5: R-4808A and R-4808S ground/airspace are not available through the NTTR Scheduling Process.
Note 6: R-4808B/C/D/E ground space is not available through the NTTR Scheduling Process; limited R-4808B/C/D/E airspace is normally available through the NTTR Scheduling Process.
Note 7: For scheduling purposes, EC West includes R-4809B ground/airspace.
Note 8: Reveille North above 30,000ft MSL, is only available through the NTTR Scheduling Process when released by SLC ARTCC.
Note 9: Sally Corridor is normally assigned to NATCF. Use of this airspace requires special coordination. Note 10: Ground parties are not scheduled for R-77B.

**Table A3.2. Scheduled Air & Ground Activities Upon NTTR Additional Specified Areas.**

| FAA Airspace Designation | NTTR Sub-Range Designation | Additional Specified Areas | Air | Ground |
|---|---|---|---|---|
| R-4806 | Alamo A/B/C | Alamo Corridor | Yes | No |
| | R-63D | R-63D Low | Yes | NA |
| | | R-63 D High | Yes | Yes |
| | R-64C | CTA | Yes[2] | No |
| | R-64A[1] | Air-to-Air Gunnery Area | Yes | Yes |
| R-4808 | R-4808D | Gomer Triangle | Yes | No |
| R-4809 | R-4809A | STA | Yes[3] | No |
| | | Shelf | Yes | No |
| Reveille South MOA | Reveille South MOA | Dorito | Yes | No |
| Desert MOA | Desert MOA and ATCAA | X-Ray Extension | No[4] | No |

Note 1: Although centered within R-64A, the Air-to-Air Gunnery Area also includes portions of R-61A, R-62A/B, R-64A/B, and R-65A/B/C.

Note 2: CTA is assigned to Creech Tower. Use of the airspace requires special coordination. Note 3: STA is assigned to Silverbow Tower. Use of the airspace requires special coordination.

Note 4: The X-Ray Extension is only available when released for use by NATCF. The request is made by the user directly to NATCF.

A3.1.1. NTTR Range Scheduling utilizes the Center Scheduling Enterprise (CSE), a computerized man-in-the-loop system, for the official scheduling of mission activities and generation of the daily range schedule. The daily range schedule reflects assigned range time in 15-minute increments.

A3.1.2. All users coordinating CT sorties, will initiate their desired usage through the NTTR Scheduling Process, rather than the BMP. New CT Customers will coordinate through the Projects office initially.

A.3.1.3. All users conducting activities other than CT will initiate their desired usage through the BMP, rather than the NTTR Scheduling Process.

**A3.2. Blackjack.** NTTR Range Monitoring provides 24-hour support. Missions requiring NTTR Range Monitoring support beyond 10 hours (Sat-Sun) require ROO approval and may incur additional costs to the user.

**A3.3. NTTR Scheduling Requirements.** All units must submit their NTTR scheduling requirements to NTTR Range Scheduling through CSE or PEX squadron mission reports (also referred to as "pencils"). Units may modify pencils according to unit scheduling requirements but must include, as a minimum, the following:

**Table A3.3. NTTR Scheduling Requirements.**

| |
|---|
| • Range(s), range time(s)<br>• JON<br>• Unit Designation<br>• Scheduler's name and contact number<br>• Adversary participants<br>• Number and type of aircraft<br>• Callsign(s)<br>• Identification friend or foe code<br>• Ordnance type(s) and quantity (including dynamic retargeting requirements), target if known<br>Desired foreign participation |
| NOTE: NTTR Projects Office will provide scheduling requirements for those units with active projects that do not have access to CSE or PEX. |

A3.3.1. CSE and GRR access is available upon request. Requests for new CSE user accounts may be submitted to NTTR Range Scheduling. The CSE website is located here: https://csenellis2.eglin.af.mil/cse/home.aspx. The GRR website is an aircrew quick-reference mission planning tool with limited information that utilizes a latest version imperative for up-to- date coordination. Access to the GRR website does not require a CSE account. The GRR website is located here: https://csenellis2.eglin.af.mil/cse/GRR/GRRMain.aspx and CSE is located here: https://csenellis2.eglin.af.mil/cse/Home.aspx-.

A3.3.2. Laser operations. When combat mode laser operations are requested, NTTR Range Scheduling and NTTR Range Monitoring (as applicable) will annotate it in the daily range schedule. Additionally, a "No Combat Laser" mission restriction may be added to the daily range schedule as applicable. During combat laser operations, NTTR ground parties are required to wear laser eye protection (LEP) within all subranges where hazardous laser energy may terminate. "LEP required" will be annotated on the daily range schedule for the affected subranges (i.e. Range 62A, Range 63B, Range 77, etc.).

A3.3.3. Support requirements: NACTS; GCI; ITAS; TOSS; EC requirements; JDNC; HSMT; threat emitters; etc.

A3.3.4. Activities that include ground parties (i.e., JTACs, convoys, and mobile threats), require specific coordination and deconfliction.

A3.3.4.1. Items essential to the schedule building process (i.e., target identification and ordnance quantity and type) may realistically not be known and/or available at the weekly scheduling meeting. It is paramount that range users inform NTTR Range Scheduling as soon as possible, or NTTR Range Monitoring (within 48 hours of execution) so the daily range schedule can be updated and deconflicted.

A3.3.4.2. All personnel requiring NTTR range access for ground party activity must comply with established guidelines. Ground party activities on NTTR subranges that require scheduling (as indicated in Tables A3.1 and A3.2) must be approved through the NTTR Scheduling Process.

**A3.4. Adjudication.** The NTTR Scheduling Adjudication Process provides a standardized framework for thorough evaluation and administration of range user requirements of all air and ground activities conducted on the range whenever a range scheduling conflict exists.

A3.4.1. The NTTR Chief of Range Scheduling and Resources is responsible for the initial administration of the range adjudication process and is the first step for resolving scheduling conflicts. The Chief of Range Scheduling and Resources will conduct a comprehensive review of all dispute-related information; refer to standard range priorities and information such as established TPOs, operational tasking information, syllabi, tactics development, training requirements, and timeliness of information. The Chief of Range Scheduling and Resources may request an impact statement from disputed parties to facilitate conflict resolution and mitigation.

A3.4.2. If either of the disputed parties is not satisfied with the adjudication decision, they may request an adjudication review at the group (or equivalent) level. All information gathered in the initial review will be forwarded and the process will continue, following the same adjudication procedures, until the issue is resolved. In the unlikely event that the issue cannot be resolved at that level, it will be referred to the wing (or equivalent) level and so forth until resolution is achieved.

A3.4.3. R65D Deconfliction. The Manned Beno Line (MB) is available as a reference tool to deconflict manned aircraft and RPA operations within R65D. The MB is not a formal boundary; its usage and implementation is based on the mutual agreement between participating units via direct coordination. When utilized, the MB is defined as a line connecting N36 40.39 W115 56.05 to N36 37.62 W115 47.40 to N36 37.62 W115 43.05 The agreements must be documented and sent to the NTTR Scheduling Office and Range Scheduling (Blackjack) prior to the submission of the schedule into CSE. CSE missions will reflect the agreement with both units. Each unit will be assigned the airspace with the appropriate agreed deconfliction. A restriction will be entered into the “remarks” section that contains the concise, agreed-upon deconfliction plan. It will be the responsibility of the participating units to adhere to their written agreement which are reflected in the mission restrictions in CSE.

**A3.5. Timeline.** The NTTR scheduling timeline is organized into three progressive phases: planning, processing, and execution.

A3.5.1. The NTTR schedule planning phase is a collaborative and integrated undertaking that is led by the Chief of Range Scheduling and Resources. The NTTR schedule planning phase provides a method for consolidating long-term NTTR scheduling issues and requirements from all range users. The planning phase ranges from 12 months to 90 days prior to the month of execution.

A3.5.2. The NTTR schedule processing phase translates long-term requirements into an executable schedule.

A3.5.2.1. The NTTR schedule processing phase begins with the release of the scheduling "shell" by the NTTR Chief of Range Resource and Scheduling to DOE, no later than 90 days prior to the month of execution.

A3.5.2.2. Once DOE receives the shell, DOE range requirements will be collectively submitted to NTTR Chief of Range Resource and Scheduling no later than 60 days prior to the month of execution.

A3.5.2.3. Once DOE mission requirements are reflected on the shell, the NTTR Chief of Range Resource and Scheduling will collaborate with 57 OSS/OSO to identify any scheduling conflicts. NTTR Range Scheduling will release the shell to 57 OSS/OSO no later than 60 days prior to the month of execution.

A3.5.2.4. Once 57 OSS/OSO receives the shell, requirements for Nellis, Creech, and ACC will be coordinated by 57 OSS/OSO. The 57 OSS/OSO will return the updated spreadsheets to the NTTR Chief of Range Scheduling and Resources no later than the Tuesday prior to Range Integration (approximately 45 days prior to the month of execution.)

A3.5.2.5. 57 WG Range Integration meeting (45-day). The 57 WG Range Integration meeting is conducted by the 57 OSS/OSOS approximately 45 days prior to the month of execution, on the last Monday of each month (unless published otherwise). The purpose of this meeting is for Nellis, Creech, and ACC range users to adjust the shell, and coordinate and confirm individual range requirements with 57 OSS/OSOS, while identifying and attempting to resolve potential scheduling conflicts prior to the 30-day NTTR Range War meeting. Once all requirements are reflected on the shell, the product is collectively referred to as the "spreadsheets." Any changes to the spreadsheets post Range Integration and Pre-Range War will be identified to the NTTR Chief of Resource and Scheduling.

A3.5.2.6. NTTR Range War meeting (30-day). The NTTR Range War meeting is conducted by NTTR Range and Resource Scheduling and conducted in the HQ NTTR building approximately 30 days prior to the month of execution on the first Friday of each month (unless published otherwise). The purpose of this meeting is for all NTTR range users and associated project managers to adjust the spreadsheets and coordinate individual range requirements with NTTR Range and Resource Scheduling, while

identifying and attempting to resolve potential scheduling conflicts prior to the NTTR Pencil meeting. Although the spreadsheet is matured through the NTTR Range War meeting, it does not yet reflect the official schedule. It is further refined through the NTTR Pencil meeting before final entry into CSE as the official schedule.

A3.5.2.6.1. Post NTTR Range War meeting, previously unidentified user requirements are scheduled on a first-come, first-served basis.

A3.5.2.6.2. The unit scheduler will provide any RPA full-motion video (FMV) requests to NTTR Range Scheduling and any foreign participation will be identified no later than the NTTR Range War meeting.

A3.5.2.6.3. NTTR Projects Office or Unit Scheduler will identify any foreign participation no later than the NTTR Range War meeting.

A3.5.2.7. Special Use Airspace and GPS testing meeting (21-day). When required, the NTTR Special Use Airspace and GPS testing meeting is scheduled and conducted by the NTTR Project Officer prior to the execution of Large Force Exercises (LFE) utilizing GPS testing, R-4808B/C/D/E and R-4809A. The purpose of the Special Use Airspace and GPS meeting is to conduct interagency coordination and deconfliction for those areas. Representatives from the LFE host, the NTTR Projects Office, NTTR Range Scheduling, NNSS/NFO, and DOE must attend. Special Use Airspace and GPS testing coordination must be complete prior to the NTTR Pencil meeting.

A3.5.2.8. NTTR Pencil meeting (14-day). The NTTR Pencil meeting is conducted by NTTR Range Scheduling and conducted in the HQ NTTR building approximately two weeks prior to the week of execution at 1000L each Tuesday. The purpose of the Pencil meeting is for all NTTR range users and associated project managers to adjust the spreadsheets and schedule individual requirements through NTTR Range and Resource Scheduling for the week beginning 13 days later. Additionally, resource allocation(s) is/are solidified per unit prioritization posture.

A3.5.2.8.1. All users with scheduling requirements must provide a representative to the Pencil meeting to confirm their requirements and coordinate the required changes. Non-local users may be represented by NTTR Projects Office and/or NTTR Range Scheduling. Nellis/Creech Unit representatives who cannot be present at the meeting will coordinate with NTTR Range Scheduling NLT 0900L the day of the meeting to confirm range requirements. If users do not represent their requirements at NTTR Pencil meeting nor coordinate their lack of attendance with NTTR Range Scheduling their user data cannot be entered into CSE and will not be reflected on the GRR.

A3.5.2.8.2. Users will submit a mission request via CSE or provide a pencil to ensure the mission requirements are processed into CSE after this meeting. Users will access CSE to review the daily range schedule for accuracy and coordinate corrections, if necessary, through NTTR Range Scheduling.

A3.5.2.8.3. NTTR Range Scheduling will distribute the final version of the spreadsheets and load that scheduling data into CSE NLT 1000L on the Thursday following the Pencil meeting. NTTR Range Scheduling will not accept changes to the post-NTTR Pencil meeting spreadsheets prior to that time. The scheduling spreadsheets remain a planning tool only; NTTR users must refer to CSE for official schedule information.

A3.5.2.8.4. Costs incurred due to user-requested schedule changes made after 1200L on the Tuesday following the release of the post-Pencil meeting spreadsheets may be charged to the scheduled user.

A3.5.2.9. The NTTR Resource Scheduling Timeline: The Resource Schedulers distribute a Monthly Threat Schedule to annotate all threat missions identified at the Range War. Monthly Threat Schedule will be distributed to Range Mangers/Supervisors NLT Thursday the week following Range War.

A3.5.2.9.1. The Resource Schedulers distribute a Weekly Threat Schedule from the two-week pencil meeting to identify specific threats scheduled for each threat mission. The Threat Schedule will be distributed NLT 1300 on Thursday following the Pencil Meeting.

A3.5.2.9.2. Threat Schedule Confirmation. Thre requesting unit has a final responsibility to ensure the Threat Schedule reflects their requirements. The units will confirm all elements of their scheduled mission activity with NTTR Resource Scheduling no later than 48 hours prior to mission execution.

A3.5.2.9.3. The daily threat schedule is made available to NTTR users and Roulette no later than 1500L the day prior to mission execution. NTTR Roulette is responsible for the same day and next day threat schedule changes.

A3.5.2.9.4. Items essential to the resource schedule building process (i.e., threats, mobile threat locations, jammers, deployment forms, and Mission Support Request/TOSS forms) may realistically not be known and/or available at the weekly scheduling meeting. It is paramount that range users inform NTTR Resource Scheduling of changes or cancellations as soon as possible. If changes are not communicated prior to 24 hours before mission execution, units will contact NTTR Roulette with daily resource schedule updates.

A3.5.2.9.5. Threat Scheduling Matrix. NTTR Resource Scheduling manages the Threat Matrix to optimize limited threat availability in a safe and efficient manner, coordinating optimal range usage.

A3.5.3. Schedule Confirmation. The requesting unit has final responsibility to make sure the spreadsheets and CSE reflect their requirements and unit schedulers will confirm all elements of their scheduled mission activity and airspace with NTTR Range Scheduling no earlier than three days prior to mission execution and no later than 1200L two days prior to mission execution. Host units will verify foreign participation as part of the schedule confirmation.

A3.5.4. The NTTR schedule execution phase allows for the execution of the published daily range schedule. The daily range schedule is made available to NTTR Range Monitoring (Blackjack) 48 hours prior to execution. NTTR Range Monitoring (Blackjack) is responsible for same day and next day schedule changes.

**A3.6. NTTR Resource and Threat Emitter Scheduling.** Threat emitter support presents a specific scheduling challenge that prospective users must understand to obtain desired support. The threat emitter requirements must be submitted to begin the process and may change as the event planning matures. HQ NTTR will support these changes as able. A significant limitation exists requiring lead time to relocate mobile threat emitters and must be coordinated through the BMP.

A3.6.1. Threat resource scheduling requirements, as a minimum, include the information in Table A3.4.

**Table A3.4. Threat Resource Scheduling Requirements.**

- Threat Emitters
- Date(s) required
- Emitter type(s) to include personality and location, if known or required
- Support coordination: Smokey Sams, Surrogates, SAS
- Deployment forms (**deadline is Wed prior to week of mission execution)
- Mission Support Request Form/TOSS request
- Contact’s name and number

A3.6.2. Threat Deconfliction. Threat resources are scheduled and deconflicted by following guidelines specified in the Threat Matrix. The Threat Matrix identifies NTTR threat systems and their weekly availability. It establishes ground rules for threat resource allocation supporting standard and non-standard operations.

A3.6.2.1 Standard operations. Threat support windows are established in the Threat Matrix (i.e. 8.5 hours per threat system), excluding startup and shutdown. The threat systems will have a max crew seat time defined in the Threat Matrix (i.e. 8.5 hours) without prior approval. With approval, the crew seat time may be extended up to one hour by the resource scheduler if the end point is within a mission. Threats require minimum turn times post mission support (i.e. 12 hours) per the Threat Matrix. The threat site support window is anchored by priority events, the same as the overall support window. The threat support window must be inside the overall support window for P5/ROC/Roulette support. Adjustments outside of the overall support window published post Range War are possible for critical mission requirements but will be adjudicated utilizing section A3.4.

A3.6.2.2. Non-Standard Operations. Surge operations may increase the overall support window up to 24 hours. If the overall support window surges, each individual threat support window will still not exceed maximum seat time (i.e. 12 hours) and minimum turn times (i.e. 12 hours) established in the Threat Matrix. The Threat Matrix will define max crew seat time, support windows, and minimum turn times per threat system.

**A3.7. TOSS.** Toss is currently active on several strikeable targets. Refer to the NTTR Mission Planning website for further information on TOSS capabilities and location. A completed Mission Support Request Form (TOSS request form) must be submitted through NTTR Resource Scheduling NLT 14 days prior to ordnance employment. These forms are required to be confirmed by Wednesday the week prior to mission execution.

**A3.8. All Users Coordinating CT Sorties.** All users coordinating CT sorties will initially identify their desired threat emitter support to the JT4 Threat Scheduler no later than the NTTR Range War meeting (approximately 30 days prior to the month of execution) IAW the NTTR Scheduling Process. Routine changes will be provided to JT4 Threat Scheduler through the NTTR Pencil meeting prior to the week of execution. Additional changes will be considered, but cannot be guaranteed, as follows:

A3.8.1. Relocation of transportable threat emitters: The time required to relocate transportable threat emitters can vary, however, relocation requests received no later than between four and two duty days prior to execution can normally be supported.

A3.8.2. Addition of threat emitters: No later than two duty days prior to execution.

A3.8.3. Deletion of threat emitters: Any time prior to execution.

A3.8.4. Reconfiguration of a threat emitter laydown: No later than 48 duty hours prior to execution.

**A3.9. All Users Conducting Activities Other Than CT.** All users conducting activities other than CT will initially identify their desired threat emitter support as part of the PID IAW the NTTR BMP timelines (no later than 90). Routine changes will be considered via SOC development, again at the NTTR Range War meeting, and finally at the Pencil meeting prior to the week of execution. Any desired changes identified after the NTTR Pencil meeting will be submitted to NTTR Projects Office for consideration and coordination with the JT4 Threat Scheduler. Changes will be considered but cannot be guaranteed.

A3.9.1. Any costs incurred by HQ NTTR in response to user changes after the PID is submitted will be charged to the user, including costs for threat emitters initially requested (but subsequently deleted) from the overall requirement.

**A3.10. Test and Unique Mission Scheduling.** Test or other missions requiring unique targets other than existing targets require additional coordination within the NTTR. These situations will be identified when NTTR Projects Office notifies NTTR Target Management Office of a customer's requirement through PID dissemination, and NTTR Target Management Office determines no existing targets will meet the mission requirements. In these situations, and when a target is built with the intention of being targeted in an area that is not listed as targetable on the Mission Planning website, and is intended for a single or short-term use, NTTR Target Management Office will notify NTTR/SE as well as the Geographic Information Systems (GIS) Office.

A3.10.1. NTTR Target Management Office will provide NTTR/SE with the centroid grid for the target area, munition employment information, and aircraft information. NTTR/SE will then build mission-specific profiles and forward them to the GIS Office for inclusion on the Mission Planning website.

A3.10.2. NTTR Target Management Office will provide the GIS Office with a target centroid and description. The GIS Office will create and load a new target, identified as XX-00 (XX being the range where the target is located) on the Mission Planning website. Upon completion of the mission, and at the discretion of NTTR Target Management Office, the GIS Office will delete the "00" target from the database and Mission Planning website. The GIS Office will notify the Target Management and Range Scheduling offices anytime "00" targets are loaded or deleted.

A3.10.3. NTTR/SE will load the "00" target into the CSE target database. The NTTR Range Scheduling or Projects Office will assign it to the appropriate missions. Once notified by GIS Office that the targets have been removed from the Mission Planning website, NTTR/SE will delete the target from the CSE database.

**A3.11. NTTR Range Scheduling Changes and Cancellations.** Users who desire schedule changes after the NTTR Range War meeting, which affects other scheduled users, will coordinate their desires with all affected scheduled users. If the scheduled users approve the request, both the requestor and originally scheduled user will notify NTTR Range Scheduling (if more than 48 hours prior to execution) or NTTR Range Monitoring (Blackjack) (within 48 hours of execution) of the agreed upon changes via telephone and/or email. If any of the originally scheduled users do not approve the requested change, the change requestor may pursue further adjudication through the NTTR Scheduling Adjudication Process. Planned resource cancellations must be coordinated NLT 48 hours prior to mission execution the same as airspace cancellation requirement. Unplanned resource cancellations will be coordinated with the NTTR Resource Scheduler no later than noon on the day prior to execution. Cancellation of resources after 1500 local the day prior to execution will be coordinated with NTTR Range Monitoring (Blackjack) and Roulette.

## Attachment 4

## INDIVIDUAL RANGE DESCRIPTIONS

**A4.1. NTTR Subranges.** Individual descriptions of the NTTR subranges and selected other NTTR specified areas follow. Users should review specific range information (available on the NTTR Mission Planning websites) for each subrange for a summary view and the NTTR Mission Planning websites for target availability, ordnance and delivery restrictions, etc.

**Table A4.1. FAA Designations and Associated Subranges.**

| FAA Airspace Designation | FAA Sub-Airspace Designation | NTTR Subrange Designations | Description Page(s) |
|---|---|---|---|
| R-4806 | R-4806E | Alamo A/B/C | 65 |
| | R-4806W | R-61A/B | 66-67 |
| | | R-62A/B | 68-69 |
| | | R-63A/B/C/D | 70-72 |
| | | R-64A/B/C/D | 73-79 |
| | | R-65A/B/C/D | 80-82 |
| | | R-67A/B | 83-84 |
| R-4807 | R-4807A | R-71N/S | 85-86 |
| | | R-74A/B/C | 87-88 |
| | | R-75E/W | 88-89 |
| | | R-76 | 90 |
| | | R-77A | 91 |
| | | R-77B | 92 |
| | | EC East | 92-93 |
| | | R-4807A (Subset) | 93 |
| | | TPECR | 94 |
| | | Pahute A/B | 95 |
| | | R-4808A1/B/C/D/E | 96-97 |
| | | R-4808S1 | N/A |
| R-4809 | R-4807B | EC West | 97-98 |
| | R-4808N | R-4809A | 99-100 |
| | R-4808S | R-4809B | 100 |
| | EC West | Reveille North | 101 |
| | R-4809A | Reveille South | 102 |
| Reveille North MOA | R-4809B | Coyote A/B/C/D | 102-103 |
| | | Caliente A/B/C/D | 103 |
| | | Elgin | 104 |
| | | Sally Corridor | 104-105 |
| Note 1: No descriptive table is provided. | | | |

**Table A4.2. Additional Specified Areas.**

| FAA Airspace Designation | FAA Sub-Airspace Designation | Additional Specified Areas | Description Page(s) |
|---|---|---|---|
| R-4806 | R-4806E | Alamo Corridor[1] | 62 |
| | R-4806W | R-63D High | 71 |
| | | CTA | 108 |
| | | Air-to-Air Gunnery Area | 109 |
| | | | |
| R-4808 | R-4808N | Gomer Triangle[1] | N/A |
| R-4809 | R-4809A | STA | 110 |
| | | Shelf [1] | N/A |
| Reveille South MOA | Reveille South MOA and ATCAA | Dorito[1] | N/A |
| Desert MOA | Desert MOA and ATCAA | X-Ray Extension | 111 |
| Note 1: No descriptive table is provided. | | | |

**Table A4.3. Alamo A/B/C.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | 100ft AGL to unlimited |
| **Highest Elevation** | 7,999ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains exist throughout Alamo A/B/C. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Sally Corridor or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in Alamo A/B/C. |
| **Ordnance Restrictions** | There are no strikeable targets in Alamo A/B/C; aircrews are prohibited from expending ordnance on Alamo A/B/C. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Alamo A/B/C. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Alamo A/B/C overlies the DNWR. Aircraft will remain above 2,000ft AGL in Alamo A/B/C unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrew will avoid noise sensitive areas IAW NAFBI 11- 250. |
| **Additional Info** | 1. Alamo A/B/C may be scheduled as a whole or by individual subranges.<br>2. The Alamo Corridor lies within Alamo A/B/C at FL190 to FL210 and is schedulable. |

**Table A4.4. R-61A.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,952ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located towards the west of R-61A. |
| **Entry/Departure Info** | 1. Fixed-wing aircraft will enter/exit R-61A along the northern border of Alamo A/R-61B or as directed by NATCF.<br>2. Helicopters will enter/exit R-61A through the valley between the northern end of the East Desert mountain range and southern end of Pahranagat mountain range (N37 13.00 W115 22.00) in R-61B. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-61A. |
| **Ordnance Restrictions** | There are no strikeable targets in R-61A; aircrews are prohibited from expending ordnance on R-61A. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-61A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-61A unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. RPAs are prohibited from flying in R-61A. |
| **Additional Info** | None |

**Table A4.5. R-61B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,096ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | A north-south mountain range bisects R-61B. |
| **Entry/Departure Info** | 1. Fixed-wing aircraft will enter/exit R-61B along the northern border of Alamo A or as directed by NATCF.<br>2. Helicopters will enter/exit R-61B through the valley between the northern end of the East Desert mountain range and southern end of Pahranagat mountain range (N37 13.00 W115 22.00) in R-61B. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-61B. |
| **Ordnance Restrictions** | There are strikeable targets in R-61B; however, aircrews are prohibited from expending ordnance on Targets 61-02, 61-98, and 61-99. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-61B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-61B unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | None |

**Table A4.6. R-62A.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,040ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains surround Dog Bone Lake on all sides and are approximately 6NM away from the target areas.<br>2. TOSS towers are located on/around the R-62A target impact area<br>3. Target 62-01 is an active CBU grid, the only ground parties authorized on this target are EOD.<br>4. Inactive target 62-07 is an inactive CBU grid. Due to multiple UXO hazards, EOD personnel are the only authorized ground parties on this target. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Alamo B or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-62A. |
| **Ordnance Restrictions** | There are strikeable targets in R-62A. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-62A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-62A unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | 1. Multiple targets on R-62A are TOSS capable.<br>2. OP C3 is located in R-62A. |

**Table A4.7. R-62B.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,627ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains surround Dog Bone Lake on all sides and are approximately 6NM away from the target areas.<br>2. TOSS towers are located on/ around the R-62B target impact area |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Alamo C or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-62B. |
| **Ordnance Restrictions** | 1. There are strikable targets in R-62B; however, aircrews are prohibited from expending ordnance on Targets 62-98, 62-99, OP Jack, OP Romeo, OP Whisky, and OP X-Ray.<br>2. Aircrew must exercise caution to avoid targeting remotely operated emitters while delivering ordnance. Emitter units are bunkered in the vicinity of strikeable targets. These emitters are painted white and have protective barriers utilizing concrete blocks to shield the systems from shrapnel.<br>3. There is a 3 mile long High-Speed Moving Target (HSMT) track that runs north to south. At the south end there is a sea- land shelter painted white that is non-strikeable.<br>4. The UOC is in the south end of range 62B. All of the structures in the UOC are non-strikeable from aircraft with the exception of targets 62-11, 62-12, and 62-13 and 62-91. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-62B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-62B unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | 1. Multiple targets on R-62B are TOSS capable.<br>2. OP Jack, OP Romeo, OP Whisky, and OP X-Ray are located in R-62B.<br>3. The UOC and UOC Airfield facilities are located on R-62B. |

**Table A4.8. R-63A.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 4,531ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located west of the dry lake. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through the southern border of R-63A or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-63A. |
| **Ordnance Restrictions** | There are no strikeable targets in R-63A; aircrews are prohibited from expending ordnance on R-63A. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in R-63A. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-63A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-63A unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrew will not overfly R-63A below 500ft AGL. |
| **Additional Info** | 1. R-63A contains manned and remotely operated threat systems.<br>2. R-63A is a ground safety zone with a manned threat site, identifiable by a flashing strobe light, with facilities painted white.<br>3. R-63A is normally scheduled in conjunction with R-62B for activity in R-62B. However, non-participating ground parties can coordinate access into R-63A with the scheduled R-62B user. |

**Table A4.9. R-63B.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited (See Additional Information block) |
| **Highest Elevation** | 6,050ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located east and west of the dry lake.<br>2. TOSS towers are located on/around R-63B target impact area.<br>3. Target 63-10 is the Depleted Uranium (DU) target. It sits at the center of a 19K acre contaminated area. Stops on roads within the DU license area must be reported to NTTR Range Monitoring. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through the southern border of R-63B or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-63B. |
| **Ordnance Restrictions** | There are strikeable targets in R-63B; however, aircrews are prohibited from expending ordnance on Targets 63-98, 63-99, OP Jill, the DU storage area, the threat pads in the southern portion of R-63B (Cine 9 at N36 32.70 W115 32.64 and Cine 7 at N36 32.13 W115 31.46), the Borrow Pit with Rock Crusher and white-painted structures in R-63B. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-63B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-63B unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircraft must be aware of the Creech corridor in R-63B when it is active and take appropriate measures to avoid transiting RPAs and CCAs.<br>3. Aircraft will not fly below 8,000ft MSL within a 2NM radius of Corn Creek Station (centered at N36 26.00 W115 22.00) within the boundaries of R-63B. |
| **Additional Info** | 1. R-63B is primarily used for operational test and evaluation (OT&E) missions, night conventional weapons training, and as a day conventional weapons delivery range. It will not normally be used for air-to-air tactics training when ordnance delivery is planned.<br>2. R-63B contains the DU target set.<br>3. OP Jill is located in R-63B.<br>4. The Cine 9 and Cine 7 threat pads are located in the southern portion of R-63B.<br>4. Multiple targets on R-63B are TOSS capable. |

**Table A4.10. R-63C.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 4050ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located in the center and east side of R-63C |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-63C. |
| **Ordnance Restrictions** | There are no strikeable targets in R-63C. Aircrew are prohibited from expending ordnance on R-63C. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. When .50 Cal is being employed in R67 (A or B), no aircraft will operatin in R63C from SFC-10K, however aircraft will still be able to use R63C above 10K MSL.<br>2. R-63C overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-63C unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | None |

**Table A4.11. R-63D Low.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to 7,000 MSL. |
| **Highest Elevation** | 7050ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located in the on the east side of R-63D Low |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-63D Low. |
| **Ordnance Restrictions** | There are no strikeable targets in R-63D Low. Aircrew are prohibited from expending ordnance on R-63D Low. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-63D Low overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-63D unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | R63D Hi sits above R63D Low and R67A/B. |

**Table A4.12. R-63D Hi.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | 7,0001 MSL to unlimited |
| **Highest Elevation** | 7050ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located in the center and east side of R-63D Hi. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-63D Hi. |
| **Ordnance Restrictions** | There are strikeable targets in R-63D Hi. Aircrew are prohibited from expending ordnance on R-63D Hi. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-63D Hi overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-63D Hi unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | None |

**Table A4.13. R-64A.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,558ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located on the eastern border of R-64A. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-64A. |
| **Ordnance Restrictions** | There are strikeable targets in R-64A; however, aircrews are prohibited from expending ordnance on OP Sierra and OP Viv. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-64A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-64A unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | OP Sierra and OP Viv are located in R-64A. |

**Table A4.14. R-64B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,772ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located on the eastern and western borders of R-64B.<br>2. The Thunderbirds routinely practice in R-64B from surface to FL180. Other aircraft may be approved use of R-64B at FL190 and above IAW the NTTR Scheduling Process. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-64B. |
| **Ordnance Restrictions** | 1. There are strikeable targets in R-64B; however, aircrews are prohibited from expending ordnance on Targets 64-98, 64-99, white-painted structures, the Checkerboard Building, and the Thunderbirds practice runway.<br>2. Aircrew will not expend ordnance on the Pintwater Cave native historical site at N36 47.72 W115 34.03. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-64B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-64B unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrews must check NOTAMs to confirm the status of Thunderbirds training activity. |
| **Additional Info** | None |

**Table A4.15. R-64C.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited (See Additional Information block) |
| **Highest Elevation** | 5,879ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located on the northeastern and western borders of R-64C.<br>2. RPAs routinely operate in The Creech Corridors and CTA.<br>3. The Thunderbirds routinely practice in R-64C from surface to FL180. Other aircraft may be approved use of R-64C at FL190 and above IAW the NTTR Scheduling Process. |
| **Entry/Departure Info** | 1. Range entry and departure is normally from the south or as directed by NATCF.<br>2. Aircrew will avoid CTA airspace when Creech Tower pattern is active. CTA extends from surface to 7,000ft MSL. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-64C. |
| **Ordnance Restrictions** | There are strikeable targets in R-64C; however, aircrews are prohibited from expending ordnance on the Thunderbirds practice runway and OP Hill. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-64C overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-64C unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrews must check NOTAMs to confirm the status of Thunderbirds training activity. |
| **Additional Info** | None |

**Table A4.16. R-64D.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface |
| **Highest Elevation** | 4,741ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountains are located on the southern border of R-64D. |
| **Entry/Departure Info** | 64D is ground space only. |
| **Supersonic Restrictions** | 64D is ground space only. |
| **Ordnance Restrictions** | There are no strikeable targets in R-64D; aircrews are prohibited from expending ordnance on R-64D. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in R-64D. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-64D overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-64D unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrews must check NOTAMs to confirm the status of Thunderbirds training activity. |
| **Additional Info** | 1. The southern portion of R-64C lies above CTA from 7,001 MSL to unlimited.<br>2. Aircraft using R-64C may not penetrate R-64D/E/F/G or CTA without prior authorization from Creech Tower. CTA is active from 0600 Monday through 0630 Saturday. For access outside these hours, contact BLACKJACK or NATCF.<br>3. The range schedule will reflect “Creech Corridor Active” when scheduled.<br>OP Hill is located in R-64C. |

**Table A4.17. R-65A.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,985ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountainous terrain exists throughout R-65A with peaks in the east and west. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-65A. |
| **Ordnance Restrictions** | There are no strikeable targets in R-65A; aircrews are prohibited from expending ordnance on R-65A. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-65A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-65A unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. RPAs are prohibited from flying in R-65A. |
| **Additional Info** | None |

**Table A4.18. R-65B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,237ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountainous terrain exists throughout R-65B with peaks rising in the southwest. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-65B. |
| **Ordnance Restrictions** | There are no strikeable targets in R-65B; aircrews are prohibited from expending ordnance on R-65B. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted<br>on the NTTR Mission Planning website. |
| **Other Restrictions** | R-65B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-65B unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | None |

**Table A4.19. R-65C.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,300ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountainous terrain exists throughout R-65C with peaks rising in the southeast.<br>2. NNSS/NFO operates the Liquefied Gaseous Fuel Spill Test Facility and HAZMAT Spill Center located on Frenchman Flat dry lakebed along the western border of R-65C.<br>3. The Thunderbirds routinely practice in R-65C from surface to FL180, east of W115 46.84. Other aircraft may be approved use of R-65C at FL190, east of W115 46.84, IAW the NTTR Scheduling Process.<br>4. Other aircraft may be approved use of R-65C from surface to unlimited west of W115 46.84 IAW the NTTR Scheduling Process. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-65C. |
| **Ordnance Restrictions** | There are strikeable targets in R-65C; however, aircrews are prohibited from expending ordnance on Targets 65-98 and 65- 99. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-65C overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-65C unless a lower altitude is required for tactical training and IAW the daily range schedule.<br>2. Aircrews must check NOTAMs to confirm the status of Thunderbirds training activity. |
| **Additional Info** | None |

**Table A4.20. R-65D.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited except the southeast portion of R65D which overlays CTA which is 7,001 MSL to unlimited. |
| **Highest Elevation** | 6,050ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountainous terrain exists throughout R-65D with peaks rising in the northwest and southeast. |
| **Entry/Departure Info** | Range entry and departure is normally from the south or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-65D. |
| **Ordnance Restrictions** | There are no strikeable targets in R-65D; aircrews are prohibited from expending ordnance on R-65D. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-65D overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-65D unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | 1. R-65D is routinely used by for RPAs and helicopters and provides a transition area for aircraft into R-64C and R-65C during Thunderbirds training activity.<br>2. Aircraft using R-65D may not penetrate CTA without prior authorization from Creech Tower. CTA is active from 0600 Monday through 0630 Saturday. For access outside these hours, contact BLACKJACK or NATCF.<br>3. The Manned Beno Line (MB) is available as a reference tool to deconflict manned aircraft and RPA operations within R65D. The MB is not a formal boundary; its usage and implementation is based on the mutual agreement between participating units via direct coordination. When utilized, the MB is defined as a line connecting N36 40.39 W115 56.05 to N36 37.62 W115 47.40 to N36 37.62 W115 43.05The agreements must be documented and sent to the NTTR Scheduling Office and Range Scheduling (Blackjack) prior to the submission of the schedule into CSE. CSE missions will reflect the agreement with both units. Each unit will be assigned the airspace with the appropriate agreed deconfliction. A restriction will be entered into the “remarks” section that contains the concise, agreed-upon deconfliction plan. It will be the responsibility of the participating units to adhere to their written agreement which are reflected in the mission restrictions in CSE. |

**Table A4.21. R-67A**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to 7,000ft MSL |
| **Highest Elevation** | 3,400ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Essentially flat terrain exists throughout R-67.<br>2. R-67 contains live, small arms ranges . Restrictions will be in place when the ranges are active.<br>3. When .50 Cal is being employed in R-67A, no aircraft may operate in in R 67-A or R-63C below 10K MSL to account for the SDZ of the weapon. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through the southern border of R-67 or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-67. |
| **Ordnance Restrictions** | There are no strikeable targets in R-67A; aircrews are prohibited from expending ordnance on R-67A. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-67A overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-67A unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | R-67A is located beneath R-63D High. |

**Table A4.22. R-67B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to 7,000ft MSL |
| **Highest Elevation** | 3,200ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Essentially flat terrain exists throughout R-67B.<br>2. R-67B contains live, small arms ranges. Restrictions will be in place when the ranges are active.<br>3. When .50 Cal is being employed in R-67B, no aircraft may operate in in R-63C below 10K MSL to account for the SDZ of the weapon. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through the southern border of R-67 or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R-67. |
| **Ordnance Restrictions** | There are no strikeable targets in R-67B; aircrews are prohibited from expending ordnance on R-67B. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | R-67B overlies the DNWR. Aircraft will remain above 2,000ft AGL in R-67B unless a lower altitude is required for tactical training and IAW the daily range schedule. |
| **Additional Info** | R-67B is located beneath R-63B High. |

**Table A4.23. Weapons and Munitions Approved for R-67A Use.**

| Weapon (Caliber) | Munition | Surface Danger Zone (SDZ) Distance (radius) | Vertical Danger Zone (VDZ) Elevation |
|---|---|---|---|
| Handgun (.38 cal) | M41 PGU-12/B | 6,036ft / 0.99NM | 2,215ft AGL |
| Handgun (9mm) | M882 M1153 | 7,198ft / 1.18NM | 2,520ft AGL |
| Rifle (5.56mm) | M855 M856 | 12,661ft / 2.08NM | 3,674ft AGL |
| Machine gun (7.62mm) | M80 M118 | 17,349ft / 2.86NM | 4,856ft AGL |
| Machine gun (.50 cal) | M2 M33 M17 M8 API M20 APIT | 21,325ft / 3.51NM | 6,365ft AGL |

**Table A4.24. R-71N.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,110ft MSL |
| **C2/Communication** | 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located on the east and west borders of R-71N.<br>2. "Double Track," a contaminated radiological area located at N37 42.36 W116 59.30, is posted with warning signs and must be avoided by ground parties. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from the north or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-71N. |
| **Ordnance Restrictions** | There are no strikeable targets in R-71N; aircrews are prohibited from expending ordnance on R-71N. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. Aircrew will not drop or jettison any ordnance/ launchers on "Double Track" (N37 42.36 W116 59.30).<br>2. Other aircraft must be aware of the RPA transit corridor in R-71N when it is active and take appropriate measures to avoid transiting RPAs. |
| **Additional Info** | Viewpoint Alpha (N37 41.15 W116 53.81) is located within R-71N. |

**Table A4.25. R-71S.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,591ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located towards the east of R-71S.<br>2. TOSS towers are located on and around R-71S target impact area. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from the north or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-71S. |
| **Ordnance Restrictions** | 1. There are strikeable targets in R-71S; however, aircrews are prohibited from expending ordnance on Target 71-99.<br>2. A NIZ within R-71S is protected from all weapons profiles. The NIZ will be active when ground parties are present. The R-71S NIZ coordinates are:<br>a. N37 34.00 W116 58.00<br>b. N37 34.00 W116 56.00<br>c. N37 33.00 W116 56.00<br>d. N37 33.00 W116 58.00<br>e. N37 34.00 W116 58.00 |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. Aircrew will not drop or jettison any ordnance/launchers on "Double Track."<br>2. Other aircraft must be aware of the RPA transit corridor in R-71S when it is active and take appropriate measures to avoid transiting RPAs. |
| **Additional Info** | Multiple targets on R-71S are TOSS capable. |

**Table A4.26. R-74A.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 8,471ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | A north-south mountain range bisects R-74A. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from adjacent range airspace or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-74A. |
| **Ordnance Restrictions** | There are no strikeable targets in R-74A; aircrews are prohibited from expending ordnance on R-74A. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | RPAs are prohibited from flying in R-74A. |
| **Additional Info** | None |

**Table A4.27. R-74B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 8,547ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | A north-south mountain range bisects R-74B. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from adjacent range airspace or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-74B. |
| **Ordnance Restrictions** | There are strikeable targets in R-74B; however, aircrews are prohibited from expending ordnance on Target 74-04. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | None |
| **Additional Info** | None |

**Table A4.28. R-74C.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,369ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | A north-south mountain range bisects R-74C. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from adjacent range airspace or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-74C. |
| **Ordnance Restrictions** | There are strikeable targets in R-74C; however, aircrews are prohibited from expending ordnance on Targets 74-27, 74-29, 74-98, and 74-99. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | None |
| **Additional Info** | None |

**Table A4.29. R-75E.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,795ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located on the east border of R-75E.<br>2. Several contaminated radiological areas in Pahute B adjacent to the southern border of R-75E add an additional hazard during ejections.<br>3. TOSS towers are located on/around R-75E target impact area |
| **Entry/Departure Info** | Range entry and departure is conducted from adjacent range airspace or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-75E. |
| **Ordnance Restrictions** | There are strikeable targets in R-75E; however, aircrews are prohibited from expending ordnance on Target 75-99 and OP Steeler. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | None |
| **Additional Info** | 1. Multiple targets on R-75E are TOSS capable.<br>2. OP Steeler is located in R-75E. |

**Table A4.30. R-75W.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,109ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located on the south and north borders of R-75W.<br>2. Several contaminated radiological areas in Pahute B adjacent to the southern border of R-75E add an additional hazard during ejections.<br>3. TOSS towers are located on/around R-75W target impact area. |
| **Entry/Departure Info** | Range entry and departure is conducted from adjacent range airspace or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-75W. |
| **Ordnance Restrictions** | There are strikeable targets in R-75W; however, aircrews are prohibited from expending ordnance on Target 75-98 and OP Delta. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | None |
| **Additional Info** | 1. Multiple targets on R-75W are TOSS capable.<br>2. OP Delta is located in R-75W. |

**Table A4.31. R-76.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 8,287ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). For missions involving EC simulators, contact ROULETTE on UHF 293.5 or UHF 383.3. |
| **Hazards** | 1. Mountains are located throughout R-76.<br>2. TOSS towers are located on and around R-76 target impact area. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from the west or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 100ft AGL in R-76. |
| **Ordnance Restrictions** | 1. There are strikeable targets in R-76; however, aircrews are prohibited from expending ordnance on Targets 76-98, 76-99, OP Seahawk, OP Viking, and OP Zulu.<br>2. Aircrew must exercise caution to avoid targeting remotely operated emitters on R-76. Emitter units are bunkered in the vicinity of strikeable targets. These emitters are painted white and have protective barriers utilizing concrete blocks to shield the systems from shrapnel.<br>3. A NIZ within R-76 is protected from all weapons profiles. The NIZ will be active when ground parties are present. The R-76 NIZ coordinates are:<br>a. N37 33.00 W116 58.00<br>b. N37 33.00 W116 56.00<br>c. N37 31.00 W116 56.00<br>d. N37 31.00 W116 58.00<br>e. N37 33.00 W116 58.00 |
| **Flare Restrictions** | 4. Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | 5. Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 6. Other aircraft must be aware of the RPA transit corridor in R- 76 when it is active and take appropriate measures to avoid transiting RPAs. |
| **Additional Info** | 1. Multiple targets on R-76 are TOSS capable.<br>7. OP Seahawk, OP Viking, and OP Zulu are located in R-76. |

**Table A4.32. R-77A.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,411ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). |
| **Hazards** | Mountainous terrain exists in the northern and southern areas. |
| **Entry/Departure Info** | Entry and exit is normally conducted from the south and west of the range or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R77. |
| **Ordnance Restrictions** | There is one strikeable target in R77A. There are several non strikeable targets in R77A, aircrew are prohibited from employing on targets 77-01, 77-03, 77-04, 77-05, 77-06, 77-07, 77-08, 77-09, 77-10, 77-11, 77-12, and 77-13. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of R77. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight.<br>2. There is an RSE site (pads ECS05A and 2-T1) located approximately 2 NM south west of target 77-02 in R77A. Each UMTE/RSE site is surrounded by white concrete bunkers or white sea-land containers. Aircrew should exercise extreme caution not to expend ordnance on UMTEs/RSEs.<br>3. Other aircraft must be aware of the RPA transit corridor in R77A when it is active and take appropriate measures to avoid transiting RPAs. |
| **Additional Info** | R77A is located along the southwest border of R-4807. |

**Table A4.33. R-77B.**

| | |
|---|---|
| **Range Type** | Class B |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,785ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). |
| **Hazards** | Mountainous terrain exists throughout the area. |
| **Entry/Departure Info** | Entry and exit is normally conducted from the south and west of the range or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in R77B. |
| **Ordnance Restrictions** | There are no strikeable targets in R-77B; aircrews are prohibited from expending ordnance on R-77B. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are restricted to laser operations approved IAW the NTTR Laser Range Certification issued by the 711 HPW. Target and subrange specific laser restrictions are noted on the NTTR Mission Planning website. |
| **Other Restrictions** | 1. R-77B includes the Tolicha Peak access gate, buildings, and road running between highway 95 and TPECR. Aircrew with live/inert/training ordnance will have their master arm switches in SIM/SAFE/OFF anytime they are within R77B<br>2. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of R77B. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight. |
| **Additional Info** | R77B is located in the southernmost portion of R-4807. |

**Table A4.34. EC East.**

| | |
|---|---|
| **Range Type** | EC |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 8,809ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). Aircrew may contact ROULETTE for threat coordination on UHF 293.5 or UHF<br>383.3. Authorized ground parties are not required to obtain additional clearance for movement once within EC East. |
| **Hazards** | A north-south mountain range bisects EC East with peaks in the south and center. |
| **Entry/Departure Info** | Entry and exit is normally conducted from adjacent ranges or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in EC East. |
| **Ordnance Restrictions** | There are no strikeable targets in EC East; aircrews are prohibited from expending ordnance on EC East. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Minimum altitude in EC East is 500ft AGL, except for USAFWS syllabus sorties and specifically approved test sorties requiring 100ft AGL maneuvers.<br>2. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of EC East. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight. |
| **Additional Info** | 1. There are multiple manned threat emitters along the eastern perimeter bordering R-74 and R-75.<br>2. OP Charlie is located in EC East. |

**Table A4.35. R-4807A (Subset).**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,424ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountainous terrain exists to the north and east of R-4807A (Subset).<br>2. Multiple towers are located throughout R-4807A. |
| **Entry/Departure Info** | Aircrews will contact Silverbow Tower for entry into R- 4807A (Subset). |
| **Supersonic Restrictions** | Supersonic flight in R-4807A (Subset) may be requested through the NTTR Scheduling Process and will be considered on a case-by-case basis. Once approved, supersonic flight will only be conducted above 5,000ft AGL. Other flight restrictions may apply and will be passed at the time of approval. |
| **Ordnance Restrictions** | There are no strikeable targets in R-4807A (Subset); aircrews are prohibited from expending ordnance on R-4807A (Subset). |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in R-4807A (Subset). |
| **Laser Restrictions** | Air/ground users are prohibited from employing lasers in R- 4807A (Subset). |
| **Other Restrictions** | Minimum altitude over R-4807A (Subset) is 500ft AGL. |
| **Additional Info** | 1. R-4807A (Subset).contains numerous buildings and ground party activity.<br>2. R-4807A (Subset) is scheduled as part of EC East. |

**Table A4.36. TPECR.**

| | |
|---|---|
| **Range Type** | EC |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,228ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). Aircrew may contact ROULETTE for threat coordination on UHF 293.5 or UHF<br>383.3. Authorized ground parties are not required to obtain clearance for movement within TPECR. |
| **Hazards** | Mountains are located on the western and southern borders. |
| **Entry/Departure Info** | Entry and exit is normally conducted from adjacent ranges or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in TPECR. |
| **Ordnance Restrictions** | There are no strikeable targets in TPECR; aircrews are prohibited from expending ordnance on TPECR. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Minimum altitude in TPECR is 500ft AGL, except for USAFWS syllabus sorties and specifically approved test sorties requiring 100ft AGL maneuvers.<br>2. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of TPECR. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight. |
| **Additional Info** | 1. TPECR is a manned and remotely operated electronic combat threat simulator range.<br>2. TPECR includes the FAC A/B ground/air space.<br>a. OP Alpha is located in the FAC A ground space.<br>b. OP Bravo is located in the FAC B ground space. |

**Table A4.37. Pahute A/B.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,415ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountains are located in the central and northern areas of Pahute A/B.<br>2. Tall towers are located throughout the Pahute A/B. Extensive construction and scientific work takes place on towers.<br>3. Helicopters routinely transit the range and flight patterns may extend up to 500ft AGL.<br>4. Contaminated radiological areas should be avoided to the maximum extent possible and foot traffic must heed posted warning signs. Contaminated radiological areas are located at the following coordinates:<br>a. Palaquin: N37 16.81 W116 31.47.<br>b. Cabriolet: N37 16.85 W116 30.93.<br>c. Schooner: N37 20.60 W116 34.00.<br>4. Pahute A/B contains live, small arms ranges. Restrictions will be in place when the ranges are active. |
| **Entry/Departure Info** | Entry and exit is normally conducted from the south and west of the range or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in Pahute A/B. |
| **Ordnance Restrictions** | There are no strikeable targets in Pahute A/B; aircrews are prohibited from expending ordnance on Pahute A/B. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Pahute A/B. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Minimum altitude in Pahute A/B is 500ft AGL, except for USAFWS syllabus sorties and test sorties requiring 100ft AGL maneuvers.<br>2. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of Pahute A/B. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight. |
| **Additional Info** | 1. NNSS/NFO has been granted use of airspace from surface to 500ft AGL by LOA.<br>2. The USAF retains overflight rights above 500ft AGL and may schedule from surface to unlimited if coordinated with NNSS/NFO.<br>3. Pahute B has an electronic combat threat simulator which may be manned and remotely operated. |

**Table A4.38. R-4808.**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface to unlimited. |
| **Highest Elevation** | 9,365ft MSL |
| **C2/Communication** | DML or UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountainous terrain exists throughout R-4808.<br>2. Radioactive areas exist throughout R-4808.<br>2. Helicopters, RPAs, nonparticipating, and aircraft not under ATC control routinely operate in R-4808 surface to 10,000ft MSL.<br>3. Temporary restrictions are routinely scheduled throughout R-4808. |
| **Entry/Departure Info** | Aircrews will contact DML or NATCF prior to entering. |
| **Supersonic Restrictions** | Supersonic flight below 20,000ft MSL in R-4808 is prohibited. Supersonic flight, if authorized, will be noted with minimum altitudes in the GRR |
| **Ordnance Restrictions** | There are no strikeable targets in R-4808.<br>With actual live/inert ordnance, Master Arm HOT in R-4808 is prohibited.<br>Expending ordnance on R-4808 is prohibited.<br>Expending chaff in R-4808 is prohibited.<br>Carriage of chaff in R-4808 is prohibited if not behind a door.<br>Overflight of R-4808 with hung stores is prohibited. |
| **Flare Restrictions** | Employing flares in R-4808 is prohibited. |
| **Laser Restrictions** | Employing lasers in R-4808 is prohibited. |

| | |
|---|---|
| **Other Restrictions** | 1. Unlimited maneuvering (IAW AFMAN 11-214) is prohibited.<br>2. Turning merges in R-4808 are prohibited.<br>3. An operational transponder to include Mode C is required.<br>4. IFF OFF missions over R-4808 are prohibited.<br>5. Lights-Out missions over R-4808 are prohibited.<br>6. Air refueling missions over R-4808 are prohibited.<br>7. Imaging in R-4808 or aircraft within is prohibited.<br>8. Targeting of ground sites is prohibited.<br>9. R-4808 B/C/D/E No Fly areas include:<br>a. Device Assembly Facility (DAF) (N36 53.89 W116 02.88)<br>b. surface to 14,000ft MSL within a 1.25NM radius.<br>c. Nonproliferation Test Evaluation Complex (NPTEC) (N36<br>d. 48.15 W115 57.28) surface to 1,500ft AGL within 1NM radius.<br>e. Radioactive Waste Management Complex (RWMC) (N36<br>f. 51.53 W115 57.29) surface to 1,500ft AGL within 1NM radius.<br>g. Mercury Controlled Firing Area (CFA) (N36 39.31 W116<br>h. 00.45) surface to 6,000ft MSL within 1NM radius.<br>10. General minimum altitudes for R-4808 B/C/D/E are:<br>a. 2,500ft AGL for fixed-wing aircraft.<br>b. 500ft AGL for helicopters.<br>11. Other aircraft must be aware of the RPA transit corridor in R-4808 C/D/E when it is active and take appropriate measures to avoid transiting RPAs. |

| | |
|---|---|
| **Additional Info** | 1. IAW FAA Order JO 7400.8Y, NNSS/NFO has been designated as the Using Agency of R-4808, which overlies the NNSS. Additionally, IAW the 1981 MOU between DAF and DOE, the agencies agree to the cooperative use of the NNSS land and airspace.<br>2. NTTR users who wish to use R-4808 airspace will coordinate their desires through the NTTR Scheduling Process to obtain NNSS/NFO and DOE approval.<br>3. Foreign users are generally prohibited.<br>4. US-assigned foreign personnel, assigned to fly US equipment where the US controls the data on that equipment are permitted overflight of R-4808 after coordination with NNSS/NFO and DOE.<br>5. Minimum schedulable altitude for NTTR users in R-4808 A is normally 19,000ft MSL.<br>a. Available to AFWC sponsored activities on a non-interference<br>a. basis.<br>b. Documented via "D-19" on GRR.<br>c. Standard altitude is 19,000ft MSL, but higher may be noted.<br>6. Minimum schedulable altitude for NTTR users in R-4808 B is normally 19,000ft MSL for traditional aircraft; RPAs may not operate in R-4808 B.<br>7. Minimum schedulable altitude for NTTR users in R-4808 C is normally 19,000ft MSL.<br>8. Minimum schedulable altitude for NTTR users in R-4808 D/E is normally 15,000ft MSL.<br>9. Waiver authority for R-4808 restrictions require NNSS/NFO and DOE approval. |

**Table A4.39. EC West.**

| | |
|---|---|
| **Range Type** | EC |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,602ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK). Aircrew may contact ROULETTE for threat coordination on UHF 293.5 or UHF 383.3. |
| **Hazards** | 1. Mountains are located on the western border.<br>2. High-trajectory field artillery firing is occasionally conducted on the range.<br>4. Contaminated radiological areas should be avoided to the maximum extent possible and foot traffic must heed posted warning signs. Three "Clean Slate" contaminated radiological areas are located at the following coordinates:<br>a. N37 42.50 W116 39.45<br>b. N37 45.67 W116.36.87<br>c. N37 45.55 W116 40.87 |
| **Entry/Departure Info** | Entry and exit is normally conducted from adjacent ranges or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight in EC West may be requested through the NTTR Scheduling Process and will be considered on a case- by-case basis. Once approved, supersonic flight will only be conducted above 5,000ft AGL. Other flight restrictions may apply and will be passed at the time of approval. |
| **Ordnance Restrictions** | There are no strikeable targets in EC West; aircrews are prohibited from expending ordnance on EC West. |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Minimum altitude in EC West is 500ft AGL, except for USAFWS syllabus sorties and specifically approved test sorties requiring 100ft AGL maneuvers.<br>2. All manned sites have white support buildings and white strobe lights (day) and steady white lights (night) to distinguish them from strikeable targets located outside of EC West. Aircrew will avoid overflight of manned sites consistent with aircraft maneuverability and safety of flight. |
| **Additional Info** | 1. Both manned and unmanned threat emitters operate in EC West.<br>2. IAW the DAF permit, the NNSA/SFO has been granted use, operation, and occupancy of specified property within EC West. |

**Table A4.40. R-4809A.**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 7,447ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountainous terrain exists along the western border.<br>2. Multiple towers are located throughout R-4809A.<br>3. A small arms, live-fire range, approximately 4NM southwest of the TTRA, exists in R-4809A at N37 44.25 W116 47.34. Use of the small arms range will be coordinated through NTTR/DO and will be scheduled in CSE when active. Once scheduled, aircrew will contact TTRA Base Operations or Silverbow Tower prior to entering STA to determine the live-fire range status. |
| **Entry/Departure Info** | Aircrews will contact NATCF for entry into R-4809A. |
| **Supersonic Restrictions** | Supersonic flight in R-4809A may be requested through the NTTR Scheduling Process and will be considered on a case- by-case basis. Once approved, supersonic flight will only be conducted above 5,000ft AGL. Other flight restrictions may apply and will be passed at the time of approval. |
| **Ordnance Restrictions** | There are no strikeable targets in R-4809A; aircrews are prohibited from expending ordnance on R-4809A. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares over TTRA. |
| **Laser Restrictions** | Air/ground users are prohibited from employing lasers in R- 4809A. |
| **Other Restrictions** | 1. Aircrew will not fly directly over TTRA with live ordnance.<br>2. Overflight of R-4809A by non USAFWC-assigned, Creech AFB-based RPAs is not authorized.<br>3. Aircraft must have an operational transponder to include Mode C. IFF OFF missions are prohibited without specific DOE approval (coordinated through the NTTR Scheduling Process).<br>4. Imaging of R-4809A is normally limited to traditional, USAFWC-assigned, Nellis AFB-based aircraft.<br>5. Imaging by USAFWC-assigned, Creech AFB-based RPAs must obtain specific approval.<br>6. Temporary Duty aircrews, excluding USAF Weapons School students, and visiting traditional aircraft are not authorized to image in R-4809A without specific approval. Visiting RPAs are not authorized to image in R-4809A. Foreign users are prohibited without DOE approval (coordinated through the NTTR Scheduling Process). |

| | |
|---|---|
| **Additional Info** | 1. Silverbow Tower controls airspace within a 5NM radius of TTRA up to, but not including, 2,500ft AGL. Aircrew will coordinate with NATCF for advisory to 500ft AGL when the tower is inactive.<br>2. NATCF directs all IFR traffic into TTRA.<br>3. Aircraft will not overfly the MSA nor the explosive material storage facilities in the vicinity of N37 50.77 W116 42.62 in R-4809A.<br>4. Requirements for R-4809A are scheduled through the NTTR Scheduling Process IAW applicable Memorandums of Understanding (MOU) (located on the NTTR Mission Planning websites).<br>5. IAW the DAF permit, the NNSA/SFO has been granted use, operation, and occupancy of specified property within R- 4809A. |

**Table A4.41. R-4809B.**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 5,938ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | 1. Mountainous terrain exists to the north and east of R- 4809B.<br>2. Multiple towers are located throughout R-4809B. |
| **Entry/Departure Info** | N/A |
| **Supersonic Restrictions** | Supersonic flight in R-4809B may be requested through the NTTR Scheduling Process and will be considered on a case- by-case basis. Once approved, supersonic flight will only be conducted above 5,000ft AGL. Other flight restrictions may apply and will be passed at the time of approval. |
| **Ordnance Restrictions** | There are no strikeable targets in R-4809B; aircrews are prohibited from expending ordnance on R-4809B. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in R-4809B. |
| **Laser Restrictions** | Air/ground users are prohibited from employing lasers in R- 4809B. |
| **Other Restrictions** | Minimum altitude over R-4809B is 500ft AGL. |
| **Additional Info** | 1. R-4809B contains numerous buildings and ground party activity.<br>2. R-4809B is scheduled as part of EC West. |

**Table A4.42. Reveille North.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL ATCAA: FL180 to FL600 |
| **Highest Elevation** | 10,164ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountain ranges exist throughout Reveille North. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from Caliente or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in Reveille North. |
| **Ordnance Restrictions** | There are no strikeable targets in Reveille North; aircrews are prohibited from expending ordnance on Reveille North. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Reveille North. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Aircrew will avoid noise sensitive areas IAW NAFBI 11- 250.<br>2. Aircrew will avoid the towns of Reveille (N38 02.00 W116 10.17) and Adaven (N38 08.00 W115 36.00) by 1.5NM when below 1,500ft AGL. |
| **Additional Info** | 1. Airspace requirements above FL300 must be specifically coordinated through the Nellis Airspace Management Office (57OSS/OSM) per the current FAA LOA.<br>2. Aircrew using Reveille North must remain alert for transiting civilian aircraft regardless of the altitudes scheduled. When operating in Reveille North, aircraft will squawk Mode 3/C at all times.<br>3. To ensure safety of non-participating aircraft, specific procedures for GPS testing within Reveille North must be coordinated between the requestor, the GPS test provider, and the FAA.<br>a. Participating aircraft are normally authorized SFC-as scheduled from 2231L-0529L when GPS testing is active.<br>b. Participating aircraft are normally authorized SFC-FL300 from 0530-2230L when GPS testing is active.<br>a. c. If participating aircraft are to operate above FL300 from 0530-2230L when GPS testing is active, the GPS testing agency, event sponsoring agency, and FAA must coordinate specific procedures for the transfer of airspace between ATC and ATCAA during the event. |

**Table A4.43. Reveille South.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL ATCAA: FL180 to FL600 |
| **Highest Elevation** | 9,393ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountain ranges exist throughout Reveille South. |
| **Entry/Departure Info** | Range entry and departure is normally conducted from Caliente or as directed by NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in Reveille South. |
| **Ordnance Restrictions** | There are no strikeable targets in Reveille South; aircrews are prohibited from expending ordnance on Reveille South. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Reveille South. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | Aircrew will avoid noise sensitive areas IAW NAFBI 11-250. |
| **Additional Info** | Aircrew using Reveille South must remain alert for transiting civilian aircraft regardless of the altitudes scheduled. When operating in Reveille South, aircraft will squawk Mode 3/C at all times. |

**Table A4.44. Coyote A/B/C/D.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL ATCAA: FL180 to unlimited |
| **Highest Elevation** | 8,934ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountain ranges exist throughout Coyote A/B/C/D. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Coyote B via the north end of Sally Corridor (VFR) or as directed by NATCF. Aircraft that exit the range via Sally are not VFR until they are cleared for and accept one of the VFR recoveries from NATCF. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft AGL in Coyote A/B/C/D. |
| **Ordnance Restrictions** | There are no strikeable targets in Coyote A/B/C/D; aircrews are prohibited from expending ordnance on Coyote A/B/C/D. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Coyote A/B/C/D. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Aircrew will avoid noise sensitive areas IAW NAFBI 11-250.<br>2. RPAs are prohibited from flying in Coyote A. |
| **Additional Info** | 1. Coyote A/B/C/D may be scheduled as a whole or by individual subranges.<br>3. UMTEs may operate in Coyote A/B/C/D. |

**Table A4.45. Caliente A/B/C/D.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL ATCAA: FL180 to unlimited |
| **Highest Elevation** | 9,370ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountain ranges bisect Caliente Bravo from Caliente Charlie. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Caliente via the north end of Sally Corridor (VFR) or as directed by NATCF. Aircraft that exit the range via Sally are not VFR until they are cleared for and accept one of the VFR recoveries from NATCF. |
| **Supersonic Restrictions** | 1. Supersonic flight is prohibited below 5,000ft AGL in Caliente A/B.<br>2. Supersonic flight is prohibited below 30,000ft AGL in Caliente C.<br>3. The noise sensitive areas of Caliente, NV and the Lincoln County Airport are subsonic surface to unlimited. |
| **Ordnance Restrictions** | There are no strikeable targets in Caliente A/B/C; aircrews are prohibited from expending ordnance on Caliente A/B/C. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Caliente A/B/C. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Aircrew will avoid noise sensitive areas IAW NAFBI 11- 250, Attachment 13.<br>2. Other aircraft must be aware of the RPA transit corridor in Caliente C when it is active and take appropriate measures to avoid transiting RPAs. The RPA COA approved altitudes on the Eastern border of the NTTR (Elgin and Caliente C) are FL210 and FL220. |
| **Additional Info** | 1. Caliente D ATCAA goes from FL310 to FL450.<br>2. If Caliente D is in use by ZLC, NATCF will set the highest operating ceiling below Caliente D based on the local Nellis Air Force Base altimeter. Caliente C airspace continues above Caliente D from FL450 to unlimited. ZLC will always own Caliente D ATCAA unless otherwise scheduled by the NTTR. |
| **Additional Info (cont.)** | 3. Caliente may be scheduled as a whole or by individual subranges. |

**Table A4.46. Elgin.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL ATCAA: FL180 to unlimited |
| **Highest Elevation** | 7,805ft MSL |
| **C2/Communication** | UHF 377.8 (BLACKJACK) |
| **Hazards** | Mountain ranges exist throughout Elgin. |
| **Entry/Departure Info** | Range entry and departure is normally conducted through Elgin via the east end of Sally Corridor (VFR) or as directed by NATCF. Aircrew may request direct entry into Elgin from NATCF. Aircraft that exit the range via Sally are not VFR until they are cleared for and accept one of the VFR recoveries from NATCF. |
| **Supersonic Restrictions** | 1. Supersonic flight is prohibited below 5,000ft AGL north of a line from N36 52.00 W114 50.72 to N37 04.00 W114 33.00to N37 04.00 W114 19.19.<br>2. Supersonic flight is prohibited below 30,000ft MSL south of a line from N36 52.00 W114 50.72 to N37 04.00 W114 33.00 to<br>N37 04.00 W114 19.19.<br>3. Supersonic flight south of N36 48.00 is only permitted with magnetic headings of 259°clockwise to 079°. |
| **Ordnance Restrictions** | There are no strikeable targets in Elgin; aircrews are prohibited from expending ordnance on Elgin. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Elgin. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. Aircrew will avoid noise sensitive areas IAW NAFBI 11- 250.<br>2. Other aircraft must be aware of the RPA transit corridor in Elgin when it is active and take appropriate measures to avoid transiting RPAs. |
| **Additional Info** | None |

**Table A4.47. Sally Corridor.**

| | |
|---|---|
| **Range Type** | MOA/ATCAA |
| **Airspace** | MOA: Surface to 17,999ft MSL<br>ATCAA: FL180 to unlimited |
| **Highest Elevation** | 9,885ft MSL |
| **C2/Communication** | UHF 317.525 (NATCF East) |
| **Hazards** | Mountains are located towards the south and along the western and northern borders of the Sally Corridor. |
| **Entry/Departure Info** | Range entry and departure is direct to/from Nellis AFB. |
| **Supersonic Restrictions** | 1. Supersonic flight is prohibited below 5,000ft AGL north of N36 52.00.<br>2. Supersonic flight is prohibited below 30,000ft MSL south of N36 52.00.<br>3. Supersonic flight south of N36 48.00 is only permitted with magnetic headings of 259°clockwise to 079°. |
| **Ordnance Restrictions** | There are no strikeable targets in Sally Corridor; aircrews are prohibited from expending ordnance on Sally Corridor. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares below 5,000ft AGL in Sally Corridor. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | Aircrew will avoid noise sensitive areas IAW NAFBI 11-250. |
| **Additional Info** | Sally Corridor is controlled by NATCF and is the main transition route/corridor to the north. The Sally Corridor may be scheduled as the “Sally Buffer” if no other user will be transiting between the Northern or Southern Ranges and Nellis. The Sally Corridor airspace will revert to NATCF 15 minutes prior to the end of the scheduled Elgin time. |

**Table A4.48. CTA.**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface to 7,000ft MSL |
| **Highest Elevation** | 4,750ft MSL |
| **C2/Communication** | UHF 360.625/VHF 118.3 (Creech Tower) |
| **Hazards** | 1. Mountains are located along the eastern and western borders.<br>2. The Thunderbirds occasionally practice in CTA from surface to 7,000ft MSL as their Over-the-Field (OTF) training area. |
| **Entry/Departure Info** | As directed by Creech Tower control |
| **Supersonic Restrictions** | Supersonic flight is prohibited in the CTA. |
| **Ordnance Restrictions** | There are no strikeable targets in CTA; aircrews are prohibited from expending ordnance on CTA. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in CTA. |
| **Laser Restrictions** | Air/ground users are prohibited from employing lasers in CTA. |
| **Other Restrictions** | 1. Entrance into CTA without permission from Creech Tower is prohibited.<br>2. Aircrews must check NOTAMs to confirm the status of Thunderbirds training activity. |
| **Additional Info** | 1. RPAs operate in CTA.<br>2. CTA is located beneath the southern portion of R-64C and 65D.<br>3. CTA is active from 0600 Monday through 0630 Saturday. For access outside these hours, contact BLACKJACK or NATCF. |

**Table A4.49. Air-to-Air Gunnery Area.**

| | |
|---|---|
| **Range Type** | Class C |
| **Airspace** | Surface to unlimited |
| **Highest Elevation** | 6,985ft MSL |
| **C2/Communication** | 1. UHF 319.7.<br>2. Aerial Gunnery Range Control Officer (AGRCO), normally the flight lead, will provide airborne range control. |
| **Hazards** | Mountains throughout the area. |
| **Entry/Departure Info** | 1. Aircrews will contact BLACKJACK prior to entering the<br>Air-to-Air Gunnery Area.<br>2. Range entry and departure will be IAW BLACKJACK instructions. |
| **Supersonic Restrictions** | Supersonic flight is prohibited below 5,000ft in the Air-to-Air Gunnery Area. |
| **Ordnance Restrictions** | N/A |
| **Flare Restrictions** | Aircrews are prohibited from employing self-protection flares at altitudes that prevent burnout prior to 100ft AGL and illumination flares that prevent burnout prior to 500ft AGL. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | 1. NTTR/SE will develop a RSA for all air-to-air gunnery missions/operations. Aircrew will comply with all requirements of the RSA. Aircrew will fly air-to-air engagement patterns so firing is conducted within the boundaries of the “shoot-box” as defined in the RSA.<br>2. Only 20mm Target Practice (TP) and Target Practice Tracer (TPT) munitions are authorized. |
| **Additional Info** | 1. The Air-to-Air Gunnery Area is established on the NTTR for recurring aerial gunnery requirements.<br>2. The Air-to-Air Gunnery Area is an 8NM radius circle, centered at N36 55.075 W115 38.294 (within R-64A) and includes portions of R-61A, R-62A/B, R-64A/B, and R- 65A/B/C. Users must schedule all of the preceding subranges in order to utilize the Air-to-Air Gunnery Area.<br>3. Aerial gunnery may be conducted in at 10,000-30,000ft MSL.<br>4. Aerial Gunnery Target Sets (AGTS) will be used in the Air- to-Air Gunnery Area. |

**Table A4.50. STA.**

| | |
|---|---|
| **Range Type** | N/A |
| **Airspace** | Surface up to, but not including, 2,500ft AGL |
| **Highest Elevation** | 7,448ft MSL |
| **C2/Communication** | UHF 257.95 (Silverbow Tower) |
| **Hazards** | 1. Mountainous terrain exists along the western border.<br>2. Multiple towers are located throughout R-4809A. |
| **Entry/Departure Info** | As directed by Silverbow Tower control. |
| **Supersonic Restrictions** | Supersonic flight is prohibited in the STA. |
| **Ordnance Restrictions** | There are no strikeable targets in STA; aircrews are prohibited from expending ordnance on STA. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in STA. |
| **Laser Restrictions** | Air/ground users are prohibited from employing lasers in STA. |
| **Other Restrictions** | Aircrew will not fly directly over STA with live ordnance. |
| **Additional Info** | 1. Aircrew will coordinate with NATCF for advisory to 500ft AGL when the tower is inactive.<br>2. NATCF directs all IFR traffic into TTRA. |

**Table A4.51. X-Ray Extension.**

| | |
|---|---|
| **Range Type** | ATCAA |
| **Airspace** | FL240 to unlimited |
| **Highest Elevation** | 7,391ft MSL |
| **C2/Communication** | UHF 317.525 (NATCF East) |
| **Hazards** | None |
| **Entry/Departure Info** | As directed by NATCF |
| **Supersonic Restrictions** | Supersonic flight is prohibited in the X-Ray Extension. |
| **Ordnance Restrictions** | There are no strikeable targets in X-Ray Extension; aircrews are prohibited from expending ordnance on X-Ray Extension. |
| **Flare Restrictions** | Aircrews are prohibited from employing flares in the X-Ray Extension. |
| **Laser Restrictions** | Air/ground users are prohibited from lasing any object on the ground beneath this airspace. Air-to-air lasing is authorized when laser operations are performed IAW AFMAN 11-214. |
| **Other Restrictions** | None |
| **Additional Info** | 1. X-Ray Extension may be scheduled by contacting NATCF.<br>2. Elgin must be scheduled to activate X-Ray Extension.<br>3. NATCF may decline use if circumstances warrant. |

## Attachment 5

## BOUNDARY COORDINATES (DD MM.MM)

**Table A5.1. Alamo A.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W115 18.05 | | N37 00.00 | W115 07.05 |
| N37 17.00 | W115 11.05 | | N37 00.00 | W115 18.05 |
| N37 12.00 | W115 07.05 | | | |

**Table A5.2. Alamo B.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W115 18.05 | | N36 51.00 | W115 07.05 |
| N37 00.00 | W115 07.05 | | N36 51.00 | W115 18.05 |

**Table A5.3. Alamo C.**

| | | | | |
|---|---|---|---|---|
| N36 51.00 | W115 18.05 | | N36 38.00 | W115 18.05 |
| N36 51.00 | W115 07.05 | | N36 43.00 | W115 18.05 |
| N36 48.00 | W115 07.05 | | | |

**Table A5.4. R-61A.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W115 35.05 | | N37 00.00 | W115 33.05 |
| N37 17.00 | W115 26.05 | | N37 06.00 | W115 35.05 |
| N37 00.00 | W115 26.06 | | | |

**Table A5.5. R-61B.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W115 26.05 | | N37 00.00 | W115 18.05 |
| N37 17.00 | W115 18.05 | | N37 00.00 | W115 26.06 |

**Table A5.6. R-62A.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W115 33.05 | | N36 51.00 | W115 18.05 |
| N37 00.00 | W115 26.06 | | N36 51.00 | W115 33.05 |
| N37 00.00 | W115 18.05 | | | |

**Table A5.7. R-62B.**

| | | | | |
|---|---|---|---|---|
| N36 51.00 | W115 33.05 | | N36 45.01 | W115 25.47 |
| N36 51.00 | W115 18.05 | | N36 45.00 | W115 29.44 |
| N36 43.00 | W115 18.05 | | N36 43.00 | W115 29.44 |
| N36 43.00 | W115 25.50 | | N36 43.00 | W115 33.05 |

**Table A5.8. R-63A.**

| | | | | |
|---|---|---|---|---|
| N36 45.00 | W115 29.44 | | N36 43.00 | W115 25.50 |
| N36 45.01 | W115 25.47 | | N36 43.00 | W115 29.44 |

**Table A5.9. R-63B.**

| | | | | |
|---|---|---|---|---|
| N36 43.00 | W115 33.05 | | N36 32.01 | W115 24.89 |
| N36 43.00 | W115 29.44 | | N36 33.62 | W115 24.89 |
| N36 43.00 | W115 25.50 | | N36 33.62 | W115 30.00 |
| N36 43.00 | W115 18.05 | | N36 32.00 | W115 30.00 |
| N36 38.00 | W115 18.05 | | N36 30.48 | W115 30.00 |
| N36 36.69 | W115 18.05 | | N36 33.43 | W115 34.61 |
| N36 36.69 | W115 19.56 | | N36 37.57 | W115 33.94 |

**Table A5.10. R-63C.**

| | | | | |
|---|---|---|---|---|
| N36 33.62 | W115 30.00 | | N36 32.01 | W115 24.89 |
| N36 33.62 | W115 24.89 | | N36 32.00 | W115 30.00 |

**Table A5.11. R-63D Low.**

| | | | | |
|---|---|---|---|---|
| N36 36.69 | W115 19.56 | | N36 27.19 | W115 24.89 |
| N36 36.69 | W115 18.05 | | N36 29.81 | W115 24.89 |
| N36 26.00 | W115 18.05 | | N36 32.01 | W115 24.89 |
| N36 26.00 | W115 23.05 | | | |

**Table A5.12. R-63D Hi.**

| | | | | |
|---|---|---|---|---|
| N36 36.69 | W115 19.56 | | N36 28.50 | W115 26.93 |
| N36 36.69 | W115 18.05 | | N36 30.48 | W115 30.00 |
| N36 26.00 | W115 18.05 | | N36 32.00 | W115 30.00 |
| N36 26.00 | W115 23.05 | | N36 32.01 | W115 24.89 |
| N36 27.19 | W115 24.89 | | | |

**Table A5.13. R-64A.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W115 43.00 | | N36 51.00 | W115 33.05 |
| N37 00.00 | W115 33.05 | | N36 51.00 | W115 43.05 |

**Table A5.14. R-64B.**

| | | | | |
|---|---|---|---|---|
| N36 51.00 | W115 43.05 | | N36 43.00 | W115 33.05 |
| N36 51.00 | W115 33.05 | | N36 42.98 | W115 43.05 |

**Table A5.15. R-64C.**

| | | | | |
|---|---|---|---|---|
| N36 42.98 | W115 43.05 | | N36 35.00 | W115 37.05 |
| N36 43.00 | W115 33.05 | | N36 35.00 | W115 43.05 |
| N36 37.57 | W115 33.94 | | N36 40.37 | W115 43.05 |
| N36 33.43 | W115 34.61 | | | |

**Table A5.16. R-64D.**

| | | | | |
|---|---|---|---|---|
| N36 38.95 | W115 39.95 | | N36 38.94 | W115 38.08 |
| N36 39.11 | W115 39.28 | | N36 37.57 | W115 38.23 |
| N36 39.10 | W115 38.75 | | N36 37.62 | W115 40.08 |

**Table A5.17. R-65A.**

| | | | | |
|---|---|---|---|---|
| N37 06.00 | W115 56.05 | | N37 00.00 | W115 43.00 |
| N37 06.00 | W115 35.05 | | N37 00.00 | W115 56.05 |
| N37 00.00 | W115 33.05 | | | |

**Table A5.18. R-65B.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W115 56.05 | | N36 51.00 | W115 43.05 |
| N37 00.00 | W115 43.00 | | N36 51.00 | W115 56.05 |

**Table A5.19. R-65C.**

| | | | | |
|---|---|---|---|---|
| N36 51.00 | W115 56.05 | | N36 40.37 | W115 43.05 |
| N36 51.00 | W115 43.05 | | N36 40.39 | W115 56.05 |
| N36 42.98 | W115 43.05 | | N36 41.00 | W115 56.05 |

**Table A5.20. R-65D.**

| | | | | |
|---|---|---|---|---|
| N36 40.39 | W115 56.05 | | N36 35.00 | W115 53.05 |
| N36 40.37 | W115 43.05 | | N36 36.00 | W115 56.05 |
| N36 35.00 | W115 43.05 | | | |

**Table A5.21. R-67A.**

| | | | | |
|---|---|---|---|---|
| N36 32.00 | W115 30.00 | | N36 28.50 | W115 26.93 |
| N36 32.01 | W115 24.89 | | N36 30.48 | W115 30.00 |
| N36 29.81 | W115 24.89 | | | |

**Table 5.22. R-67B.**

| | | | | |
|---|---|---|---|---|
| N36 28.50 | W115 26.93 | | N36 27.19 | W115 24.89 |
| N36 29.81 | W115 24.89 | | | |

**Table A5.23. R-71N.**

| | | | | |
|---|---|---|---|---|
| N37 53.00 | W117 05.68 | | N37 40.00 | W116 49.05 |
| N37 53.00 | W116 55.05 | | N37 40.00 | W117 05.68 |
| N37 47.00 | W116 55.05 | | | |

**Table A5.24. R-71S.**

| | | | | |
|---|---|---|---|---|
| N37 40.00 | W117 05.68 | | N37 33.00 | W116 43.00 |
| N37 40.00 | W116 49.05 | | N37 33.00 | W117 05.68 |

**Table A5.25. R-74A.**

| | | | | |
|---|---|---|---|---|
| N37 33.00 | W116 00.84 | | N37 28.00 | W116 00.05 |
| N37 33.03 | W115 53.05 | | N37 16.00 | W116 00.05 |
| N37 33.05 | W115 48.05 | | N37 16.00 | W116 11.05 |
| N37 28.01 | W115 48.05 | | N37 19.98 | W116 11.03 |

**Table A5.26. R-74B.**

| | | | | |
|---|---|---|---|---|
| N37 37.96 | W116 01.48 | | N37 19.98 | W116 11.03 |
| N37 37.93 | W115 53.05 | | N37 23.00 | W116 17.05 |
| N37 33.03 | W115 53.05 | | N37 23.00 | W116 18.05 |
| N37 33.00 | W116 00.84 | | N37 23.00 | W116 19.03 |

**Table A5.27. R-74C.**

| | | | | |
|---|---|---|---|---|
| N37 42.00 | W116 11.05 | | N37 23.00 | W116 19.03 |
| N37 42.00 | W115 53.05 | | N37 32.00 | W116 19.01 |
| N37 37.93 | W115 53.05 | | N37 32.00 | W116 17.00 |
| N37 37.96 | W116 01.48 | | | |

**Table A5.28. R-75E.**

| | | | | |
|---|---|---|---|---|
| N37 33.00 | W116 26.05 | | N37 23.00 | W116 22.05 |
| N37 33.00 | W116 19.01 | | N37 21.00 | W116 27.05 |
| N37 32.00 | W116 19.01 | | N37 21.00 | W116 34.05 |
| N37 23.00 | W116 19.03 | | | |

**Table A5.29. R-75W.**

| | | | | |
|---|---|---|---|---|
| N37 33.00 | W116 42.50 | | N37 20.00 | W116 40.12 |
| N37 33.00 | W116 26.05 | | N37 22.52 | W116 41.96 |
| N37 21.00 | W116 34.05 | | N37 22.52 | W116 42.50 |
| N37 20.00 | W116 37.05 | | | |

**Table A5.30. R-76.**

| | | | | |
|---|---|---|---|---|
| N37 33.00 | W117 05.68 | | N37 20.00 | W116 49.51 |
| N37 33.00 | W116 43.00 | | N37 20.00 | W116 50.05 |
| N37 33.00 | W116 42.50 | | N37 16.01 | W116 50.05 |
| N37 22.52 | W116 42.50 | | N37 16.00 | W116 55.32 |
| N37 22.51 | W116 45.01 | | N37 26.50 | W117 04.55 |
| N37 22.52 | W116 47.38 | | | |

**Table A5.31. R-77A.**

| | | | | |
|---|---|---|---|---|
| N37 09.88 | W116 49.96 | | N37 00.00 | W116 27.05 |
| N37 16.01 | W116 46.07 | | N36 55.00 | W116 27.05 |
| N37 16.00 | W116 31.05 | | N36 55.00 | W116 33.55 |
| N37 08.00 | W116 27.05 | | N36 51.00 | W116 33.55 |

**Table A5.31. R-77B.**

| | | | | |
|---|---|---|---|---|
| N37 09.88 | W116 49.96 | | N37 16.01 | W116 50.05 |
| N37 16.00 | W116 55.32 | | N37 16.01 | W116 46.07 |

**Table A5.32. EC East.**

| | | | | |
|---|---|---|---|---|
| N37 53.00 | W116 26.05 | | N37 33.00 | W116 26.05 |
| N37 53.00 | W116 11.05 | | N37 42.11 | W116 26.05 |
| N37 42.00 | W116 11.05 | | N37 42.12 | W116 23.00 |
| N37 32.00 | W116 17.00 | | N37 45.11 | W116 23.00 |
| N37 32.00 | W116 19.01 | | N37 45.11 | W116 26.05 |
| N37 33.00 | W116 19.01 | | | |

**Table A5.33. R-4807A (Subset).**

| | | | | |
|---|---|---|---|---|
| N37 45.11 | W116 26.05 | | N37 42.12 | W116 23.00 |
| N37 45.11 | W116 23.00 | | N37 42.11 | W116 26.05 |

**Table A5.34. TPECR.**

| | | | | |
|---|---|---|---|---|
| N37 20.00 | W116 50.05 | | N37 21.00 | W116 34.05 |
| N37 20.00 | W116 49.51 | | N37 16.00 | W116 31.05 |
| N37 20.00 | W116 45.02 | | N37 16.01 | W116 46.07 |
| N37 20.00 | W116 40.12 | | N37 16.01 | W116 50.05 |
| N37 20.00 | W116 37.05 | | | |

**Table A5.35. TPECR (FAC A).**

| | | | | |
|---|---|---|---|---|
| N37 22.52 | W116 47.38 | | N37 20.00 | W116 45.02 |
| N37 22.51 | W116 45.01 | | N37 20.00 | W116 49.51 |

**Table A5.36. TPECR (FAC B).**

| | | | | |
|---|---|---|---|---|
| N37 22.51 | W116 45.01 | | N37 20.00 | W116 40.12 |
| N37 22.52 | W116 42.50 | | N37 20.00 | W116 45.02 |
| N37 22.52 | W116 41.96 | | | |

**Table A5.37. Pahute A.**

| | | | | |
|---|---|---|---|---|
| N37 23.00 | W116 18.05 | | N37 16.00 | W116 11.05 |
| N37 23.00 | W116 17.05 | | N37 16.00 | W116 15.05 |
| N37 19.98 | W116 11.03 | | N37 16.01 | W116 18.05 |

**Table A5.38. Pahute B.**

| | | | | |
|---|---|---|---|---|
| N37 21.00 | W116 34.05 | | N37 23.00 | W116 18.05 |
| N37 21.00 | W116 27.05 | | N37 16.01 | W116 18.05 |
| N37 23.00 | W116 22.05 | | N37 16.00 | W116 31.05 |
| N37 23.00 | W116 19.03 | | | |

**Table A5.39. R-4808A.**

| | | | | |
|---|---|---|---|---|
| N37 28.00 | W116 00.05 | | N37 06.00 | W115 35.05 |
| N37 28.01 | W115 48.05 | | N37 06.00 | W115 56.05 |
| N37 28.00 | W115 35.05 | | N37 06.00 | W116 00.05 |
| N37 17.00 | W115 35.05 | | N37 16.00 | W116 00.05 |

**Table A5.40. R-4808B.**

| | | | | |
|---|---|---|---|---|
| N37 16.00 | W116 15.05 | | N37 06.00 | W115 56.05 |
| N37 16.00 | W116 11.05 | | N37 00.00 | W115 56.05 |
| N37 16.00 | W116 00.05 | | N37 00.00 | W116 15.05 |
| N37 06.00 | W116 00.05 | | | |

**Table A5.41. R-4808C.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W116 15.05 | | N36 41.00 | W115 56.05 |
| N37 00.00 | W115 56.05 | | N36 41.00 | W116 01.06 |
| N36 51.00 | W115 56.05 | | N36 50.00 | W116 15.05 |

**Table A5.42. R-4808D.**

| | | | | |
|---|---|---|---|---|
| N37 16.00 | W116 31.05 | | N37 00.00 | W116 15.05 |
| N37 16.01 | W116 18.05 | | N37 00.00 | W116 27.05 |
| N37 16.00 | W116 15.05 | | N37 08.00 | W116 27.05 |

**Table A5.43. R-4808E.**

| | | | | |
|---|---|---|---|---|
| N37 00.00 | W116 27.05 | | N36 46.00 | W116 26.55 |
| N37 00.00 | W116 15.05 | | N36 51.00 | W116 26.55 |
| N36 50.00 | W116 15.05 | | N36 51.00 | W116 33.55 |
| N36 41.00 | W116 01.06 | | N36 55.00 | W116 33.55 |
| N36 41.00 | W116 14.80 | | N36 55.00 | W116 27.05 |

**Table A5.44. R-4808S.**

| | | | | |
|---|---|---|---|---|
| N36 46.00 | W116 26.55 | | N36 41.00 | W116 26.55 |
| N36 41.00 | W116 14.80 | | | |

**Table A5.45. EC West.**

| | | | | |
|---|---|---|---|---|
| N37 53.01 | W116 40.05 | | N37 42.11 | W116 26.05 |
| N37 53.00 | W116 26.05 | | N37 33.00 | W116 26.05 |
| N37 45.11 | W116 26.05 | | N37 33.00 | W116 42.50 |
| N37 45.12 | W116 30.03 | | N37 33.00 | W116 43.00 |
| N37 44.21 | W116 30.02 | | N37 40.00 | W116 49.05 |
| N37 42.15 | W116 27.99 | | N37 40.00 | W116 40.50 |

**Table A5.46. R-4809A.**

| | | | | |
|---|---|---|---|---|
| N37 53.00 | W116 55.05 | | N37 40.00 | W116 40.05 |
| N37 53.01 | W116 50.00 | | N37 40.00 | W116 49.05 |
| N37 53.01 | W116 40.05 | | N37 47.00 | W116 55.05 |

**Table A5.47. R-4809B.**

| | | | | |
|---|---|---|---|---|
| N37 45.12 | W116 30.03 | | N37 42.15 | W116 27.99 |
| N37 45.11 | W116 26.05 | | N37 44.21 | W116 30.02 |
| N37 42.11 | W116 26.05 | | | |

**Table A5.48. Reveille North.**

| | | | | |
|---|---|---|---|---|
| N38 14.00 | W116 18.87 | | N38 00.25 | W114 34.56 |
| N38 14.00 | W115 00.00 | | N37 59.96 | W116 40.02 |
| N38 01.00 | W114 12.00 | | | |

**Table A5.49. Reveille South.**

| | | | | |
|---|---|---|---|---|
| N37 59.96 | W116 40.02 | | N37 54.53 | W115 53.03 |
| N38 00.25 | W114 34.56 | | N37 53.00 | W116 11.05 |
| N37 59.33 | W114 50.72 | | N37 53.00 | W116 26.05 |
| N37 58.24 | W115 07.05 | | N37 53.01 | W116 40.05 |
| N37 56.75 | W115 25.98 | | N37 53.01 | W116 50.00 |

**Table A5.50. Coyote A.**

| | | | | |
|---|---|---|---|---|
| N37 33.05 | W115 48.05 | | N37 17.00 | W115 35.05 |
| N37 33.07 | W115 26.00 | | N37 28.00 | W115 35.05 |
| N37 17.00 | W115 26.05 | | N37 28.01 | W115 48.05 |

**Table A5.51. Coyote B.**

| | | | | |
|---|---|---|---|---|
| N37 56.75 | W115 25.98 | | N37 17.00 | W115 11.05 |
| N37 58.24 | W115 07.05 | | N37 17.00 | W115 18.05 |
| N37 17.00 | W115 07.05 | | N37 17.00 | W115 26.05 |
| N37 12.00 | W115 07.05 | | N37 33.07 | W115 26.00 |

**Table A5.52. Coyote C.**

| | | | | |
|---|---|---|---|---|
| N37 54.53 | W115 53.03 | | N37 33.03 | W115 53.05 |
| N37 56.75 | W115 25.98 | | N37 37.93 | W115 53.05 |
| N37 33.07 | W115 26.00 | | N37 42.00 | W115 53.05 |
| N37 33.05 | W115 48.05 | | | |

**Table A5.53. Coyote D.**

| | | | | |
|---|---|---|---|---|
| N37 53.00 | W116 11.05 | | N37 42.00 | W115 53.05 |
| N37 54.53 | W115 53.03 | | N37 42.00 | W116 11.05 |

**Table A5.54. Caliente A.**

| | | | | |
|---|---|---|---|---|
| N37 58.24 | W115 07.05 | | N37 17.00 | W114 50.72 |
| N37 59.33 | W114 50.72 | | N37 17.00 | W115 07.05 |

**Table A5.55. Caliente B**

| | | | | |
|---|---|---|---|---|
| N37 59.33 | W114 50.72 | | N37 27.90 | W114 34.70 |
| N38 00.25 | W114 34.56 | | N37 17.00 | W114 50.72 |

**Table A5.56. Caliente C.**

| | | | | |
|---|---|---|---|---|
| N38 00.25 | W114 34.56 | | N37 49.42 | W113 42.02 |
| N38 01.00 | W114 12.00 | | N37 27.02 | W114 00.47 |
| N37 53.73 | W113 42.07 | | N37 27.90 | W114 34.70 |

**Table A5.57. Caliente D.**

| | | | | |
|---|---|---|---|---|
| N37 58.10 | W114.00.00 | | N37 49.42 | W113 42.02 |
| N37 53.73 | W113 42.07 | | N37 43.20 | W113 47.17 |

**Table A5.58. Elgin.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W114 50.72 | | N36 43.00 | W114 36.05 |
| N37 27.90 | W114 34.70 | | N36 43.00 | W114 50.72 |
| N37 27.02 | W114 00.47 | | | |

**Table A5.59. Sally Corridor.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W115 07.05 | | N36 26.00 | W115 18.05 |
| N37 17.00 | W114 50.72 | | N36 38.00 | W115 18.05 |
| N36 43.00 | W114 50.72 | | N36 48.00 | W115 07.05 |
| N36 43.00 | W115 03.05 | | | |

**Table A5.60. Alamo Corridor.**

| | | | | |
|---|---|---|---|---|
| N37 17.00 | W115 18.05 | | N36 48.00 | W115 07.05 |
| N37 17.00 | W115 11.05 | | N36 38.00 | W115 18.05 |
| N37 12.00 | W115 07.05 | | | |

**Table A5.61. Creech Tower Airspace.**

| | | | | |
|---|---|---|---|---|
| N36 37.62 | W115 47.40 | | N36 33.43 | W115 34.61 |
| N36 37.62 | W115 40.08 | | N36 35.00 | W115 37.05 |
| N36 37.57 | W115 38.23 | | N36 35.00 | W115 43.05 |
| N36 37.57 | W115 33.94 | | N36 35.00 | W115 47.40 |

**Table A5.62. Gomer Triangle.**

| | | | | |
|---|---|---|---|---|
| N37 16.00 | W116 31.05 | | N37 08.00 | W116 27.05 |
| N37 16.00 | W116 15.05 | | N37 16.00 | W116 31.05 |

**Table A5.63. Silverbow Tower Airspace**.

| | | | | |
|---|---|---|---|---|
| STA is described as a 5NM radius circle within the boundaries of R-4809A, centered at: | | | | |
| N37 47.93 | W116 46.85 | | | |

**Table A5.64. Shelf.**

| | | | | |
|---|---|---|---|---|
| N37 42.90 | W116 51.53 | | N37 40.00 | W116 49.05 |
| N37 42.90 | W116 40.05 | | N37 42.90 | W116 51.53 |
| N37 40.00 | W116 40.05 | | | |

**Table A5.65. Dorito.**

| | | | | |
|---|---|---|---|---|
| N37 59.96 | W116 40.02 | | N37 53.01 | W116 50.00 |
| N37 53.01 | W116 40.05 | | N37 59.96 | W116 40.02 |

**Table A5.66. X-Ray Extension.**

| | | | | |
|---|---|---|---|---|
| N36 43.00 | W115 03.05 | | N36 26.00 | W114 53.00 |
| N36 43.00 | W114 36.05 | | N36 26.00 | W115 18.05 |

**Table A5.67. NTTR/UTTR Transition Corridor.**

| | | | | |
|---|---|---|---|---|
| N39 23.00 | W114 27.00 | | N38 07.92 | W114 37.37 |
| N39 23.00 | W113 19.00 | | N39 23.00 | W114 27.00 |
| N38 01.00 | W114 12.00 | | | |

## Attachment 6

## REMOTELY PILOTED AIRCRAFT (RPA) OPERATING AREAS

**A6.1. RPA Routes**. In additiona to scheduled airspace, predetermined routes have been identified for use for RPA transit:

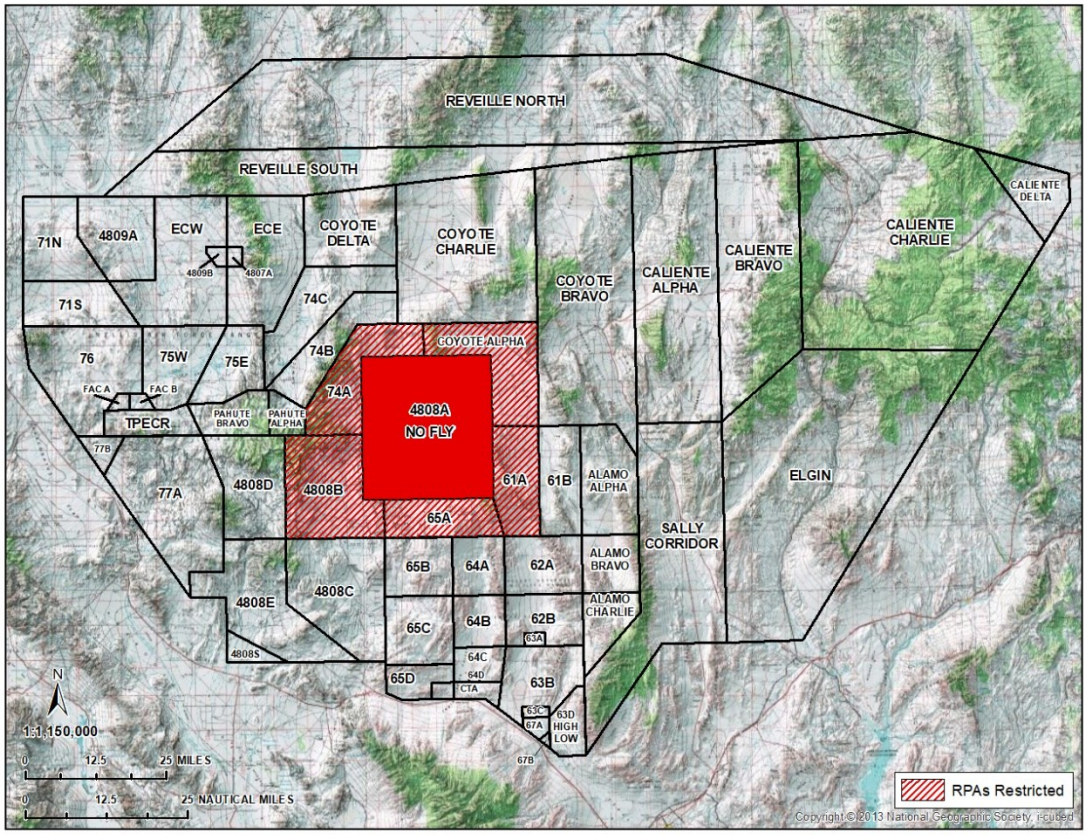
A6.1.1. Creech Corridor N/S and RPA Corridor Two. The RPA imaging sensor will be fully stowed (e.g., unable to image the surrounding air and ground space) while utilizing the corridors unless otherwise approved through the NTTR Scheduling Process.

A6.1.2. RPA Transit Routes Tan, Green, and Blue.

**A6.2. Operating Areas.** Authority to operate RPAs within R-4808 requires specific NNSS/NFO and DOE approval through the NTTR Scheduling Process. RPAs are normally restricted from operating in the contiguous airspace made up of R-74A, Coyote A, R-61A, R-65A, R-4808B.

A6.2.1. RPAs may operate in R-4808B with NNSS and DOE approval.

**Figure A6.1. RPA Contiguous Airspace Restriction.**




**A6.3. Creech Corridor and RPA Corridor Two.** These corridors are used to transit between Creech AFB and R-63D H/L and through R-4808C/D/E. When utilized, deviation from the scheduled corridor is not authorized without coordination from appropriate using agency and orbiting within corridors is not authorized.

**A6.4. Creech Corridor N/S.** This corridor is 1.5NM wide and overlies CTA from the east edge of R-65D to the southeast edge of CTA, then extends southeast to R-63D High. Creech Corridor N/S may be scheduled in whole or in segments through the NTTR Scheduling Process. The Creech Corridor N/S segments are as depicted in figure A6.2

**Figure A6.2. Creech Corridor One.**

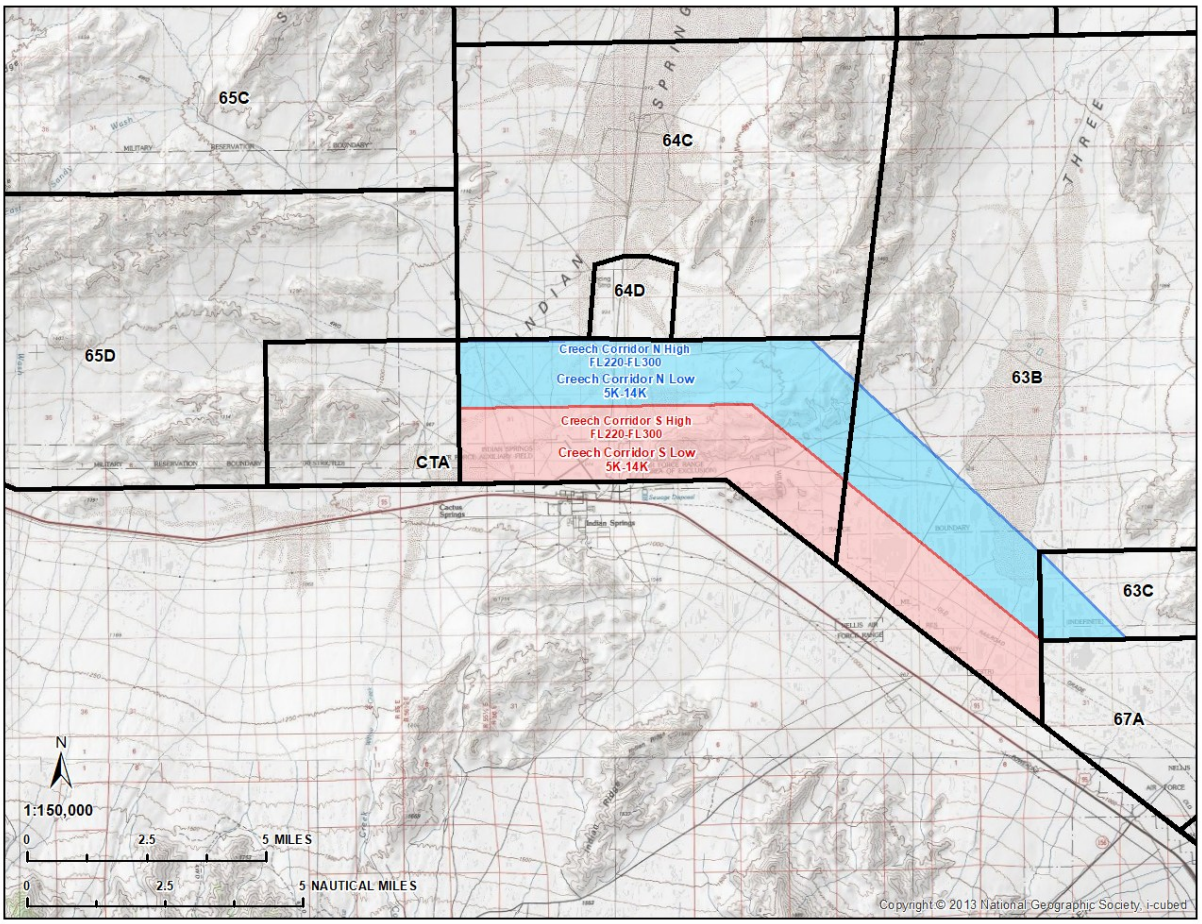
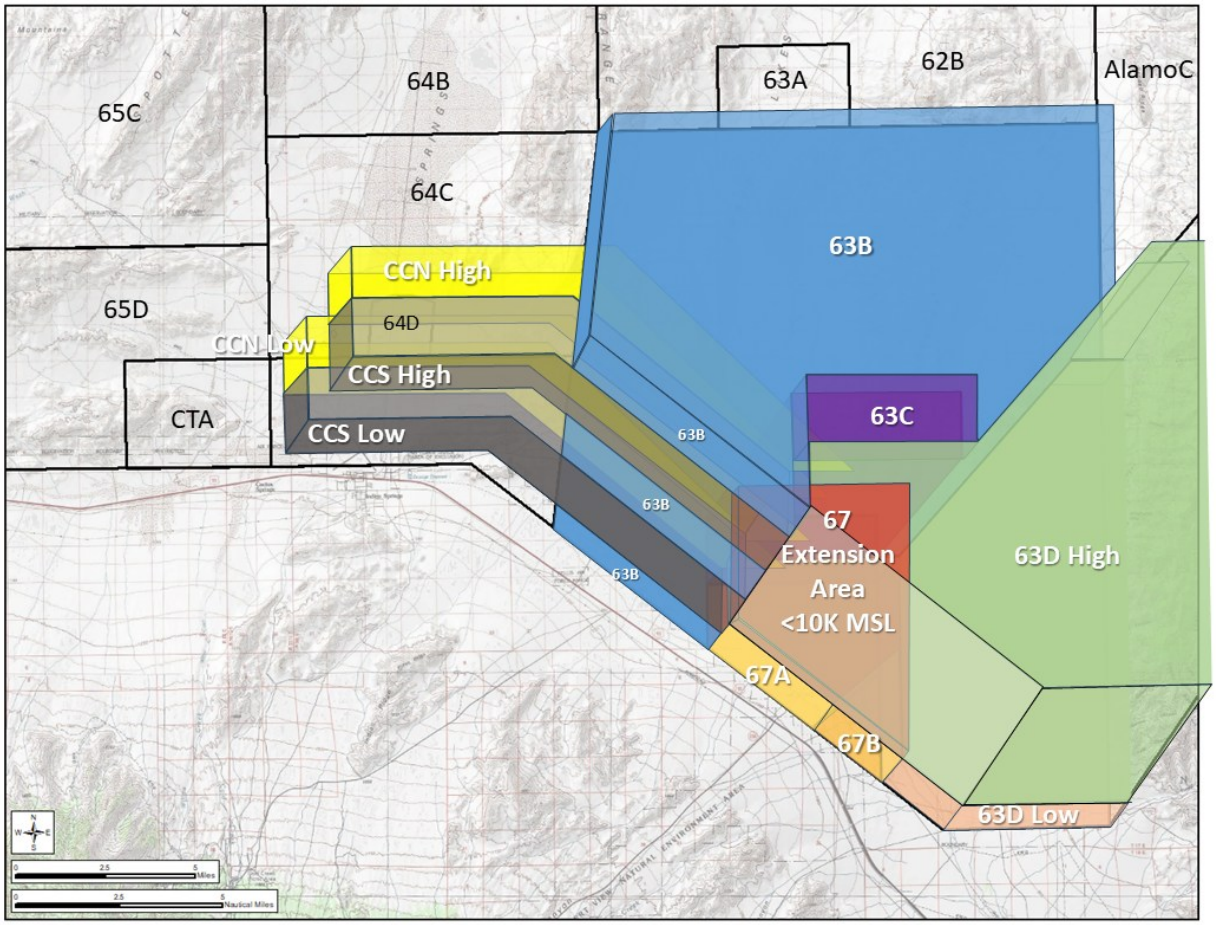


**Figure A6.3. Relationship of Creech Corridor One to R-63B and R-63D High.**




**A6.5. RPA Transit Corridor Two.** This corridor is 1NM wide between Inlet (in R-65C) and either Alpha (in R-77A/B) or Roy (in Pahute B) traversing R-4808C/D/E through Elbow, Skid, Fridmo, and Gerte. RPA Transit Corridor Two may be scheduled in whole or in segments through the NTTR Scheduling Process utilizing the following way points:

**Table A6.1. RPA Transit Corridor WayPoints.**

| Turn Point | Coordinates | | | Turn Point | Coordinates | |
|---|---|---|---|---|---|---|
| Inlet | N36 42.00 | W115 55.00 | | Gerte | N36 55.00 | W116 26.00 |
| Elbow | N36 42.00 | W116 10.00 | | Alpha | N36 55.53 | W116 27.25 |
| Skid | N36 42.00 | W116 15.00 | | Roy | N37 16.02 | W116 26.00 |
| Fridmo | N36 46.35 | W116 25.77 | | | | |

A6.5.1. The RPA Transit Corridor Two segments are:

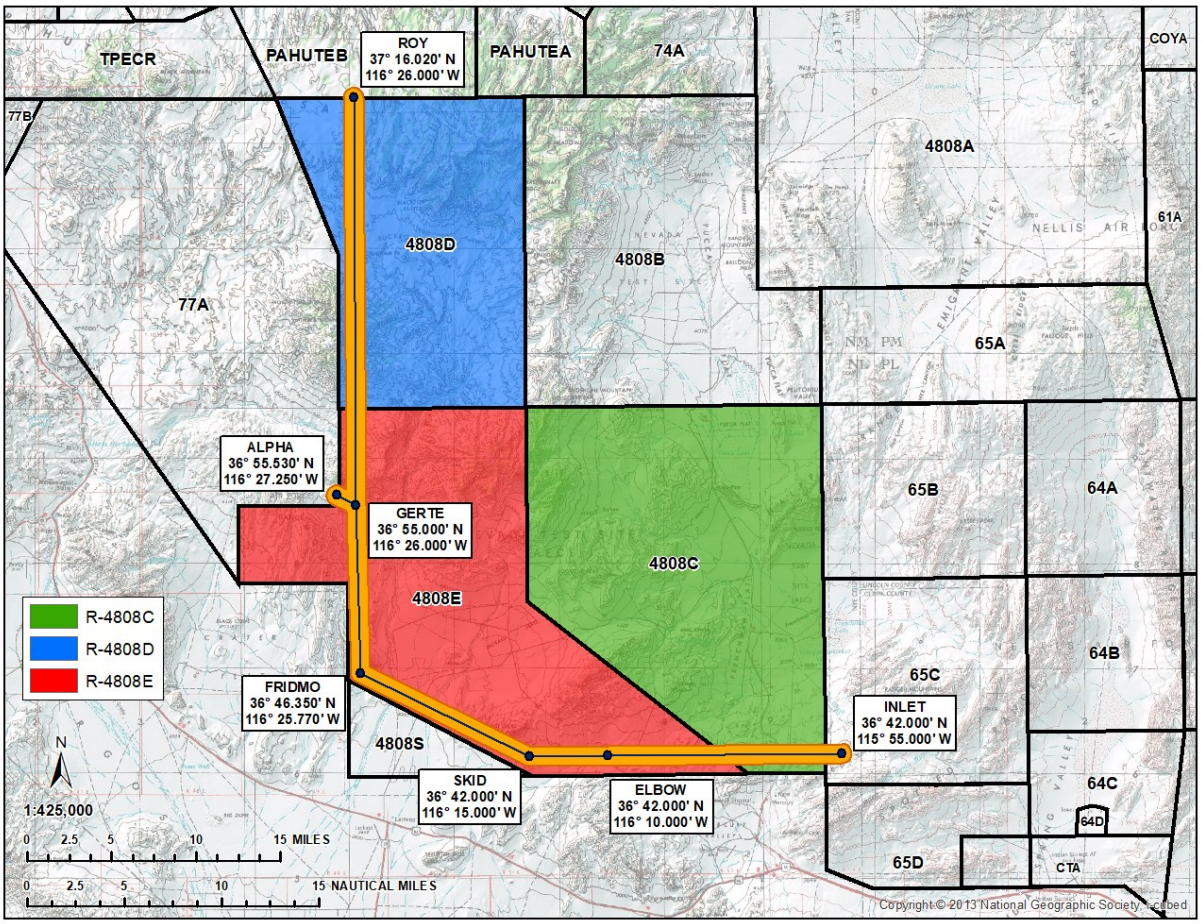
A6.5.1.1. RPA Transit Corridor Two-Segment 1: Between Inlet, Elbow, Skid, and Fridmo; 10,000-12,000ft MSL.

A6.5.1.2. RPA Transit Corridor Two-Segment 2: Between Fridmo, Gerte, and Alpha; 10,000- 18,000ft MSL.

A6.5.1.3. RPA Transit Corridor Two-Segment 3: Between Gerte and Roy; 10,000-18,000ft MSL.

A6.5.2. The RPA must contact DML and Birddog Operations to relay position/altitude NLT 5 minutes prior to entering RPA Transit Corridor Two. Additionally, the RPA must contact DML and Birddog Operations to relay position/altitude NLT 5 minutes prior to entering the Elbow to Inlet corridor segment when traveling eastbound.

**Figure A6.4. RPA Transit Corridor Two.**




**A6.6. RPA Transit Routes.** These transit routes are designed to aid RPA transit within the NTTR, to and from the north portions of the range.

A6.6.1. Tan Route. This route is 1NM wide and located on the western border of the NTTR. It starts at Alpha, located in R-77A, and ends at Foxtrot in R-71N. This route may only be scheduled in segments if the unit is not leaving the NTTR. The minimum altitude authorized is 1,000ft AGL. Use of this route must be coordinated through the NTTR Scheduling Process.

**Table A6.2. Tan Route WayPoints.**

| Turn Point | Coordinates | | | Turn Point | Coordinates | |
|---|---|---|---|---|---|---|
| Alpha | N36 55.53 | W116 27.25 | | Delta | N37 33.00 | W117 05.01 |
| Bravo | N36 55.56 | W116 36.69 | | Echo | N37 48.06 | W117 05.01 |
| Charlie | N37 26.52 | W117 03.69 | | Foxtrot | N37 52.47 | W117 00.00 |

A6.6.2. Blue Route. This route is 1NM wide and located on the western border of Caliente C and Elgin. It starts at Golf, located in Elgin, and ends at Kilo in Caliente C. This route may only be scheduled in segments if the unit is not leaving the NTTR. The minimum altitude authorized is 12,000ft MSL. Use of this route must be coordinated through the NTTR Scheduling Process.

**Table A6.3. Blue Route WayPoints.**

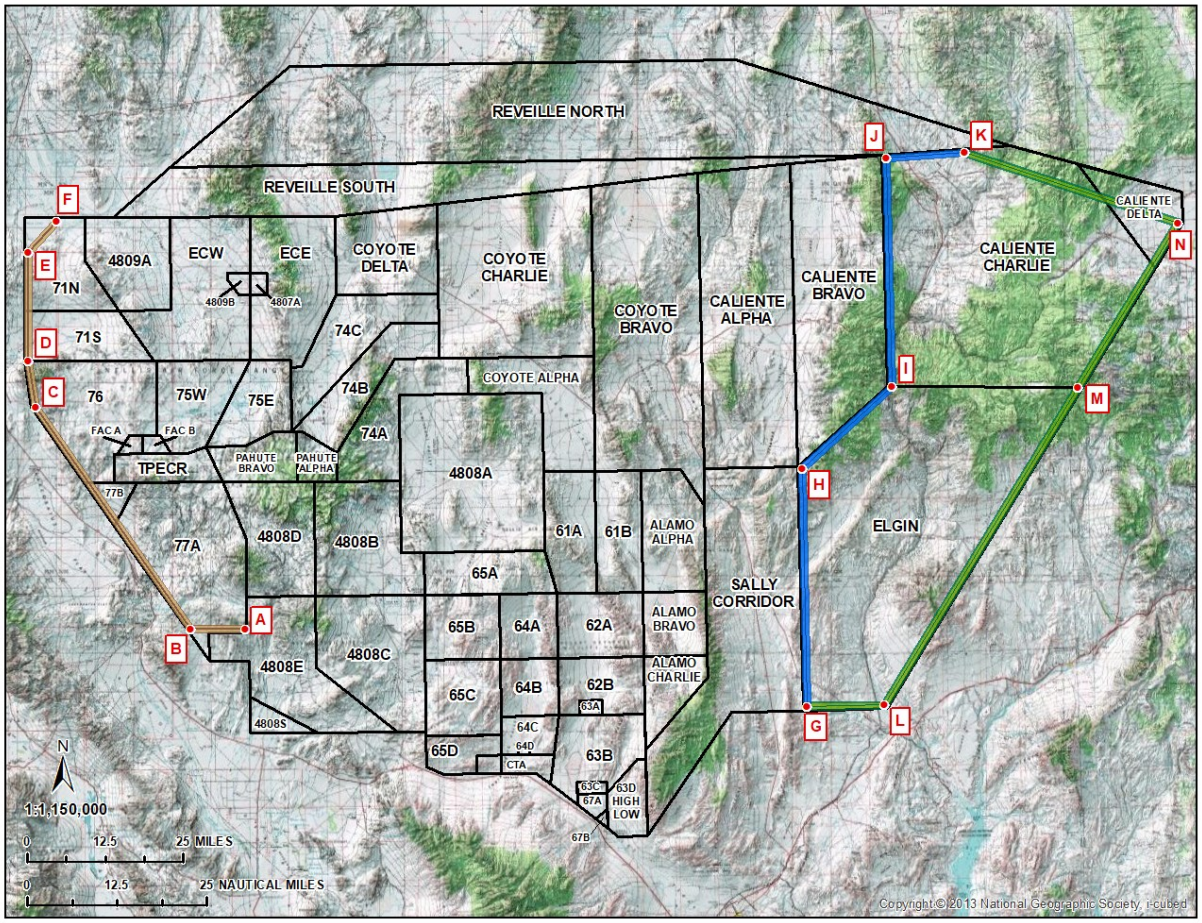
| Turn Point | Coordinates | | | Turn Point | Coordinates | |
|---|---|---|---|---|---|---|
| Golf | N36 43.57 | W114 49.98 | | Juliet | N37 59.73 | W114 33.85 |
| Hotel | N37 16.78 | W114 49.87 | | Kilo | N38 00.21 | W114 20.01 |
| India | N37 27.88 | W114 33.79 | | | | |

A6.6.3. Green Route. This route is 1NM wide and located on the eastern border of the NTTR. It starts at Golf, located in Elgin, and ends at Kilo in Caliente C. This route may only be scheduled in segments if the unit is not leaving the NTTR. The minimum altitude authorized is 12,000ft MSL. Use of this route must be coordinated through the NTTR Scheduling Process.

**Table A6.4. Green Route WayPoints.**

| Turn Point | Coordinates | | | Turn Point | Coordinates | |
|---|---|---|---|---|---|---|
| Golf | N36 43.57 | W114 49.98 | | November | N37 49.52 | W113 42.89 |
| Lima | N36 43.55 | W114 36.50 | | Kilo | N38 00.21 | W114 20.01 |
| Mike | N37 27.04 | W114 01.25 | | | | |

**Figure A6.5. RPA Transit Routes.**



## Attachment 7

## SUPERSONIC OPERATING AREAS

**Table A7.1. Authorized NTTR Supersonic Areas.**

| NTTR Subrange Designations | Supersonic Flight Authorized |
|---|---|
| R-71N/S<br>R-74A/B/C<br>R-75E/W R-76 | 100ft AGL to unlimited |
| Alamo A/B/C<br>Caliente A/B[1]<br>Coyote A/B/C/D<br>EC East R77A/B<br>Pahute A/B<br>R-61A/B<br>R-62A/B<br>R-63A/B/C/D<br>R-64A/B/C<br>R-65A/B/C/D R-67A/BTPECR[2] | 5,000ft AGL to unlimited |
| Reveille North | 5,000ft AGL to maximum scheduled altitude west of a line from:<br>N38 00.42 W114 29.53 to N38 05.90 W114 34.92 to N38 07.28 W114 35.01 |
| | 30,000ft MSL to unlimited east of a line from:<br>N38 00.42 W114 29.53 to N38 05.90 W114 34.92 to N38 07.28 W114 35.01 |
| Reveille South | 5,000ft AGL to maximum scheduled altitude |
| Sally Corridor[3] | 5,000ft AGL to unlimited north of N36 52.00 |
| | 30,000ft MSL to unlimited south of N36 52.00 |
| Elgin[3] | 5,000ft AGL to unlimited north of a line from:<br>N36 52.00 W114 50.72 to N37 04.00 W114 33.00 to N37 04.00 W114 19.19 |
| | 30,000ft MSL to unlimited south of a line from:<br>N36 52.00 W114 50.72 to N37 04.00 W114 33.00 to N37 04.00 W114 19.19 |
| Caliente C[1,4] | 30,000ft MSL to unlimited |
| Caliente D | 31,000ft MSL to unlimited |

| | |
|---|---|
| EC West[5]<br>R-4809A[5]<br>R-4809B | Above 5,000ft AGL and specifically scheduled through the NTTR Scheduling Process. |
| R-4808B/C/D/E | As approved by NNSS/NFO and DOE and specifically scheduled through the NTTR scheduling process. |

Note 1: The noise sensitive area in the vicinity of the Caliente, NV (a 5NM radius circle, centered at N37 36.83 W114 31.33, within the boundaries of Caliente B/C) is restricted to subsonic, surface to unlimited.
Note 2: For scheduling and supersonic operations purposes, TPECR includes FAC A/B ground/airspace.
Note 3: Supersonic flight south of N36 48.00 is only permitted with magnetic headings of 259°clockwise to 079°. Note 4: The noise sensitive area in the vicinity of the Lincoln County Airport (a 3NM radius circle, centered at N37 47.25 W114 25.30, within the boundaries of Caliente C) is restricted to subsonic, surface to unlimited.
Note 5: NNSS/SFO may schedule supersonic operations 100ft AGL to unlimited in the three supersonic target entry cones in EC West and R-4809A.
NNSS/SFO will coordinate their desires through NTTR Projects Office to obtain approval.

**Figure A7.1. Authorized NTTR Supersonic Areas.**

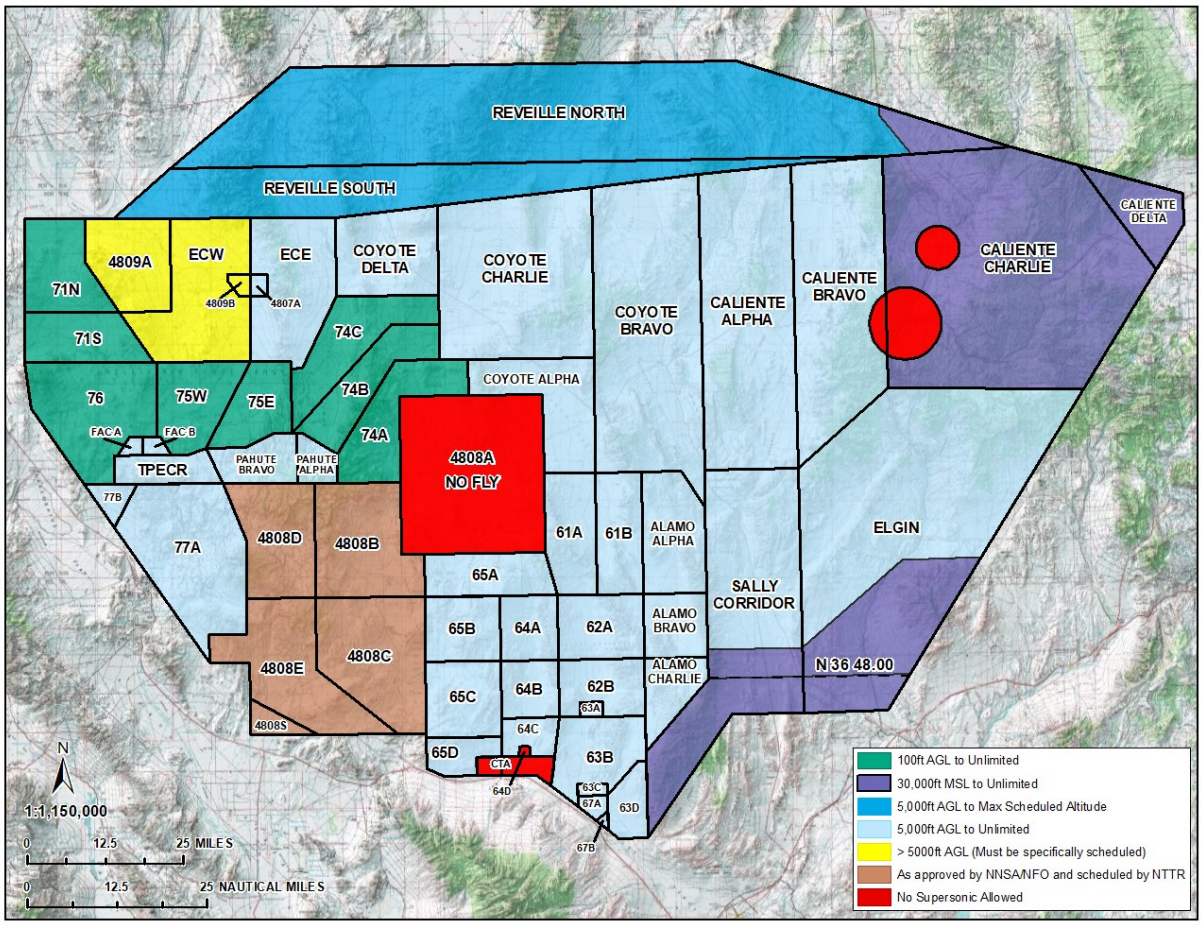


**Table A7.2. NNSA/SFO Antelope Lake Target North Supersonic Entry/Exit Cone.**

| Magnetic headings of 150° clockwise to 223° or 330° clockwise to 043°. | | | | |
|---|---|---|---|---|
| Coordinates | | | | |
| N37 53.00 | W116 45.00 | | N37 47.60 | W116 26.05 |
| N37 42.00 | W116 41.30 | | N37 40.50 | W116 39.00 |

**Table A7.3. NNSA/SFO Antelope Lake Target South Supersonic Entry/Exit Cone.**

| Magnetic headings of 339° clockwise to 007° or 159° clockwise to 187°. | | | | |
|---|---|---|---|---|
| Coordinates | | | | |
| N37 42.00 | W116 41.30 | | N37 40.50 | W116 39.00 |
| N37 34.80 | W116 44.55 | | N37 33.00 | W116 37.60 |

**Figure A7.2. NNSA/SFO Antelope Lake Target North and South Supersonic Entry/Exit Cones.**

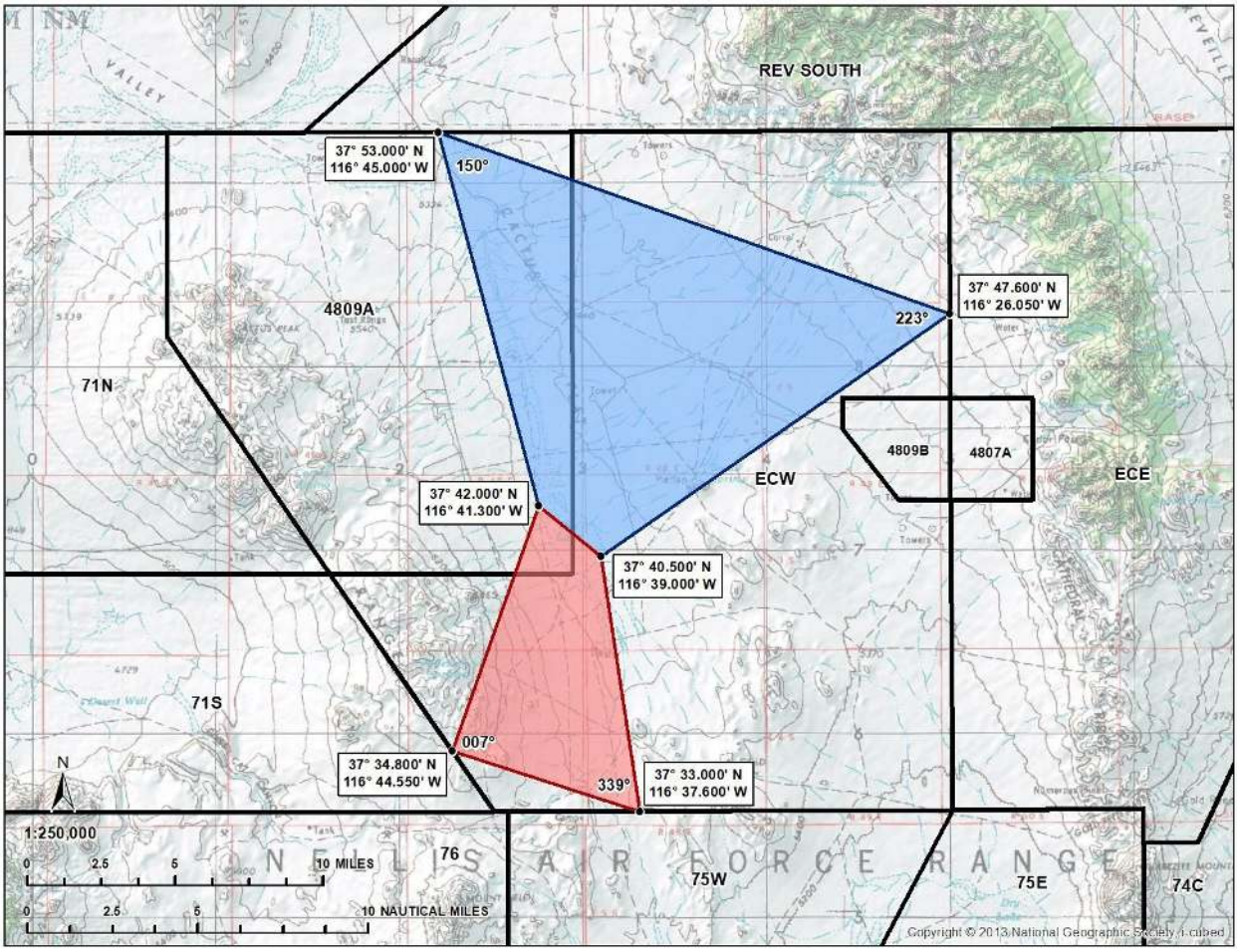
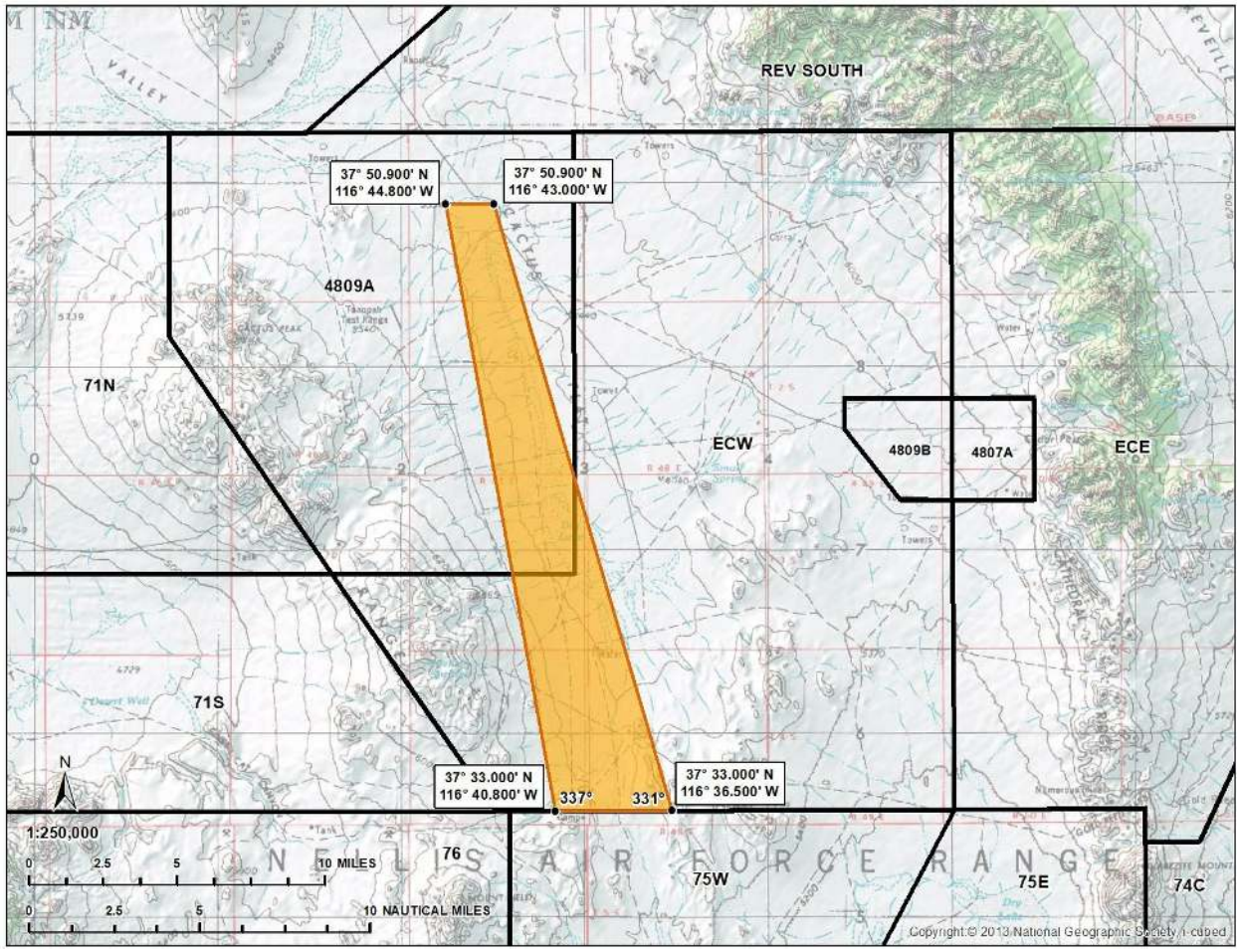


**Table A7.4. NNSA/SFO Main Lake Target South Supersonic Entry/Exit Cone.**

| Magnetic headings of 331° clockwise to 337° or 151° clockwise to 157°. | | | | |
|---|---|---|---|---|
| Coordinates | | | | |
| N37 50.90 | W116 44.80 | | N37 50.90 | W116 43.00 |
| N37 33.00 | W116 40.80 | | N37 33.00 | W116 36.50 |

**Figure A7.3. NNSA/SFO Main Lake Target South Supersonic Entry/Exit Cone.**



## Attachment 8

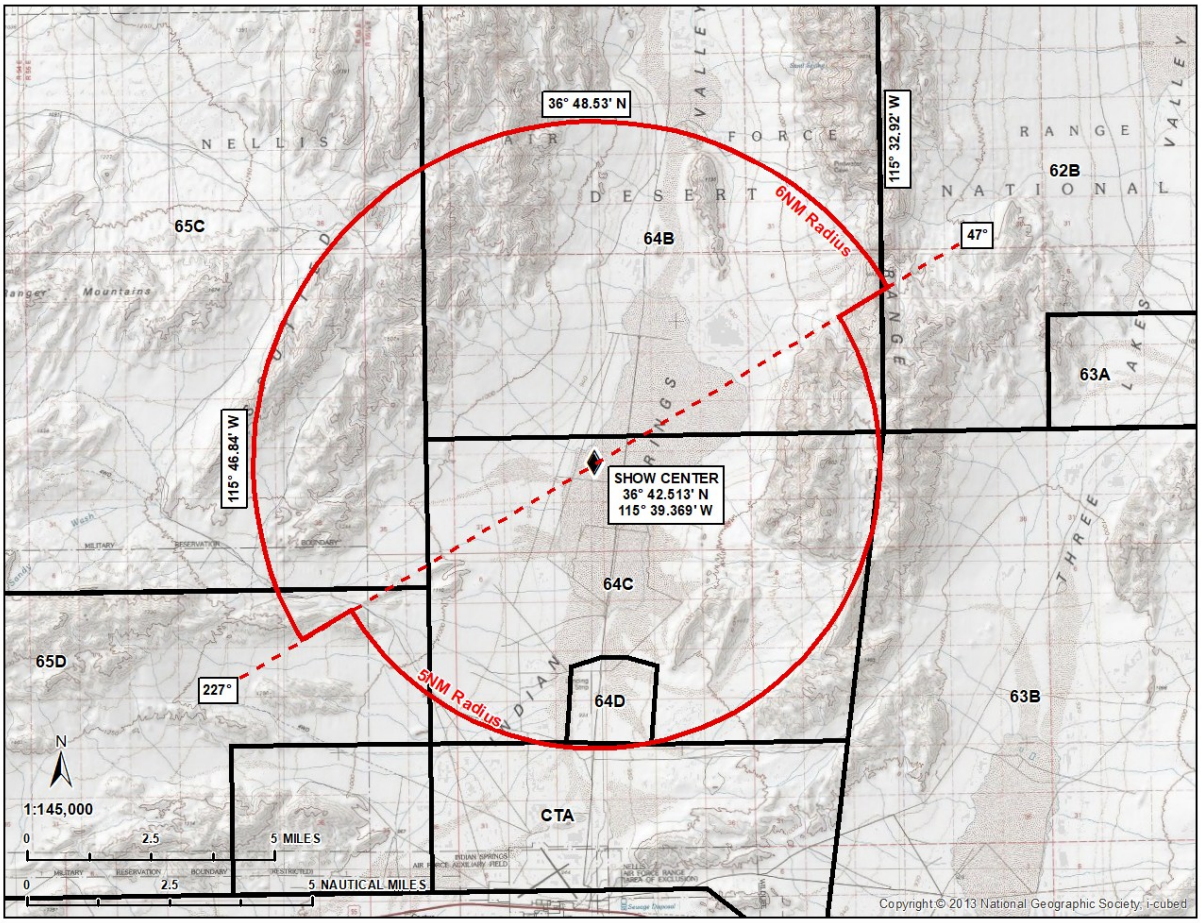
## THUNDERBIRDS TRAINING AREAS

**A8.1. Thunderbirds NTTR Range Training Area.** When active, this training area normally reaches from surface to FL180. When this training area is activated, R-64B/C/E and R-65C will be scheduled IAW the NTTR Scheduling Process.

**Table A8.1. Thunderbirds NTTR Range Training Area Boundaries.**

| Location | Coordinates | |
|---|---|---|
| **Show Center[1]** | N36 42.513 | W115 39.37 |
| **Northernmost Boundary** | N36 48.53 | |
| **Easternmost Boundary** | | W115 32.92 |
| **Southernmost Boundary** | N36 37.51 | |
| **Westernmost Boundary** | | W115 46.84 |
| Note 1: The NTTR Training Area is an approximate 6NM radius from Show Center at a magnetic heading of 227° clockwise to 047° and a 5NM radius from Show Center with a magnetic heading of 047° clockwise to 227°. | | |

**Figure A8.1. Thunderbirds NTTR Range Training Area.**

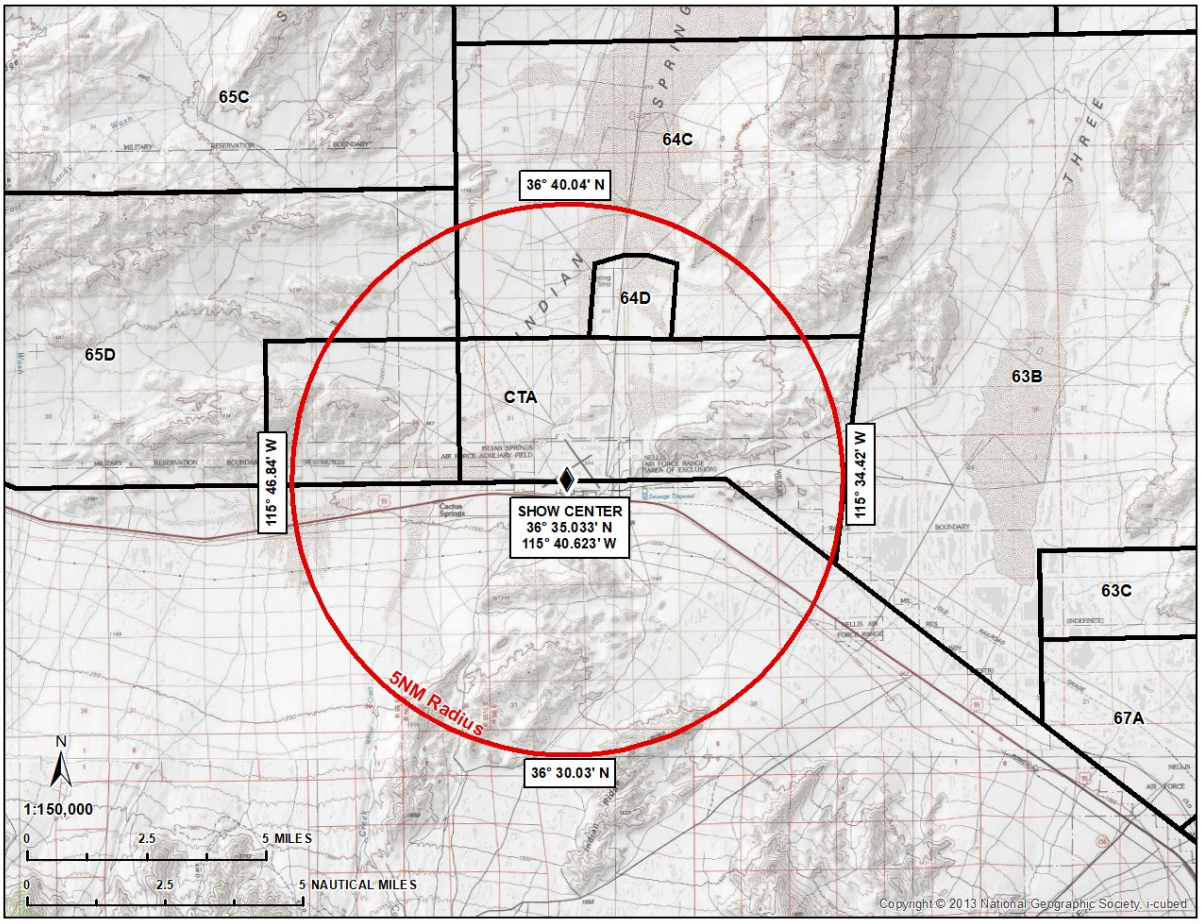



**A8.2. Thunderbirds Creech AFB OTF Training Area.** When active, this training area normally reaches from surface to FL180. To activate this training area, the Thunderbirds will coordinate directly with 432 WG for the release of CTA and R-64D/E/F/G for NTTR scheduling. Thereafter, CTA, R-64D/E/F/G, and R-64C will be scheduled IAW the NTTR Scheduling Process.

**Table A8.2. Thunderbirds Creech AFB OTF Training Area Boundaries.**

| Location | Coordinates | |
|---|---|---|
| **Show Center** | N36 35.033 | W115 40.623 |
| **Northernmost Boundary** | N36 40.04 | |
| **Easternmost Boundary** | | W115 34.42 |
| **Southernmost Boundary** | N36 30.03 | |
| **Westernmost Boundary** | | W115 46.84 |
| Note 1: The Creech AFB OTF Training Area is an approximate 5NM radius from Show Center at all headings. | | |

**Figure A8.2. Thunderbirds Creech AFB OTF Training Area.**



## Attachment 9

## AIRSPACE RECALL

**A9.1. NTTR Emergency Airspace Recall.** In the event of an NTTR emergency, range users will contact NTTR Range Monitoring (assisted by NATCF as required) and state ranges required to address the emergency.

A9.1.1. NTTR Range Monitoring will immediately announce the activation of this emergency recall on appropriate range frequencies and Guard.

A9.1.2. NTTR Range Monitoring will assist in the emergency, as required.

A9.1.3. NTTR Range Monitoring will notify NTTR/DO with details of emergency airspace recall.

**A9.2. DML Airspace Recall.** In the event of conditions impacting DML activities, DML will coordinate with NTTR Range Monitoring for the immediate use of Coyote A and/or R-61A, R-65A, and R-74A, as required. When a recall is required, these areas will be recalled with a maximum of warning time and for the minimum time necessary.

A9.2.1. Upon notification, NTTR Range Monitoring will then pass the recall information to all affected aircraft and to all aircraft that subsequently check-in.

A9.2.2. NTTR Range Monitoring will notify the ROO with details of DML airspace recall. The ROO will ensure the necessary reprioritization coordination is accomplished, particularly with regards to large, complex events (i.e., exercise RED FLAG, etc.).

**A9.3. DML Weather Recall.** In the event of severe weather phenomena (e.g., thunderstorms, etc.) impacting DML activities, DML will coordinate with NTTR Range Monitoring for the immediate use of affected areas. When a recall is required, these areas will be recalled with a maximum of warning time and for the minimum time necessary.

A9.3.1. Upon notification, NTTR Range Monitoring will announce the activation of the recall on the appropriate range frequencies and Guard. NTTR Range Monitoring will then pass the recall information to all aircraft that subsequently check-in.

A9.3.2. DML weather recall places the affected ranges under DML control at 12,000ft MSL and below. Liberal use of these ranges above that altitude is allowed, but aircraft operating on those ranges below 13,000ft MSL require clearance to enter and continuous contact with DML.

A9.3.3. If DML observes traffic entering affected ranges below 13,000ft MSL, they will repeat the recall on Guard.

**A9.4. R-4809A Airspace Recall.** In the event of conditions impacting DML activities in R-4809A, DML will coordinate with NTTR Range Monitoring for the immediate use of R-4809A. When a recall is required, R-4809A will be recalled with a maximum of warning time and for the minimum time necessary.

A9.4.1. Upon notification, NTTR Range Monitoring will announce the activation of the recall on the appropriate range frequencies and Guard. NTTR Range Monitoring will then pass the recall information to all aircraft that subsequently check-in.

A9.4.2. If DML observes traffic entering R-4809A below 15,000ft MSL, they will repeat the airspace recall announcement on Guard.

**A9.5. NNSS Airspace Recall.** In the event of conditions impacting NNSS activities, NNSS will coordinate with the appropriate controlling agency for the immediate use of R-4808 as required. When a recall is required, these areas will be recalled with a maximum of warning time and for the minimum time necessary.

A9.5.1. Upon notification, NTTR Range Monitoring will then pass the recall information to all affected aircraft and to all aircraft that subsequently check-in.

A9.5.2. NTTR Range Monitoring will notify the ROO with details of NNSS airspace recall. The ROO will ensure the necessary reprioritization coordination is accomplished, particularly with regards to large, complex events (i.e., exercise RED FLAG, etc.).

**A9.6. FAA Airspace Weather Recall.** Severe weather phenomena (e.g., thunderstorms, etc.) may occur that necessitate the routing of non-participating (civilian/airliner) aircraft through the NTTR. LA ARTCC shall request release/transit of the airspace from NATCF, as required. Reference current NAFBI 11-250.

**A9.7. NTTR Fire Airspace Recall.** In the event of a fire on the NTTR withdrawn lands/restricted ranges (e.g. R-4806, R-4807, R-4809B, ECE, ECW) NTTR Duty Officer will direct implementation of a Fire Airspace Recall. Recalls affecting R-4808 and R-4809A must be coordinated with and receive approval from DOE and NNSS. These recalls will encompass the airspace necessary to conduct safe firefighting operations. Because of the dynamic nature of such operations this recall may be as small as a 5NM radius Restricted Operating Zone (ROZ) or TFR in a MOA, or could cover the entire NTTR or designated sector (Table A9.6) below 15,000 ft MSL (Figure A9.6). This recall ensures safe integration of firefighting aviation assets, minimizes risks to range users, and complies with NTTR Wildland Fire Management Plan protocols. The recall will be activated with maximum warning time and for the minimum duration necessary to resolve the fire hazard. Reference Attachment 19 for fire, chaff, and flare restrictions, and Section 3.1.16 for seasonal fire hazards.

A9.7.1. NTTR Duty Officer will work with BLM and Blackjack to establish what airspace and altitudes firefighting aircraft need to operate safely on the NTTR. The Duty Officers will attempt to scale down airspace recalls by recalling individual ranges when able while still ensuring effective firefighting operations.

A9.7.2. Firefighting Operations in MOAs will still require a TFR for separation from civilian traffic and sUAS. NTTR customers will adhere to the more restrictive of the Airspace Recall or TFR.

**Figure A9.1. NTTR Fire Airspace Recall (FR) Map.**

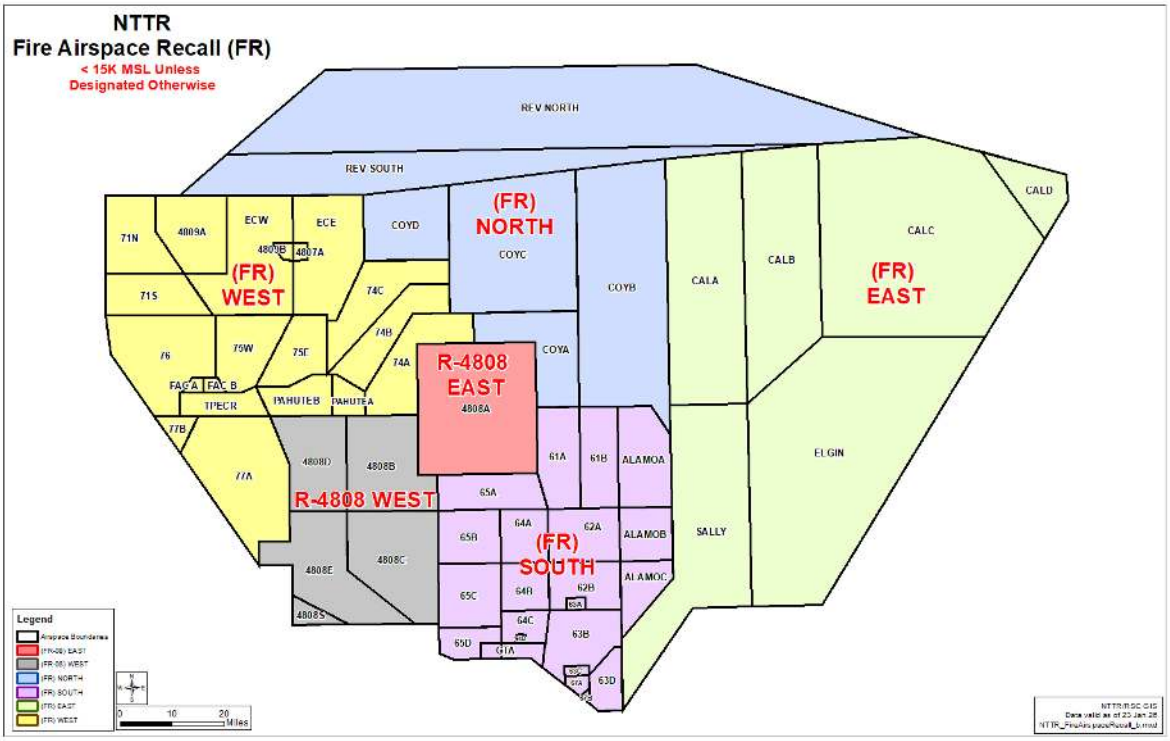



**Table A9.1. NTTR Fire Airspace Recall Sectors and Subranges.**

| Sector | Ranges |
|---|---|
| **FR NORTH** | REV NORTH, REV SOUTH, COYA, COYB, COYC, COYD |
| **FR SOUTH** | 61A, 61B, 62A, 62B, 63A, 63B, 63C, R63D High, 64A, 64B, 64C, 64D, 65A, 65B, 65C, 65D, 67A, 67B, ALAMOA, ALAMOB, ALAMOC |
| **FR WEST** | 4807A, 71N, 71S, 74A, 74B, 74C, 75E, 75W, 76, 77A, 77B, TPECR, FACA, FACB, ECE, ECW, R4809A, R4809B, PAHUTEA, PAHUṬEB |
| **FR EAST** | CALA, CALB, CALC, CALD, ELGIN, SALLY |
| **R-4808 East** | 4808A |
| **R-4808 West** | 4808B, 4808C, 4808D, 4808E, 4808S |

A9.7.3. Detection and Reporting. All range users, including aircrew and ground parties, must report any observed fires to NTTR Range Monitoring immediately via appropriate range frequencies or Guard, providing details such as location (range number, GPS coordinates, or nearest landmark), size, spread direction, assets at risk, and suspected cause (if known). NTTR Duty Officer will evaluate the report of the fire, determine airspace recall necessity, and direct Nellis Command Post notification (702-652-2446) if required for escalation per Installation Emergency Management Plan 10-2, Tab F (Wildland Fire Checklist). The NTTR Duty Officer will notify leadership and key stakeholders (e.g., NTTR/DO, Nevada National Security Site, and regional stakeholders). During high-fire periods (1 May–31 October), daily fire danger codes (Low–Extreme) issued by NTTR

Range Monitoring in coordination with the Bureau of Land Management (BLM) will be displayed on Graphical Range Restrictions (GRR) and NTTR Mission Planning sites; Very High/Extreme codes automatically prohibit flares/chaff below 5,000 ft AGL NTTR-wide and may trigger preemptive recall evaluation.

A9.7.4. Activation. Upon confirmation of a stationery or moving fire, NTTR Range Monitoring will contact the Duty Officer. The Duty Officer will coordinate fire response and determine the necessity of an airspace recall. Once determined, NTTR Range Monitoring will immediately contact all formations in the NTTR on their tactical frequency to announce the recall, specifying the affected areas, altitudes, and estimated duration. If positive acknowledgement is not received expect Range Monitoring to make an announcement on Guard. NTTR Range Monitoring will pass the recall information to all affected aircraft and to all aircraft that subsequently check-in. Assistance from NATCF may be requested for coordination if the fire impacts adjacent airspace.

A9.7.5. Operations During Recall. Operations During Recall Operations below 15,000' MSL in the recalled airspace are generally restricted to BLM firefighting aircraft. Liberal use of airspace above 15,000 ft MSL is permitted, but aircraft intending to operate below 16,000 ft MSL require explicit clearance from NTTR Range Monitoring and must maintain continuous contact. If unauthorized traffic is observed entering the recalled airspace, NTTR Range Monitoring will repeat the recall announcement on Guard. Range operations unrelated to fire response (e.g., ordnance delivery, low-level training) must cease immediately in the affected airspace until cleared.

A9.7.5.1 Firefighting aircraft may transit airspace that has not been recalled. Range Monitoring will use normal procedures to contact users and alert them to the presence of BLM aircraft with a range reference or geo ref, altitude, track direction, and destination.

A9.7.5.2 NTTR customers wishing to operate below 15,000' MSL should contact Range Monitoring. If able Range Monitoring will deconflict real time between low altitude users and BLM aircraft. All aircraft operating below 15,000' MSL in recalled airspace must maintain two-way communications with Range Monitoring.

A9.7.5.3. When necessary, NTTR Duty Officer may direct a multi-agency response per Interagency Standards for Fire and Fire Aviation Operations (NWCG guidelines), including temporary flight restrictions (TFRs) if escalated by BLM or FAA.

A9.7.6. Notifications. NTTR Range Monitoring will notify the NTTR Duty Officer with details of the recall, including fire location, scope, and impacts on scheduled events or Large Force Exercises (e.g., RED FLAG). The NTTR Duty Officer will coordinate reprioritization as needed. If the fire involves hazardous materials or unexploded ordnance, reference Section 5.4.2 for EOD and safety briefings, and Paragraph 4.17.4.5 for ongoing hazard monitoring.

A9.7.7. Termination. The recall will terminate once the fire is contained or no longer poses an airspace hazard, as determined by NTTR Range Monitoring in coordination with on-scene fire response leads (e.g., Air Force Fire Emergency Services, BLM). NTTR Range Monitoring will announce termination on all appropriate range frequencies and Guard, and notify all relevant parties (NTTR Duty Officer, NATCF). Post-recall, NTTR Range Monitoring will document the event per Paragraph 4.17.4.6, ensuring no distractions during critical tasks, and conduct a review for lessons learned.

## Attachment 10

## AIRCRAFT ENTRY AND EXIT PROCEDURES

**A10.1. Aircraft Entry.** To request range entry, aircrew will contact both NATCF and NTTR Range Monitoring (in order). To request range exit, aircrew will contact both NTTR Range Monitoring and NATCF (in order). Refer to the NTTR Mission Planning websites for assigned frequency clearances. Additionally, users will adhere to NAFBI 11-250, CAFBI 11-250, and local operating procedures, as applicable.

**A10.2. Helicopters.** Helicopters should contact NTTR Range Monitoring prior to losing line of sight communications for clearance onto the range.

**A10.3. Contingencies.** Due to limited line-of-sight communications capability on many NTTR ranges, it may not be possible to contact NTTR Range Monitoring or NATCF prior to exiting the range. Users may exit the range VFR and contact NTTR Range Monitoring and NATCF as soon as communications can be established.

**A10.4. Aircraft Range Entry Handoff.** NATCF will transfer communications to NTTR Range Monitoring by voicing, "Cleared scheduled, contact BLACKJACK, Range Altimeter XX XX."

A10.4.1. Mission flight leads will then check-in with NTTR Range Monitoring prior to range entry. Every flight lead, regardless of scheduled mission type, must check-in with NTTR Range Monitoring. If the flight is delivering live, inert, or training ordnance aircrew must check in with Blackjack via phone call prior to departure. For NTTR concurrent-use operations with multiple flights, one flight lead may check-in for all flights. Failure to check-in with NTTR Range Monitoring prior to range entry may result in lack of awareness to the presence of ground parties on range. Check-in with the information listed in table A10.1.

**Table A10.1. Range Check-in Data.**

- Call Sign.
- Number of aircraft in flight.
- Scheduled range(s) and times.
- GRR version.
- Working frequency.
- Concurrent-Use Operations (if applicable).
- Number/type of ordnance to be dropped/fired (live, inert, gun rounds).
- Target(s) to include dynamic retargeting.
- Laser operations.
- Intent to conduct lights-out training.
- Any mission changes/restrictions (i.e., chaff, fire condition, and flares).

**A10.5. LFEs.** For LFEs, the exercise POC will coordinate information with NTTR Range Monitoring, after which LFE participant units/flights are exempt from the requirement to check-in with NTTR Range Monitoring.

**A10.6. Blackjack Comms.** NTTR Range Monitoring will advise aircrew of changes to range restrictions/conditions and clear aircraft to their area working frequency by voicing "Cleared Tactical."

**A10.7. Exit.** Prior to range exit, mission flight leads will check-out with NTTR Range Monitoring. Every flight lead, regardless of scheduled mission type (including LFE events), must check-out with NTTR Range Monitoring. For NTTR concurrent-use operations with multiple flights, one flight lead may check-out for all flights. Check-out with the data listed in table A10.2.

**Table A10.2. Range Check-out Information.**

| |
|---|
| • Call sign.<br>• Number of aircraft in flight.<br>• Total number and type of ordnance expended on each target (i.e., "Strike 01, 12 x Mk-82 on 76-10"). |

**A10.8. Clearance off Range.** Prior to range exit, mission flight leads will contact NATCF to coordinate for RTB/follow-on clearance.

## Attachment 11

## GROUND PARTY PROCEDURES

**A11.1. General.** Ground access to the NTTR holds inherent risks. As such, access is carefully controlled and it is incumbent on all ground parties to understand and adhere to NTTR ground party procedures. HQ NTTR limits ground access to certain areas due to UXO and explosive residue throughout the NTTR which can present a hazard to personnel as well as in support of environmental stewardship efforts.

A11.1.1. NTTR Range Monitoring authorizes entry to the NTTR subranges that include strikeable targets, monitors ground parties on those subranges, and monitors exit from those subranges. To ensure ground parties depart the range at their scheduled time prior to an ordnance-release (live or inert) mission, NTTR Range Monitoring will contact ground parties 15 minutes prior to their scheduled exit time.

A11.1.2. Remaining overnight (RON) on the NTTR for convenience is not authorized. Activity that spans the night period may be authorized if properly coordinated through HQ NTTR.

A11.1.3. Sensitive Equipment. Sensitive equipment consists of any personal device capable of enhancing vision (i.e., NVDs, binoculars, and targeting scopes) recording video or electronically storing data (i.e., cameras, cell phones with camera capability, memory cards, computers, PDAs) and weapons and ammunition. Sensitive equipment is not permitted on range without specific approval from NTTR/SD no later than 45 days prior.

A11.1.4. In no case will there be more than six persons, including range crewmembers, allowed in any range tower during active range operations.

**A11.2. Associated vs Non-Associated.** Ground parties are identified as either “associated ground parties” or “non-associated ground parties:

A11.2.1. Associated ground parties consist of personnel and vehicles directly involved in tactical mission execution, such as field artillery fire teams, special operations teams, or JTACs. Video recording is routinely employed by Thunderbirds personnel.

A11.2.2. Non-associated ground parties consist of personnel and vehicles not directly involved in mission execution, such as HQ NTTR staff, range maintenance contractors, EOD technicians, Public Affairs “Combat Camera,” and range tour groups.

**A11.3. Requests.** All ground party access requests must be coordinated through NTTR/SD using the NTTR Visit Request/Authorization. An authorized escort must accompany all non- cleared visitors on the NTTR ranges. The escort will brief visitors on range procedures and restrictions. Escorts will coordinate range entry and departure routes, times and procedures with NTTR/SD to determine security requirements.

**A11.4. Vehicle Ops.** Prior to operation of vehicles on the NTTR, ground parties must complete the requirements listed in table A11.1.

**Table A11.1. NTTR Vehicle Operator Requirements.**

- Obtain appropriate safety training from NTTR/SE.
- Possess valid government vehicle certification training and licensing.
- Ensure every vehicle is in safe operating condition prior to entering the NTTR.
- Possess range access badges.
- Have an NTTR approved callsign from the NTTR/RS-managed NTTR Master Callsign List.
- Review the airspace schedule and restrictions on mission execution day prior to entering the range.

**A11.5. Equipment.** Ensure each vehicle is equipped with equipment as listed in table A11.2.

**Table A11.2. Vehicle Equipment Requirements.**

- LMR (available for check-out at HQ NTTR building) for contacting the controlling agency.
- LEP (available for check-out at HQ NTTR building) for each occupant, if required by the range schedule.
- NTTR driving maps with mandatory reporting points and procedures annotated.
- Auto-kit with basic repair equipment (e.g., serviceable spare tire, jack, etc.).
- A basic first aid kit.
- Vehicle Tracking System (VTS) (if available). Strikeable ranges are first priority for VTS use.
- At least three liters of water per person.
- Vehicles assigned to HQ NTTR and operated by HQ NTTR personnel will also be equipped with a minimum of: one pair of leather gloves, one flashlight with strobe, four emergency blankets, one rain poncho, one tow rope, one glow stick, one warning triangle, and a set of jumper cables.

**A11.6. Convoys.** Ground party vehicles may deploy individually or in convoys (two or more vehicles that depart at the same time from the same origin to a destination and are organized for the purpose of control under a single convoy lead). When in convoy, the convoy lead must operate in accordance with table A11.3.

**Table A11.3. Convoy Requirements.**

- Determine and notify all convoy personnel of marshaling/assembly area and departure time.
- Inspect vehicles and ensure all personnel have required equipment.
- Brief convoy personnel on range schedule and restrictions.
- Follow the proper route from the departure point to the destination point.
- Observe all range restrictions, safety policies, and regulations at all times.
- Ensure that all convoy drivers maintain a safe distance from each other.
- Request/receive range access clearance for the entire convoy and maintain constant radio contact with the controlling agency.
- Ensure all occupants have and use adequate LEPs for scheduled laser activities.

**A11.7. General Restrictions.** NTTR roads are a mix of asphalt, improved dirt/gravel, and unimproved dirt. Off-road operations are strictly prohibited without specific approval. Headlights must be on at all times while on range. All vehicles must strictly adhere to posted speed limits. Unless otherwise posted, maximum speed limits on NTTR roads are as follows:

**Table A11.4. Maximum Speed Limits on NTTR**

- Asphalt and Improved Roads = 45 MPH
- Unimproved Roads = 35 MPH
- Other = 25 MPH

**A11.8. Travel times and routes.** The following estimated driving times assume a single vehicle driving at the speed limit, daytime, clear weather, road familiarity and good road conditions. Distances are provided in statute miles.

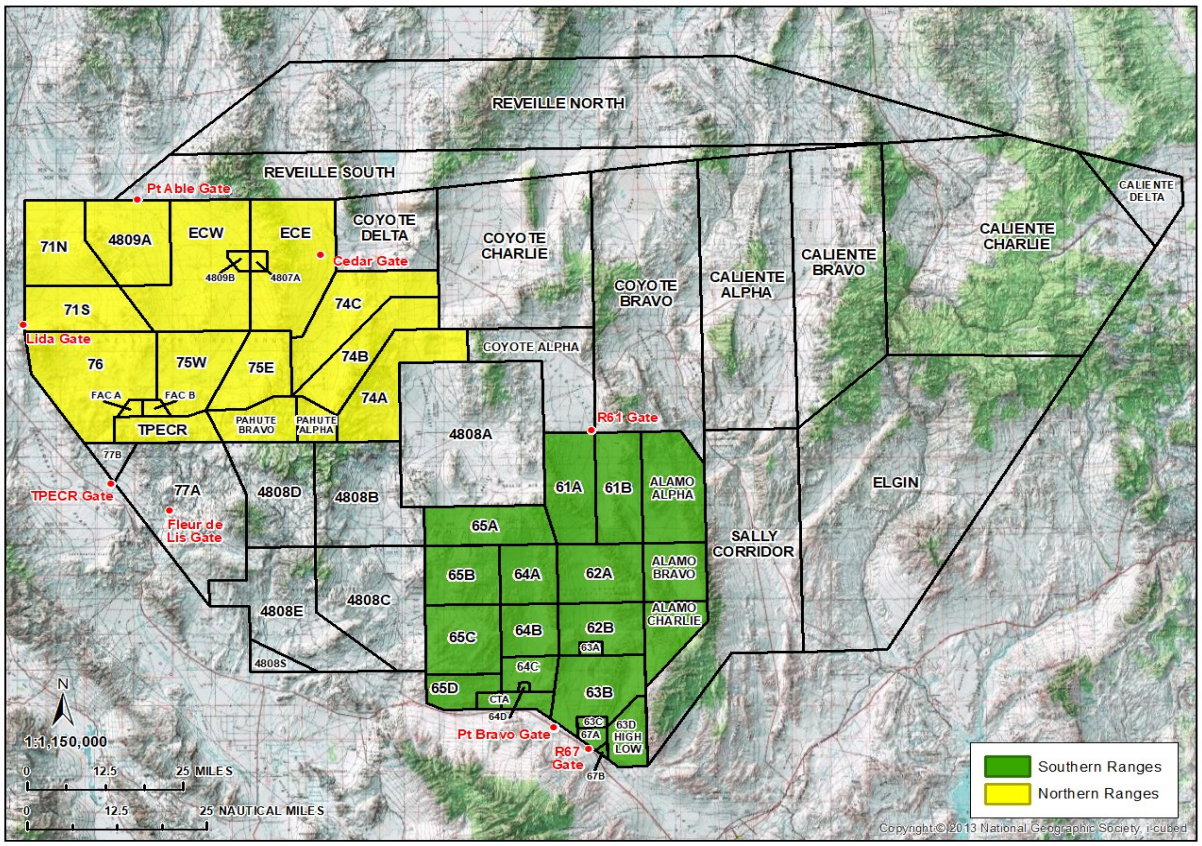
**Table A11.5. Nellis AFB to Range Entry Point Driving Times.**

| Nellis AFB to Range Entry Points | Distance/Speed | Travel Time |
|---|---|---|
| Via I-215 to US-95 to Point Bravo | 40 miles/55-70 mph | 57 min |
| Via I-215 to US-95 to Creech AFB | 52 miles/55-70 mph | 1 hr 15 min |
| Via I-215 to US-95 to TPECR Gate | 145 miles/55-70 mph | 2 hrs 59 min |
| Via I-215 to US-95 to Lida Gate | 180 miles/55-70 mph | 3 hrs 29 min |
| Via I-215 to US-95 to Tonopah to Able Gate | 246 miles/55-70 mph | 4 hrs 52 min |
| Via I-15 to US-93 to SR-375 to Cedar Gate | 175 miles/55-70 mph | 3 hrs 17 min |
| Via I-15 to US-93 to SR-375 to SR-6 to Able Gate | 262 miles/55-70 mph | 4 hrs 38 min |

**Table A11.6. Range Crossing Times.**

| Range Crossing Time / Roads Used | Distance/Speed | Travel Time |
|---|---|---|
| R-62B/Three Lakes Road | 10 miles/25-45 mph | 20 min |
| R-63B/Three Lakes Road | 13 miles/25-45 mph | 25 min |
| R-64B/Mile Range Road | 10 miles/25-40 mph | 20 min |
| R-64C/Spotted Range Road (to R-65C) | 7-8 miles/25-45 mph | 15 min |
| R-64C/Mile Range Road (to R-64B) | 7-8 miles/25-45 mph | 15 min |
| R-71/Sleeping Column Road/East Stonewall Flat Road/Civet Cat Canyon Road | 18 miles/15-40 mph | 30 min |
| R-71/Sleeping Column Road/E. Stonewall Flat Road/Stonewall Flat Road | 26 miles/25-45 mph | 45 min |
| R-75W/Cross Range Road | 18 miles/25-45 mph | 35 min |
| R-75W/Wilson Camp Road/Cross Range Road | 15 miles/15-40 mph | 30 min |
| R-76/Civet Cat Canyon Road | 17 miles/15-40 mph | 45 min |

**Figure A11.1. NTTR Gate Locations.**



**A11.9. Entry/Exit procedures.**

A11.9.1. Communications. Ground parties will use the range safety net (Fox 4) to contact NTTR Range Monitoring for range access. If NTTR Range Monitoring is not available, contact contractor security operations on range safety net (Fox 4) at (702) 404-0556 for access. Creech North and South gates no longer use/allow access from the North or South gates of Creech AFB to the southern ranges. Access is through the by-pass road by contacting the southern range BDOC at (702) 653-4693 for access.

A11.9.2. Ground parties entering and exiting the NTTR must utilize authorized entry points at the following gate locations:

**Table A11.7. Authorized Range Entry Points.**

| Gate | Type | Access To | Contact |
|---|---|---|---|
| Range 67 Gate | Unmanned | Firing Ranges 1-13 in R-67A | NTTR Range Monitoring |
| Range 61 Gate | Unmanned | Range 61A/B | NTTR Range Monitoring |
| Creech Bypass Gate | Unmanned | Southern subranges | NTTR Range Monitoring[1] |
| Point Bravo Gate | Manned | Southern subranges | Scorpion |
| Able Gate | Manned | Northern subranges | Cactus |
| TPECR Gate | Manned | Northern subranges | Cactus |
| Cedar Gate | Manned | Northern subranges | Cactus |
| Lida Gate | Unmanned | Northern subranges | Cactus |
| By-pass Road to Mile Range Road | Unmanned | Transit between by-pass road and Southern subranges | Southern Range BDOC |
| By-pass road at Box Canyon Road | Unmanned | Transit between by-pass road and Southern subranges | Southern Range BDOC |
| Fleur De Lis Gate | Unmanned | Range 77A | NTTR Range Monitoring |

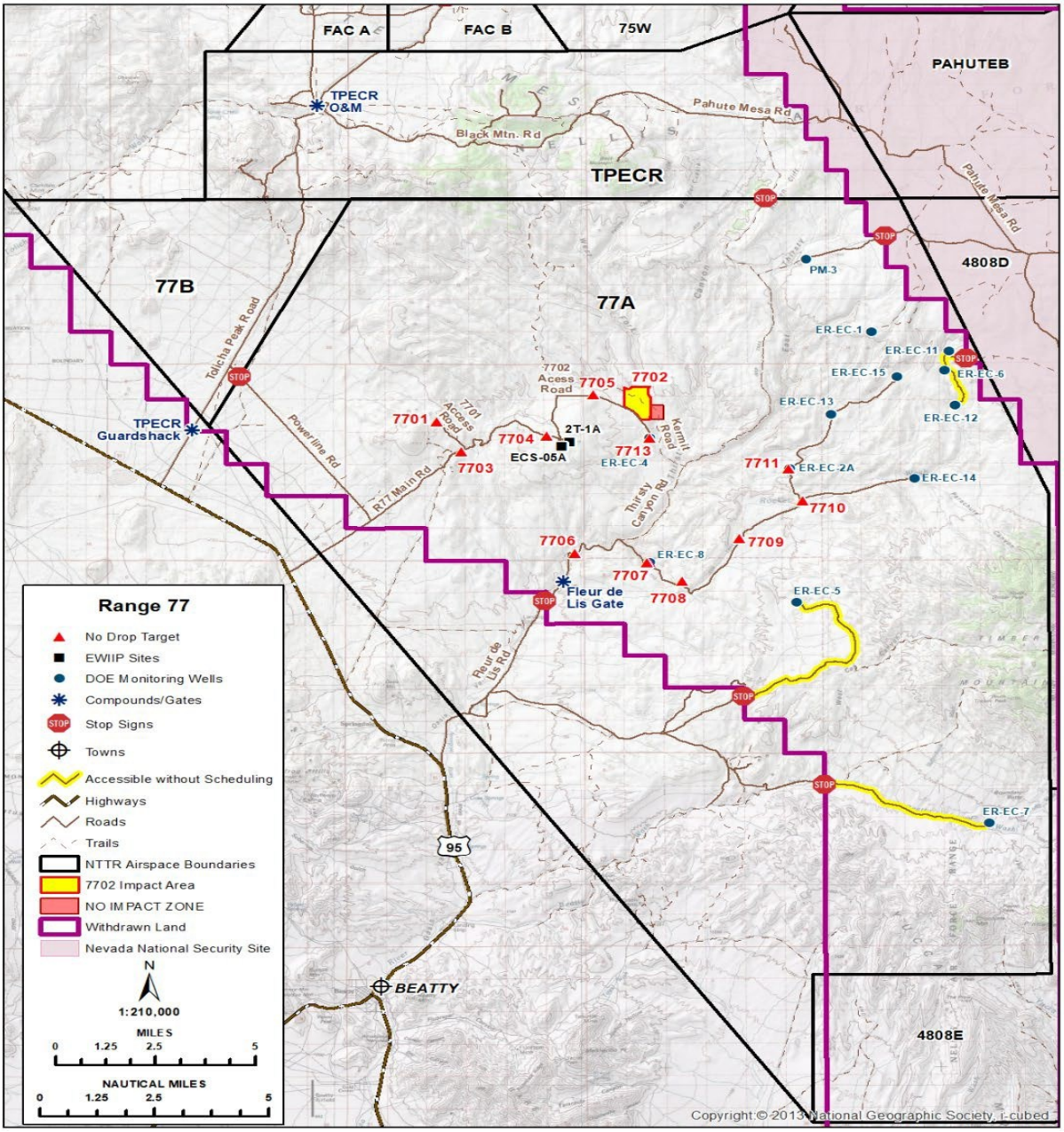
A11.9.3. Having entered the range, further entry into any other subrange requires a radio check-in/check-out on Fox 4 with NTTR Range Monitoring.

A11.9.4. Safe-to-Pass Roads. There are several sections of range roads that are authorized for travel without specific scheduling of the subrange. Contact with NTTR Range Monitoring IS required for entry and exit authorization for these “safe-to-Pass Roads” unless otherwise posted. Contact with NTTR Range Monitoring is required if departing the “safe-to-pass” road into a bombing range regardless if a sign is posted stating contact with NTTR Range Monitoring is not required. Safety of these roads is maintained by ensuring no weapons safety footprints overlay them:

**Table A11.8. Safe-to-Pass Roads.**

| Road | Subrange | Description |
|---|---|---|
| Blockhouse Road, Stagecoach Road | R-63B | Transit from Point Bravo along Blockhouse Road up to Stagecoach Road, and then along Stagecoach Road to R67. Blockhouse Road beyond Stagecoach Road IS NOT SAFE TO PASS. |
| All roads | R-64D | Transiting R-64D |
| Mile Range and Tank Table Road | CTA | CTA |
| Box Canyon Road | CTA | Transit between Creech AFB and R-63B |
| Sleeping Column Road | R-71N | Transit between R-4809A and View Point Alpha |
| All roads north of N37 45.00 | R-71N | Transit to/from Mud Lake starting N of the N37 45 line. |
| All roads east of the W116 00.0 | R-74A/B/C | Traversing R-64D east of the W116 00.00 |
| Frans Hamel Road | R-75W | Transit between R-76 and EC West |
| Highlighted access routes on R-77A | R-77A | Access to ER-EC-5, 6, 11 and 12 |
| Tolicha Peak Access Road | R-77B | Transition between HWY-95 and TPECR |
| Spotted Range Road | R-64D, R-65C | Transition between R-64D and R-65C. |

**Figure A11.2. Range 77A Road Accessibility.**




A11.9.4.1. Except as authorized by Table 3.1, "No ground party" restrictions will not be assigned to R-77B without prior coordination and approval from the NTTR/DO.

**A11.10. Vehicle Breakdown**. Advise NTTR Range Monitoring immediately of the location of a vehicle breakdown and intentions. If the vehicle breakdown occurs within the R-63B DU licensed area, do not exit the vehicle until receiving instructions from NTTR Range Monitoring. If the vehicle can be repaired quickly (i.e., a flat tire), attempt to repair it. If it cannot be repaired quickly, or there is doubt, notify NTTR Range Monitoring.

**A11.11. Emergencies.** In the case of a vehicle accident or medical emergency, render first aid to the victims, notify the NTTR Range Monitoring immediately of the location, and provide status of all personnel and nature of emergency response required if necessary. After returning from the range, report to NTTR/SE to accomplish the SF 91, Motor Vehicle Accident Report or AF Form 978, Supervisor's Mishap Report as applicable.

**A11.12. Ground Weapons Employment**. Ground parties are authorized to fire surface-to-surface ordnance at targets as listed on the NTTR Mission Planning websites. The following restrictions apply:

A11.12.1. All Firing Points (FPs) must be approved by NTTR/SE prior to weapons employment.

A11.12.2. No surface-to-surface weapons employment is authorized against any target unless specifically approved by NTTR/SE with an approved SDZ plot. Approved surface-to-surface targets and weapons can be found on the Nellis Mission Planning websites.

A11.12.3. Prior to weapon employment ground parties must contact NTTR Range Monitoring (Blackjack) with target and weapon pairing.

A11.12.4. Ground weapons fire will not be aimed towards fixed, high-value sites (TOSS, UMTE locations).

A11.12.5. Gun-to-Target Lines (GTLs) will not cross over any personnel.

A11.12.6. Surface-to-surface weapons will not be fired when the target is obscured unless tracking radar is used. All impacts must be visually observed or tracked with radar.

A11.12.7. Brass/Dunnage Cleanup. All dunnage and brass debris left on range is the responsibility of the ground party that generated the debris. In most cases, additional range time is required to accomplish brass/dunnage cleanup. Prior coordination can be made to preposition 55-gallon barrels at firing locations to collect brass and dunnage, or the unit may opt to pay for the range maintenance contractor to clean up debris. A standard charge will be applied to the unit for brass/dunnage cleanup IAW the NTTR BMP.

## Attachment 12

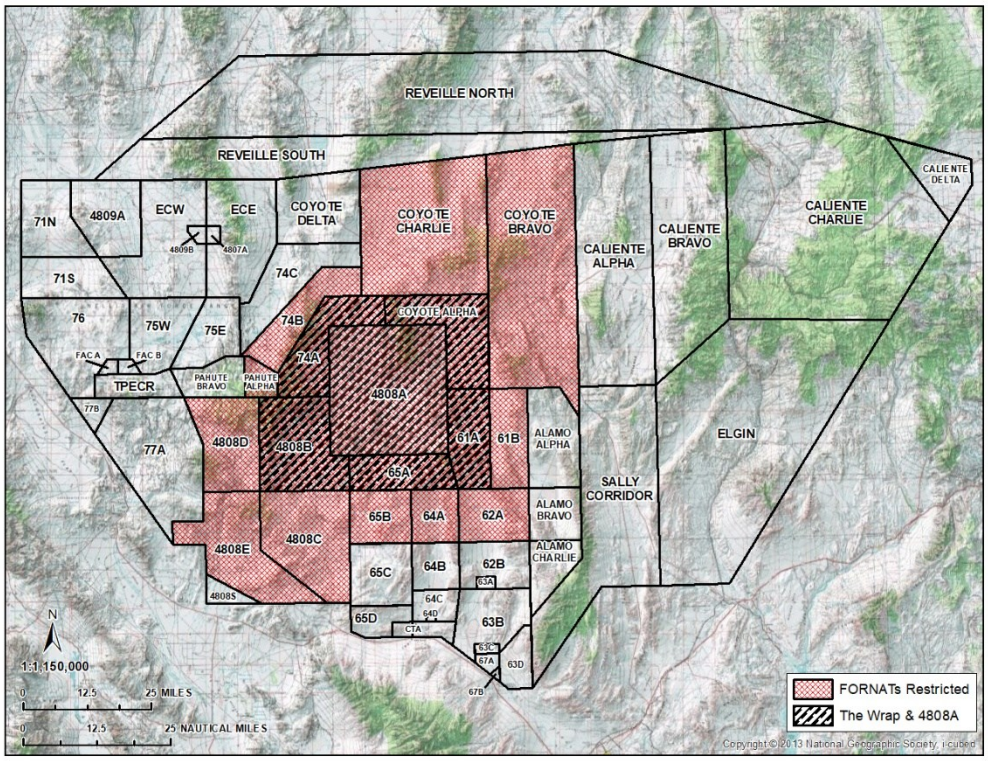
## FOREIGN USERS

**A12.1. Foreign Air and Ground Users**. Foreign air and ground users of the NTTR are normally categorized in one of two groups.

A12.1.1. Foreign Exchange Officers (FEOs). Regardless of Military Personnel Exchange Program Status, foreign aircrew fall into this category when operating U.S. owned aircraft where the U.S. controls the post flight data, and when their names have been submitted to NNSS Senior Director for Security and Emergency Services (NNSS SES) at least 7 days prior. For the purposes of flying in R-4808 A/B/C/D/E, Range 74A, Range 61A, Range 65A, and Coyote A (The Wrap and R-4808 A), FEOs will be treated as U.S. aircrew. For R-4808 restrictions, reference Table A4.38. NOTE: Personnel receiving Familiarization or Incentive flights, and personnel who are passengers on U.S. aircraft do not meet the requirement to be an FEO, and FORNAT restrictions will apply.

A12.1.2. Foreign National (FORNAT) users in foreign aircraft. These personnel do not meet the requirements of para A12.1.1, and are typically restricted from flying in the wrap and R-4808 A. When required and identified on the GRR, FORNATs in foreign aircraft are restricted from the contiguous airspace as provided in figure A12.1.

**Figure A12.1. FORNAT Contiguous Airspace Restriction.**



**A12.2. Foreign Air and Ground Participation.** Foreign air and ground participation in activities conducted on the NTTR are scheduled through the NTTR Scheduling Process and reflected on the GRR. For daily information, CSE can be used to generate a Foreign Participant Report that reflects all known and scheduled foreign participation in NTTR activities.

**A12.3. Foreign Visit Request (FVR) Process**. The FVR process is initiated by the partner nation's embassy in Washington, DC. The host embassy inputs applicable data on each individual into the Foreign Visit System (FVS). FVS then notifies ACC/IAG. In turn, ACC/IAG sends the FVR to all affected agencies (including HQ USAFWC) for action.

A12.3.1. FORNATs. USAFWC Foreign Disclosure office (FDO) notifies NTTR/SD of approved FORNAT FVRs. A host unit for the FORNAT must be identified and will fulfill all host unit responsibilities.

A12.3.2. Military Personnel Exchange Program Officers (MPEPOs). USAFWC/FDO notifies NTTR/SD of approved MPEPO FVRs and US unit of assignment as well as MPEPO assignments that have ended. NTTR/SD will provide NTTR/DO a consolidated listing of all current MPEPOs when MPEPOs are added or removed from approved status. NTTR/DO then sends names of MPEPOs to NNSS SES, allowing them to qualify for FEO status.

A12.3.3. For units TDY to Nellis or off-station units using NTTR airspace, FORNATs and MPEPOs should be coordinated through the host Unit or NTTR PROJO prior to departing home station. If due to unforeseen circumstances this coordination does not occur or if coordination status is unknown, unit Commanders should immediately call the NTTR/DOXS office at DSN: 348-4935 for coordination and confirmation of airspace restrictions.

**A12.4. Non-Local Foreign Participants**. Foreign aircraft and/or aircrews launching from/recovering to Nellis AFB or Creech AFB, but not operating on the NTTR, are not scheduled through the NTTR Scheduling Process nor reflected on the GRR.

**A12.5. Host Unit Scheduling Responsibilities for Foreign Aircraft, Aircrews, and Ground Participants.**

A12.5.1. Activities conducted on the NTTR. Nellis AFB or Creech AFB units hosting foreign aircraft, aircrews, and/or ground participants operating on the NTTR will accomplish the following:

A12.5.1.1. Ensure hosted foreign participant requirements are provided to 57 OSS/OSOS no later than 60 days prior to the month of execution and submitted to NTTR Range Scheduling no later than 45 days prior to the month of execution.

A12.5.1.2. Annotate "FOREIGN PARTICIPANTS" and the country of origin on weekly pencils scheduling requests and submit IAW the NTTR Scheduling Process.

A12.5.1.3. Ensure "FOREIGN PARTS" have been entered into the CSE scheduling database and accurately reflected on the GRR.

A12.5.1.4. If greater than 48 hours prior to the day of scheduled execution, coordinate all foreign participant scheduling changes through NTTR Range Scheduling.

A12.5.1.5. As part of the schedule confirmation process (no earlier than three days prior to the day of scheduled execution and no later than 1200L two days prior to the day of scheduled execution), verify foreign participation with NTTR Range Scheduling.

A12.5.1.6. If within 48 hours of the day of scheduled execution, coordinate all foreign participant scheduling changes with NTTR Range Monitoring. If foreign participation is cancelled, notify NTTR Range Monitoring for removal of "FOREIGN PARTS" from the GRR.

A12.5.2. Activities conducted on other than the NTTR. Nellis AFB or Creech AFB units hosting foreign aircraft and/or aircrews launching from/recovering to Nellis AFB or Creech AFB, but not operating on the NTTR, will accomplish the following:

A12.5.2.1. Ensure hosted foreign participation requirements are provided to 57 OSS/OSOS no later than 60 days prior to the month of execution. 57 OSS/OSOS will inform NTTR Range Scheduling of the foreign participation no later than 45 days prior to the month of execution.

A12.5.2.2. Ensure foreign participation information is entered into the PEX system, as required.

A12.5.2.3. If within 48-hours of the day of scheduled execution, coordinate all foreign participation scheduling changes with the 57 WG SOF.

**Figure A12.1. Example Consolidated MPEPO Memorandum.**

DEPARTMENT OF THE AIR FORCE
NEVADA TEST AND TRAINING RANGE (ACC)
NELLIS AIR FORCE BASE, NEVADA

*DD MMM YY*

MEMORANDUM FOR NTTR/DO

FROM: NTTR/IP

SUBJECT: Consolidated Foreign Exchange Officer (FEO) Memorandum

1. The following FEOs are authorized to participate in events taking place on the NTTR:

2. Unit: *XX TES*
   a. Name: *Last, First M.I.*
      1) Rank:
      2) Service #:
      3) Nationality:
      4) MDS:
      5) Aircrew Position: *Pilot, Navigation, Weapon System Operator, Sensor Operator, etc.*
      6) Clearance:

   b. Name: *Last, First M.I.*
      1) Rank:
      2) Service #:
      3) Nationality:
      4) MDS:
      5) Aircrew Position: *Pilot, Navigation, Weapon System Operator, Sensor Operator, etc.*
      6) Clearance:

3. Unit: *XX AGRS*
   a. Name: *Last, First M.I.*
      1) Rank:
      2) Service #:
      3) Nationality:
      4) MDS:
      5) Aircrew Position: *Pilot, Navigation, Weapon System Operator, Sensor Operator, etc.*
      6) Clearance:

4. FEOs are prohibited from R-4808 without express NNSA/NFO and/or DOE approval (coordinated through the NTTR Scheduling Process). This memorandum supersedes all previous versions, same subject.

*SIGNATURE*
*NAME*
Information Protection Office

*Testing - Tactics - Training- Innovation - Integration*

## Attachment 13

## LOW OBSERVABLE (LO) DECONFLICTION.

**A13.1. Deconfliction.** Throughout the NTTR BMP and Scheduling Process, host units must be prepared to coordinate shifts to desired range times to ensure deconfliction with LO activities of higher priority. If the LO unit scheduler determines that deconfliction is required between LO aircraft and foreign participation, the necessary deconfliction will be incorporated into the daily range schedule.

**A13.2. LO Unit Scheduler Responsibilities.**

A13.2.1. Comply with foreign participant deconfliction requirements IAW applicable security classification guides and signature protection concept of operations (CONOP).

A13.2.2. Through the NTTR Scheduling process, schedule appropriate time buffers between foreign participants and LO operations.

A13.2.3. Foreign aircraft and/or aircrews launching from/recovering to Nellis AFB or Creech AFB, but not operating on the NTTR, are not scheduled in the CSE nor reflected on the GRR. LO unit schedulers should take appropriate measures to ensure deconfliction with foreign aircraft and/or aircrews IAW applicable security classification guides and signature protection CONOP.

**A13.3. Electronic Intelligence (ELINT) Deconfliction.** The coordination and deconfliction of concurrent operations by LO aircraft and radar equipped or ELINT gathering aircraft of foreign origin and/or foreign aircrews operating similar US aircraft require careful coordination between the host unit and the LO unit schedulers.

**A13.4. NTTR LO-Foreign Scheduling Restrictions.**

A13.4.1. NTTR-LO Foreign Aircrew Test Missions. Only foreign aircrew members who hold the appropriate security clearances will be scheduled to operate upon the NTTR as participants in LO test missions. The designated Responsible Test Organization is responsible for ensuring proper clearances.

A13.4.2. Foreign Aircrew Scheduling Exceptions. Foreign participants will not be scheduled to operate upon the NTTR during LO training missions, with the following exceptions:

A13.4.2.1. B-2. As directed by the B-2 scheduling representative.

A13.4.2.2. F-22A. Aircrews from Australia and the United Kingdom who hold the appropriate security clearances are authorized. The F-22A scheduling representative is responsible for ensuring proper clearances and/or authorizing exceptions.

A13.4.2.3. F-35A. Aircrews from Australia, Canada, Denmark, Italy, the Netherlands, Norway, and the United Kingdom who hold the appropriate security clearances are authorized. The F-35A scheduling representative is responsible for ensuring proper clearances.

A13.4.2.4. RQ-170. As directed by the RQ-170 scheduling representative.

## Attachment 14

## ASSIGNED FREQUENCIES.

**A14.1. Area Frequencies.** Each agency, numbered range, and MOA is assigned a discrete tactical frequency for communications. Users should confirm frequencies prior to operations. Current frequencies are listed in the tables below:

**Table A14.1. Tactical Frequencies (MHz).**

| Range/Function | Primary | Secondary | Tertiary |
|---|---|---|---|
| R-61 | 341.925 | | |
| R-62 | 234.25 | | |
| R-63 | 361.6 | 274.875 | 122.9 |
| R-64 | 288.8 | 274.875 | 122.9 |
| R-65 | 225.45 | | |
| R-71 | 335.45 | 288.225 | |
| R-74 | 228.0 | 288.225 | |
| R-75 | 363.9 | 288.225 | |
| R-76 | 354.3 | 288.225 | |
| R-77, ECE, ECW and TPECR | 293.5 | 379.65 | |
| Alamo | 238.8 | | |
| Caliente | 289.3 | 294.9 | |
| Coyote | 234.325 | 370.825 | |
| Elgin | 281.025 | 268.2 | |
| NTTR (Spare) | 357.1 | 255.95 | |
| HAVE QUICK Time of Day (TOD) | 369.0 | | |

**Table A14.2. Agency Frequencies (MHz).**

| Agency | Primary | Secondary |
|---|---|---|
| NATCF East (Sally) | 317.525 | 126.65 |
| NATCF West (Lee) | 254.4 | 119.35 |
| NTTR Range Monitoring (BLACKJACK) | 377.8 | 123.55 |
| DML | 261.1 | 126.15 |
| Creech AFB SOF | 148.175 | |
| Creech AFB SFA | 285.525 | 118.3 |
| Creech Tower (KINS) | 360.625 | 118.3 |
| Silverbow Tower (KTNX) | 257.95 | 124.75 |
| NTTR Threat Systems (ROULETTE) | 293.5 | 379.65 |
| JICO “Juice” | 231.100 | |

## Attachment 15

## GLOBAL POSITIONING SYSTEM (GPS) TESTING

**A15.1. GPS Testing.** NTTR events involving GPS interference (referred to as "GPS testing") must be conducted while maintaining the safety of all range users. GPS testing is approved and conducted IAW CJCSM 3212.03. The requestor, NTTR SMO, DOE SMO, and the Nevada Area Frequency Coordinator (NAFC) will collaborate, as required, to ensure GPS testing is effectively deconflicted.

**A15.2. GPS Testing Process.** Review, coordination, and approval of GPS testing normally requires direct coordination between the requestor and the GPS test provider and will proceed along IAW CJCSM 3212.03 timelines (which may require in excess of 90 days). In parallel, the requestor will pursue local coordination through the NTTR BMP as part of the overall project. Requestors must submit a GPS testing description to HQ NTTR NLT 90 days prior to the first day of execution, including:

A15.2.1. Overall GPS testing POC information.

A15.2.2. Desired GPS testing dates/times (in local date/time). The requestor must account for Coordinated Universal Time (UTC) and daylight savings time conversions.

A15.2.3. Desired transmitter locations.

A15.2.4. GPS testing "Cease Buzzer" procedures and POC information.

A15.2.5. Copies of submitted GPS testing requests.

**A15.3. GPS Testing Timeline.** NLT 30 days prior to the first day of execution, requesting units must submit GPS Interference and Navigation Tool (GIANT) Reliability Prediction Model (RPM) analyses from the following altitudes: 60,000ft MSL, 40,000ft MSL, 25,000ft MSL, 15,000ft MSL, 10,000ft MSL, and 4,000ft AGL.

**A15.4. GPS Testing in Reville North.** To ensure safety of non-participating aircraft, specific procedures for GPS testing within Reveille North must be coordinated between the requestor, NTTR SMO, and the NAFC.

**A15.5. GPS Testing Approval.** Written approval must be provided by the ROO prior to the conduct of GPS testing on the NTTR. Approval includes the date(s), time(s), and transmitter location(s) as well as and any necessary amplifying information. Any necessary deconfliction with other operations (DOE, NNSA, etc.) will be managed through the NTTR BMP.

**A15.6. GPS Testing NOTAMS.** Requesting units will ensure all GPS activity NOTAMs are filed and published through the FAA, including those associated with classified GPS testing.

**A15.7. GPS Testing Montitoring.** In addition to CJCSM 3212.03 monitoring requirements, the GPS testing unit must monitor the range safety network (Fox 4) during GPS testing execution.

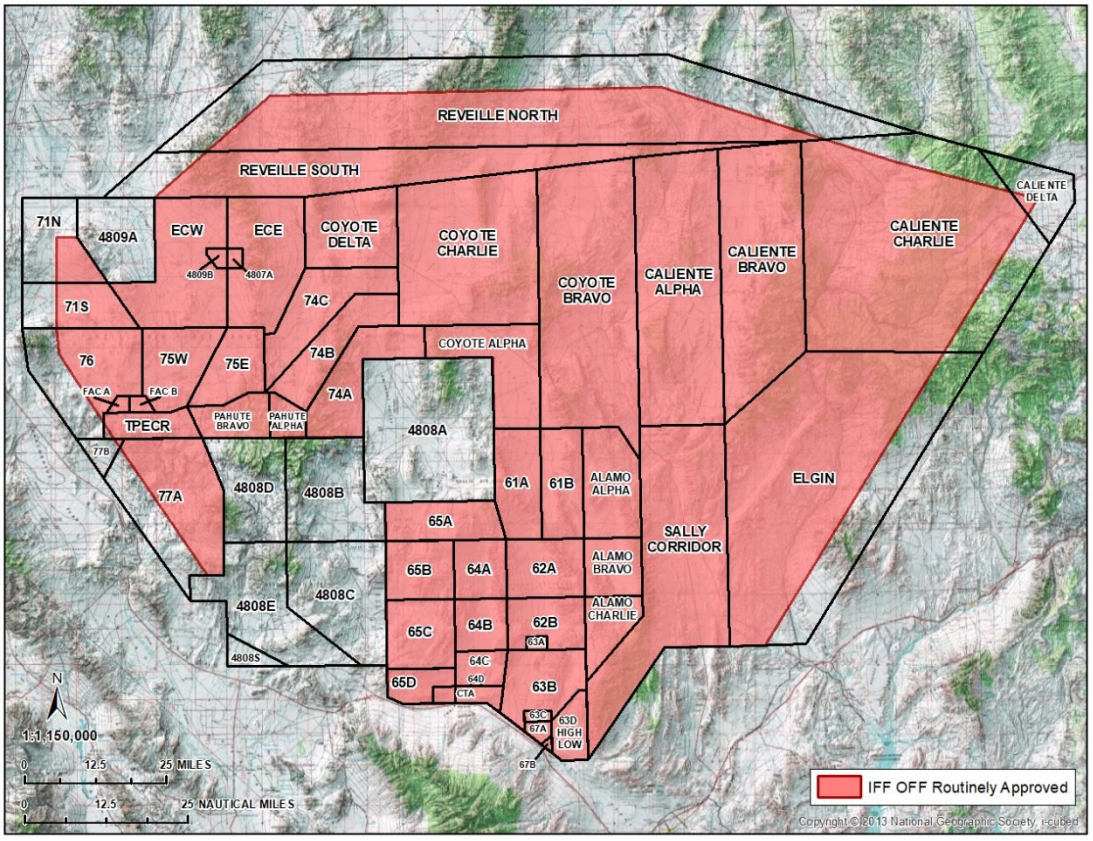
## Attachment 16

## TRANSPONDER OPERATIONS

**A16.1. Mode 3C Requirements.** Unless specifically authorized and scheduled, all aircraft will have an operational identification friend or foe (IFF) system and will squawk Mode C and a tactical Mode 3 (non-1200 squawk) on all missions on range. Normally, only military aircraft operating in support of USAFWS, 64th Aggressor Squadron and 53d Test and Evaluation Group may conduct IFF OFF operations.

**A16.2. IFF Off operations.** IFF OFF operations, when authorized through the NTTR Scheduling Process, are permitted within the boundaries of Elgin, Caliente, Coyote, Reveille, R-4806, R-4807, and portions of R-4809. Use of these areas must also include a NTTR boundary buffer as follows: a 3NM northern buffer, a 5NM eastern buffer, and a 5NM western buffer where applicable. The Coyote northern buffer would not apply when the user has also scheduled Reveille. Aircraft will not operate IFF OFF in these buffer zones. IFF OFF missions in R-4808 and R-4809A are prohibited without specific approval of NNSS/NFO and DOE through the NTTR Scheduling Process. Aircrew must ensure to schedule Caliente Delta airspace when conducting IFF off operations within Caliente Charlie airspace.

**Figure A16.1. NTTR IFF OFF Area.**



## Attachment 17

## LIGHTS-OUT FLYING OPERATIONS

**A17.1. MARSA.** MARSA operations apply for all participating aircraft conducting covert/lights-out operations on the NTTR.

A17.1.1. Covert/lights-out operations are allowed in R-4806; R-4807; ECW; ECE; and above FL180 in Reveille, Coyote, Caliente, and Elgin, without any coordination.

A17.1.2. Covert/lights-out operations in R-4808 and R-4809A are prohibited without specific approval of NNSA/NFO and DOE through the NTTR Scheduling Process.

A17.1.3. Covert/lights-out operations between the hours of sunset and sunrise below FL180 in the Reveille, Coyote, Caliente, and Elgin MOAs are approved if the mission is scheduled at least 96 hours in advance. Missions scheduled within 96 hours may be approved covert/lights-out on a case-by-case basis. Missions scheduled inside 48 hours will not be approved covert/lights out.

A17.1.3.1. Lights out will appear in CSE/GRR if covert/lights out are approved below FL180 in the Reveille, Coyote, Caliente, and Elgin MOAs.

A17.1.3.2. NTTR Range Scheduling will send changes to ZLA and ZLC ARTCC's Military Operations desk up to 96 hours prior to the scheduled mission.

A17.1.3.3. NTTR Range Monitoring will process the appropriate ZLC NOTAMs if the mission is scheduled at least 96 hours in advance. NTTR Range Monitoring will notify the ZLC ARTCC Military Operations desk as appropriate of changes and cancellations affecting lights-out times.

A17.1.4. Covert/lights-out operations will be terminated immediately if advised by NATCF, NTTR Range Monitoring, DML, or if the aircrew observes any of the following:

A17.1.4.1. Conflicting traffic.

A17.1.4.2. A participating aircraft spills out of the designated airspace.

A17.1.4.3. Loss of positive control (i.e., NATCF, NTTR Range Monitoring, or DML radar and/or other associated monitor equipment outage).

A17.1.4.4. Loss of communication (i.e., aircraft, NATCF, or DML radio outage).

## Attachment 18

## LASER AND DEW OPERATIONS

**A18.1. Approval.** Contact NTTR/DO to coordinate use of laser or DEW systems on the NTTR; NTTR/DO will in-turn coordinate with NTTR/SE. NTTR/SE has the responsibility to evaluate and monitor the use of laser and DEW systems on the NTTR and to enforce control of associated hazards. A list of approved laser and DEW systems and targets on the NTTR is available on the NTTR Mission Planning website. Reference the NTTR Mission Planning website for a current list of authorized lasers and ranges certified for their use.

**A18.2. Safety Training.** Personnel who operate or who may be exposed to laser or DEW radiation will accomplish appropriate safety training and wear NTTR-approved laser eye protection (LEP) IAW command directives and NTTR safety policies.

A18.2.1. Personnel who operate or who are intentionally exposed to laser or DEW systems will provide a copy of the system-specific safety training to NTTR/SE and complete the NTTR Laser Awareness Training prior to being granted access to any range where lasers are being employed. System-specific training should meet the requirements of AFI 48-139, Laser and Optical Radiation Protection Program and ANSI Z136.1, American National Standards Institutes Standard for the Safe Use of Lasers.

A18.2.2. Personnel who are only incidentally exposed to laser or DEW systems while operating on the NTTR will complete the NTTR Laser Awareness Training prior to being granted access to any range where lasers are being employed.

A18.2.3. Only NTTR/SE-approved LEP may be used and must be checked out from NTTR/SE; except, for laser and DEW systems that are not authorized for training purposes and for which NTTR/SE-approved LEP is insufficient. In these cases, the visiting unit will provide adequate LEP to NTTR/SE for approval prior to the use of any of these systems. The unit must furnish enough LEP for all individuals involved in the test, as well as additional pairs for NTTR/SE to utilize for safety assessment.

A18.2.4. Personnel operating within enclosures where no direct or reflected laser energy can enter (for example, inside a building with covered windows) are exempt from the requirement to wear LEP.

A18.2.5. During laser operations, ground personnel will not use optics (i.e. binoculars, optical sighting equipment, telescopes, etc.) unless the device incorporates appropriate optical density/filters for the laser in use. These devices must be evaluated and approved by NTTR/SE prior to their use.

**A18.3. Laser Targets.** The NTTR Mission Planning website identifies which range targets may be lased. Targets will be inspected and approved by the NTTR/SE, IAW AFMAN 13-212 and AFI 48-139.

**A18.4. Environmental Hazards.** Environmental factors (e.g., standing water, ice, snow, etc.) or other reflective objects within 200 meters of a laser target may create a reflectivity hazard which will preclude the use of "combat" lasers while the condition is present. The flight lead is ultimately responsible for this determination and for the decision to operate a laser or not. A dry pass to determine target area conditions may be required.

**A18.5. System Approvals.** Laser/DEW systems approved for training operations. Requests will be accomplished through the NTTR Scheduling Process and must include (as applicable):

A18.5.1. Common and/or Military nomenclature for the laser system (i.e., AN/AAS-52 MTS-A, AN/AAQ-33 Sniper, etc.).

A18.5.2. Mode of Operation (Combat or Training).

A18.5.3. Range(s) and target(s) requested for laser operations.

A18.5.4. Minimum and maximum distance from laser to desire target(s).

A18.5.5. Minimum and maximum lasing altitudes.

**A18.6. Tests/Demonstrations.** Laser/DEW systems for tests and demonstrations. Requests will be accomplished through the NTTR Projects Office and must include (as applicable):

A18.6.1. Approvals/authorizations from the Air Force Directed Energy Safety Board and/or the 711 HPW IAW AFI 48-139 and AFI 91-501, Directed Energy System Safety.

A18.6.2. Common and/or Military nomenclature for the laser system (i.e., AN/AAS-52 MTS-A, AN/AAQ-33 Sniper, etc.);

A18.6.3. Mode of Operation (Combat or Training).

A18.6.4. Platform Type (handheld, weapons mounted, vehicle mounted, fixed wing, or rotary wing);

A18.6.5. Laser System Parameters:

A18.6.5.1. All laser wavelength(s).

A18.6.5.2. Beam profile (i.e., Gaussian, Top hat, etc.).

A18.6.5.3. Beam diameter at 1/e (both axes, if applicable).

A18.6.5.4. Beam divergence (full angle at 1/e – both axes, if applicable).

A18.6.5.5. Laser Type (Continuous Wave or Pulsed) and maximum beam irradiance (Watts/cm2) for Continuous Wave systems or maximum beam radiant exposure (Joules/cm2) for Pulsed systems.

A18.6.6. Range(s) and target(s) requested for laser operations.

A18.6.7. Minimum and maximum distance from laser to desire target(s).

A18.6.8. Minimum and maximum lasing altitudes.

**A18.7. Airborne Laser Employment.**

A18.7.1. Aircrews will only lase targets that have been specifically approved for laser operations IAW with the NTTR Laser Range Certification issued by the 711 HPW (refer to the NTTR Mission Planning website). Requests for exceptions must be submitted to NTTR/DO, who will coordinate with NTTR/SE.

A18.7.2. Aircrews will not activate the laser until the target is positively identified.

A18.7.3. Aircrews will not scan from one target to another with the laser.

A18.7.4. Aircrews will not fire the laser above the horizon unless authorization has been granted by the DoD Laser Clearing House and NTTR/SE.

A18.7.5. Line-of-Sight to the approved target must be maintained during lasing. Lasing may be initiated from any altitude and slant range that precludes laser energy from terminating outside of a laser certified subrange. NTTR approved weapons profiles are designed to take laser Flight Limitation Profile Curves (FLPCs) into account. Specific FLPCs may be obtained from NTTR Mission Planning website.

A18.7.6. Air-to-Air laser employment is restricted to the guidance detailed in AFMAN 11-214.

A18.7.7. Unless specifically authorized within a test/training plan and approved by NTTR/SE, Aircrews will not fire any laser at personnel, manned locations or manned vehicles anywhere on the NTTR.

A18.7.8. Lasing From 64C. For the sole purpose of using the 64G boresight, RPAs are permitted to lase from 64C into 64G even when the NTTR Graphical Range Restriction (GRR) has a “No Combat Lasers” restriction in effect in 64C under the following constraints:

A18.7.8.1. Lasing operations are performed in accordance with the following Flight Profile Limitations in figure A18.1 and table A18.1.

A18.7.8.2. The Supervisor of Flying (SOF) will coordinate with tower controllers to ensure lower aircraft remain horizontally clear of the boresighting aircraft's laser pathway. Should another aircraft enter the Multi-Spectral Targeting System's (MTS) field of view, the boresighting aircraft will immediately cease lasing.

**Figure A18.1 Certified Laser Flight Profiles.**

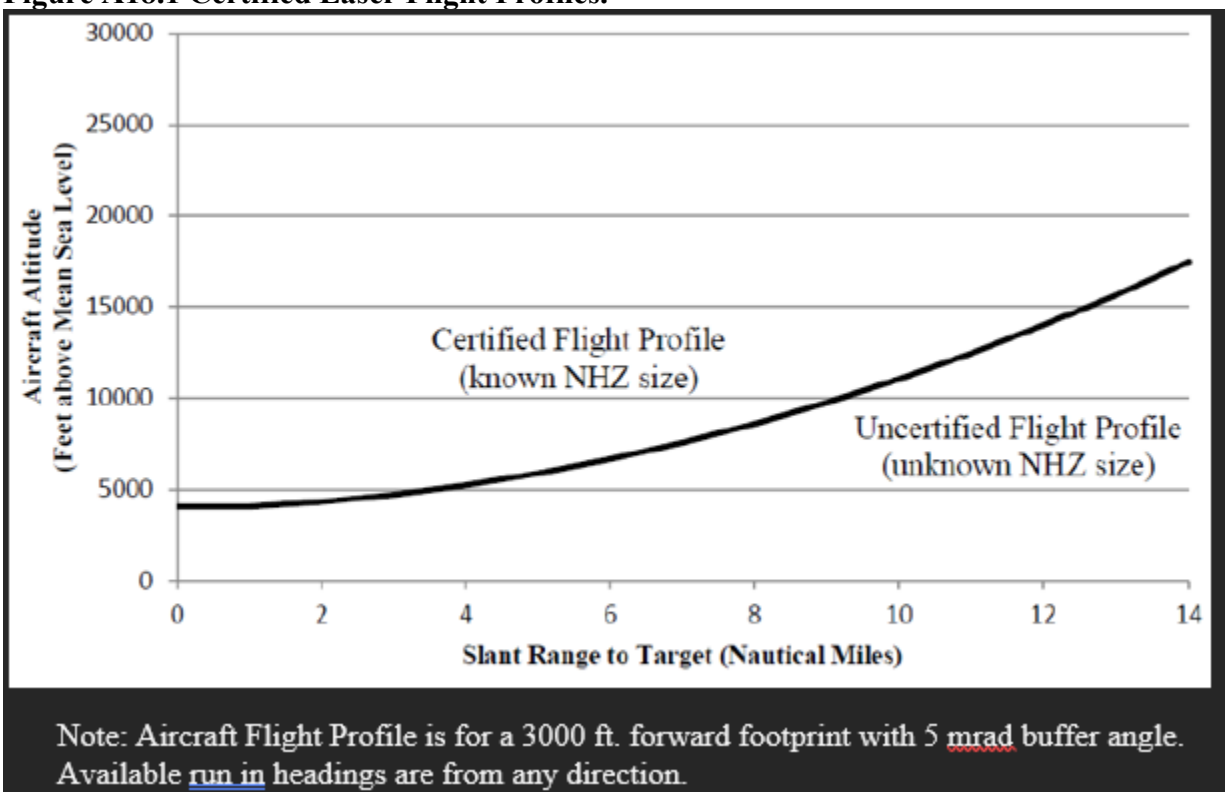


**Figure A18.2 Certified Laser Profiles.**

| Slant Range to Target (NM) | Minimum Safe Lasing Altitude (feet above MSL) | Slant Range to Target (NM) | Minimum Safe Lasing Altitude (feet above MSL) |
|---|---|---|---|
| 14 | 17,450 | 7 | 7,600 |
| 13 | 15,650 | 6 | 6,700 |
| 12 | 14,000 | 5 | 5,900 |
| 11 | 12,450 | 4 | 5,250 |
| 10 | 11,050 | 3 | 4,750 |
| 9 | 9,800 | 2 | 4,350 |
| 8 | 8,600 | ≤1 | 4,100 |

Note: The Boresight Board is certified as approved to lase when aircraft is above the minimum altitude at the indicated range from any direction from which there is line of sight.

**A18.8. Ground Laser Employment.**

A18.8.1. Personnel will only lase targets that have been specifically approved for laser operations IAW with the NTTR Laser Range Certification issued by the 711 HPW (refer to the NTTR Mission Planning website). Requests for exceptions must be submitted to NTTR/DO, who will coordinate with NTTR/SE.

A18.8.2. Personnel will not activate the laser until the target is positively identified.

A18.8.3. Personnel will not scan from one target to another with the laser.

A18.8.4. Personnel will not employ lasers from a location lower than the target's elevation; nor, above the horizon unless authorization has been granted by the DoD Laser Clearing House and NTTR/SE.

A18.8.5. Lasing may only be fired from approved NTTR OPs or designated ground firing locations authorized under the NTTR Laser Range Certification (NTTR Mission Planning website). Line of site to the approved target must be maintained during the entire activation of the laser/DEW system.

A18.8.6. Unless specifically authorized within a test/training plan and approved by NTTR/SE, personnel will not fire any laser at other personnel, manned locations, or manned vehicles anywhere on the NTTR.

## Attachment 19

## FIRE, CHAFF, FLARES, AND TOWED DECOYS

**A19.1. Fire Risk**. High-fire potential normally exists from 1 May through 31 October, yearly. Ordnance delivery and range operations may be negatively impacted by fire code restrictions. NTTR Range Monitoring will disseminate daily fire codes during the high-fire potential months. Fire codes will be depicted on the NTTR Mission Planning websites and the GRR.

**A19.2. Fire Codes**. NTTR Range Monitoring will check daily with BLM for fire codes during the high-fire potential months. NTTR Range Monitoring will disseminate daily fire codes during the high-fire potential months using the following nomenclature:

A19.2.1. Low. Fires that are easily controlled with hand tools and mechanical equipment. Fire spread is slow.

A19.2.2. Moderate. Indicates the upper limit of fires which can be controlled with hand tools because the fire is becoming too hot for crews to work safely. Fire spread is increasing.

A19.2.3. High. The prospects of controlling a fire with hand tools and mechanical equipment are poor because working safety is diminished by an intense heat, and/or the fire spreads more quickly than the suppression capability of the fire-fighting equipment. Control lines may have to be placed well beyond the fire, and air tankers with retardant will probably be needed.

A19.2.4. Very High. The heat load on people within 30ft of the fire is dangerously high. Control lines must be placed well ahead of the fire. Retardant efforts may be ineffective due to the intensity of the fire. Fire spread is very fast.

A19.2.5. Extreme. Fires are very difficult to control. Extreme fire behavior is possible. There is significant danger for the firefighters and extreme caution must be taken.

**A19.3. Expendable Restrictions**. In addition to standing restrictions, periodically, specific restrictions will be published prohibiting the use of chaff and flares. Reasons may include extreme ground fire hazards, threats to ground property, high personnel injury potential, and ATC radar interference. The ROO will establish restrictions as appropriate based on the prevailing conditions.

**A19.4. Chaff Restrictions**. Chaff restrictions on the NTTR vary daily due to atmospheric conditions. NTTR Range Monitoring validates by 0630L daily that the chaff restrictions have been uploaded as required.

A19.4.1. Rope chaff, explosive chaff and large-scale chaff corridor employments may require Environmental Impact Analysis. Reference the chaff authorization on the Mission Planning website for current authorizations. For chaff employment that falls outside the scope of the current authorization, contact the NTTR SMO and Environmental offices for coordination and approval.

A19.4.2. Chaff drops that appear as corridor or extensive area seeding in Elgin and Caliente C are not authorized.

A19.4.3. Minimum altitude for helicopter chaff deployment is 100ft AGL.

A19.4.4. Refer to the GRR for additional chaff restrictions.

## A19.5. Flare Restrictions.

A19.5.1. Minimum flare release altitude for self-protection flares is that altitude which allows the flare to burnout prior to 100ft AGL.

A19.5.2. Minimum release altitude for illumination flares, such as photo-flash and parachute flares is that altitude which allows the flare to burn out prior to 500ft AGL.

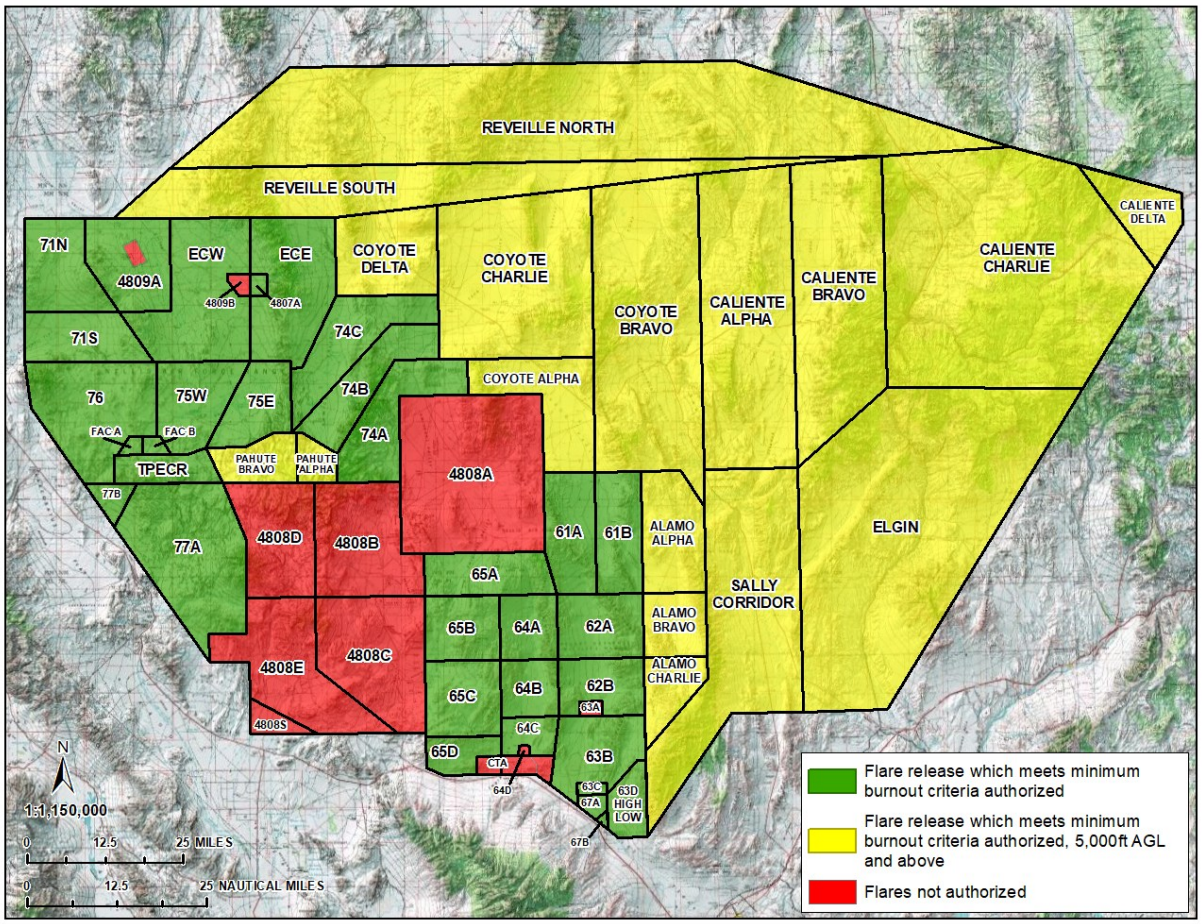
A19.5.3. Flare release altitude over manned sites, ground parties, or within 3NM of forested areas is 5,000ft AGL, provided minimum flare release altitude has been achieved.

A19.5.4. Flare release altitude in the MOAs and Pahute A/B is 5,000ft AGL and above, provided minimum flare release altitude has been achieved.

A19.5.5. During fire codes “very high” or “extreme,” no flares of any type will be permitted on the NTTR below 5,000ft AGL.

A19.5.6. Refer to the GRR for additional flare restrictions.

**Figure A19.1. NTTR Flare Restriction Areas.**




## A19.7. AN/ALE 50 Operations

A19.7.1. Any aircraft certified to carry the AN/ALE 50 CMDDS with AAED rounds is authorized to conduct AN/ALE 50 operations on the NTTR.

A19.7.2. The mission aircraft may deploy the AN/ALE 50 CMDDS over any NTTR range or MOA. The mission aircraft may only jettison the AAED round while over active bombing ranges, which have been scheduled for the aircraft's use and clear of non-associated ground parties.

A19.7.2.1 A visual inspection of the anticipated impact area must be performed immediately prior to release to ensure no ground personnel are in the projected impact area. This inspection may be performed by an associated ground party that is in communication with the aircrew. During periods of limited visibility, aircraft sensors and communication with NTTR Range Monitoring (Blackjack) can fulfill this requirement.

A19.7.2.2. If using an associated ground party, it does not need to be a JTAC ground party. The associated ground party shall be in communication with the aircraft, or be briefed on the planned time and location of the AAED release and at a safe location.

A19.7.3. The AAED round may be jettisoned in the vicinity of any live or inert target complex. If the unit wishes to have the spent rounds retrieved, an inert target complex should be used. The best jettison procedure to have a round retrieved is to fly in line with and over the simulated airfield runway on one of the following target complexes: 62-31/32, 71-21/22/23/24, 74-04, 75-47, and 76-03.

A19.7.4. When checking out with Blackjack, the aircraft shall report how many AAEDs were released, and over which range(s) they were released.

A19.7.5. In the event of an aircraft emergency, the mission aircraft may release the AAED over any location and using any means deemed necessary and responsible by the aircrew.

**A19.8. Wildland Fire Response**. In the event of fire resulting from natural causes or as the result of a major disaster on the range, an emergency response force may respond. This depends on the specific situation and jurisdiction of the response force. Refer to the Installation Emergency Management Plan (PLAN 10-2) for further details under Tab F, Wild Land Fire/Forest Fire Checklist. Report wildland fires to Nellis Command Post at (702) 652-2446. For fires that impact R4808 or Pahute A/B also contact BirdDog Ops at (702) 295-2467. Provide the following information in the report:

A19.8.1. General Location/Description and/or Range Number.

A19.8.2. GPS Point. Specify Latitude/Longitude or Universal Transverse Mercator (UTM), if possible.

A19.8.3. Size of Fire and Direction of Fire movement.

A19.8.4. Structures/Assets potentially in danger.

A19.8.5. Cause of Fire (if known).

## Attachment 20

## NTTR JOINT DATA NETWORK OPERATIONS (JDNO)

**A20.1. NTTR JDN**. The NTTR JDN is intended to satisfy the range's robust and dynamic information exchange requirements (IERs) during all NTTR activities, including day-to-day operations, LFEs, and unique test and training events. Access to the NTTR JDN is coordinated through the NTTR BMP or the NTTR Scheduling Process, as appropriate.

**A20.2. NTTR JDN Cell**. The NTTR JDN Cell (JDNC) manages the NTTR JDN and is responsible for the integration of all sub-networks, including the Multi-Tactical Data Link Network (MTN), Defense Research and Engineering Network, Defense Information Systems Agency (DISA), and the Joint Technical Enabling Network.

**A20.3. JDNC Roles and Responsibilities:**

A20.3.1. In coordination with the relevant NTTR agencies, plan, program, and budget for JDN infrastructure; maintain and sustain assigned NTTR JDN equipment; maintain configuration control over assigned NTTR JDN equipment.

A20.3.2. Collect NTTR-user IERs in support of network design.

A20.3.3. Coordinate with the Network Design Facility (NDF).

A20.3.4. Obtain FAA Link-16 Time Slot Duty Factor (TSDF) approvals and waivers.

A20.3.5. Author and publish NTTR Operational Tasking Links (OPTASKLINKs) and associated USAF Initialization Data Loads (IDLs), and updates, as required. NTTR OPTASKLINKs may be found on the NTTR Share Point.

A20.3.6. Provide the NTTR JDN Joint Interface Control Officer (JICO) (unless specifically delegated via the OPTASKLINK) to actively manage the JDN and Common Tactical Picture (CTP).

A20.3.7. Provide the Common Tactical Picture (CTP) to the Shadow Operations Center-Nellis (ShOC-N) for display and/or further distribution.

A20.3.8. Log weekly TSDF on the Joint Interoperability & Data Link Training Center (JID-TC) Link Pulse Deconfliction Server.

**A20.4. NTTR JDN User Roles and Responsibilities:**

A20.4.1. Provide recommendations for desired JDN infrastructure changes.

A20.4.2. Provide IERs in support of network design.

A20.4.3. Identify unique security concerns associated with mission-specific data shared on the JDN.

A20.4.4. Identify JDN SAR/SAA requirements pertaining to SATCOM data.

A20.4.5. Recommend desired network design changes to JDNC.

A20.4.6. Comply with published OPTASKLINKs.

A20.4.7. Recommend desired OPTASKLINK changes to JDNC.

A20.4.8. Assign a Link Unit Manager (LUM) to actively monitor the JDN and CTP and coordinate with the JICO during mission execution.

A20.4.9. NTTR users who plan and execute data link networks outside of the NTTR JDN, and without an NTTR JICO assigned, must provide the JDNC with network identifier, crypto short title and network time no later than 45 duty days before execution in order to avoid network fratricide.

## Attachment 21

## IMAGERY PROCESSES AND RESTRICTIONS

**A21.1. Imagery.** Imagery of the NTTR includes all data of the Earth's surface or structures on the Earth's surface collected, displayed, disseminated, or recorded, via any medium (digital/film/RADAR/LiDAR/other), regardless of equipment category, of the NTTR DoD withdrawn lands, restricted airspace, range equipment, or operations. Video imagery includes all compilations of moving images. The following guidance applies to aerial imagery (imagery generated by equipment aboard an aircraft) and ground imagery (imagery generated by equipment on the ground). Space-based (commonly referred to as "overhead imagery") and SIGINT collections on the NTTR and R-4808 must be coordinated at the monthly JWG. USAFWC/A2 is the coordinating authority for overhead imagery and SIGINT ops on the NTTR. NNSS/NFO and DOE are approval authorities for all imagery and SIGINT requests of R-4808N.

A21.1.1. NTTR users who wish to image on the NTTR and/or R-4808 will coordinate their desires through the NTTR Projects Office. Approval must be in place prior to scheduling range time through the NTTR Scheduling Process. Range restrictions regarding imagery are annotated in CSE, under "Reports, Block Restrictions." Ultimately, the aircraft commander is responsible for adherence to imagery standards by individuals operating aboard that aircraft. Public release of any images must be coordinated by the requestor through NTTR/SD to obtain approval.

A21.1.2. Imaging of R-4808 requires NNSS/NFO and DOE approval.

A21.1.3. Imaging of R-4809A is normally limited to USAFWC-assigned, Nellis AFB-based aircraft.

A21.1.4. Imaging of R-4809A by USAFWC-assigned, Creech AFB-based RPAs must be coordinated by the requestor through NTTR Projects Office to obtain approval.

A21.1.5. Temporary Duty aircrews (excluding USAF Weapons School students) and visiting aircraft are not authorized to image in R-4809A without coordination by the requestor through NTTR Projects Office to obtain approval via the Joint Working Group (JWG) process. Visiting RPAs and/or non-Creech assigned RPA crews are not authorized to image in R-4809A.

A21.1.6. RPA imaging of threats must be coordinated by the requestor through NTTR Projects Office to obtain approval via the JWG process.

**A21.2. NTTR/SD Coordination**. All imaging on range requires coordination with NTTR/SD through the NTTR BMP or NTTR Scheduling Process (as appropriate). Before introduction on the NTTR, NTTR/SD will initiate a review of the technical specifications of the integrated military equipment with the relevant sponsoring agencies. Use is approved in perpetuity, except as limited by the daily restrictions. Users may contact NTTR/SD to determine any preexisting approvals for use on the NTTR.

A21.2.1. Sensitive Equipment, Controlled and Prohibited Articles. NTTR, NNSS/NFO and NNSA/SFO all have processes defining Prohibited Articles and allowing Controlled Articles and Sensitive Equipment on their controlled areas.

A21.2.1.1. NTTR Sensitive Equipment Passes. The management of controlled sensitive equipment is articulated in NTTR OI 31-101. In summary, users must obtain a sensitive equipment pass prior to imaging on NTTR. Sensitive equipment passes will be issued only for a period of time commensurate with the associated activity, although annual passes may be issued to local units with a recurring requirement. A local host unit, experienced in local imagery procedures, will be identified for each sensitive equipment pass issued. Approval must be in place prior to scheduling range time through the NTTR Scheduling Process. Prior to issue of an NTTR sensitive equipment pass, a complete inventory of intended imaging equipment is required. All imaging equipment will be itemized on the NTTR sensitive equipment listing. This equipment listing will include name, brand, type, resolution, frame rate, aperture, zoom capability, etc. NTTR/SD will approve and issue NTTR sensitive equipment passes.

A21.2.1.2. NNSS/NFO Controlled Articles. NNSS/NFO has a process for approving certain controlled articles onto NNSS/NFO property (under R-4808 airspace) and restricting use of those articles. A special permit is required for all imaging equipment brought onto the NNSS. NNSS/NFO security personnel will determine approval and use of controlled articles on the NNSS.

A21.2.1.3. NNSA/SFO Sensitive Equipment Passes. Prior to issue of an NNSA/SFO sensitive equipment pass, a complete inventory of intended imaging equipment is required. All imaging equipment will be itemized by NNSA/SFO. This equipment listing will include name, brand, type, resolution, frame rate, aperture, zoom capability, etc. NNSA/SFO will coordinate with the relevant sponsoring agency(ies) to establish approval to issue NNSA/SFO sensitive equipment passes.

**A21.3. Integrated Military Imaging Equipment**. Integrated military imaging equipment that operates as an integrated sensor of a weapon system to record and/or transmit imagery. This equipment will operate IAW individual MDS security guidelines and NTTR range restrictions regarding imagery as annotated in CSE.

**A21.4. Ancillary Imaging Equipment.** Approved ancillary imaging equipment includes: government owned equipment (non-weapons system), government or commercial media equipment, and government sponsored equipment (e.g. GoPro). Ancillary imaging equipment may be used for US-only purposes with limited releasability to foreign nationals.

**A21.5. RPA Imaging.** RPAs and associated support systems must meet specific requirements to conduct imaging (still and FMV) on the NTTR.

A21.5.1. RPA Ground Control Stations (GCS) and associated data-storage and processing devices must operate and be controlled at the commensurate classification level of the imagery.

A21.5.2. All personnel operating within the GCS, or with access to any relevant imagery, must be cleared commensurate to the classified level of the GCS.

A21.5.3. Classified feeds will not be disseminated outside of the cleared GCS unless specifically coordinated with the relevant sponsoring agencies.

**A21.6. Aerial Survey.** Aerial Surveys are conducted by civilian companies, either primarily for their use or as a contracted entity for a governmental agency. Non-military imaging of DoD withdrawn lands on the NTTR is prohibited. Aerial surveys of public lands under BLM/USFWS oversight conducted from NTTR-associated MOA airspaces present a unique challenge to operations on the NTTR. Aerial surveys, when identified, will be reflected on the daily range schedule. Prior to survey execution, NTTR/DO will provide suggested operational guidelines to the survey agency to include:

**Table A21.1. Aerial Survey Guidelines.**

| |
|---|
| • Survey company name<br>• Event name<br>• Scheduled dates and times<br>• Survey aircraft type, N-number, and callsign (if applicable)<br>• Airspace to be used<br>• Intended collection areas (collection-area maps and tracks will be included, if available)<br>• Positive control measures<br>• Procedures for inadvertent overflight<br>• Procedures for inadvertent imaging of restricted areas (including imagery review)<br>• Suggested buffers<br>• Relevant radio frequencies<br>• NTTR-assigned project manager contact information |

**A21.7. FMV.** All FMV imaging conducted on the NTTR must be specifically coordinated by the NTTR Projects Office at the monthly JWG. NNSS and DOE are the approval authorities for all imagery requests of R-4808N. The requesting unit will submit a “NTTR FMV Questionnaire” (Figure A21.2) in conjunction with the PID (when utilized) IAW NTTR BMP timelines. When the PID is not utilized, the questionnaire must be submitted at the NTTR Range War meeting (approximately 30 days prior to the month of execution). Any questions that are not answered at the time of submittal may delay or preclude approval. FMVs such as GoPro cameras should be coordinated through NTTR/SD.

**Table A21.2. NTTR FMV Questionnaire.**

- Range user requesting FMV approval?
- Event name and desired dates and times (if known)?
- Event objective? For test events, provide test plan.
- Intended collection target?
- Will the aircraft use FMV enroute to the target?
- Are foreign nationals involved?
- Local host unit(s) (if any)?
- Has the requesting user operated on the NTTR and/or NNSS before? When? Why?
- Sensor platform?
- What are the desired platform operating and FMV sensor collection altitude(s)?
- Call sign?
- IFF code?
- FMV equipment to be used?
- Sensor type (optical, infra-red, electromagnetic spectrum, LIDAR, radiometer, etc.)?
- Can the sensor be turned on and off during the mission?
- Sensor field of view?
- Sensor field of regard?
- Sensor resolution?
- Sensor range?
- Will experimental equipment/software be used?
- What will be the disposition of test sensor equipment after the test is complete?
- Where does the data flow after it is collected?
- Is it a live feed or will it be recorded/processed first?
- Is the data downlinked to a ground station or retained on-board?
- If necessary, can the data be edited prior to distribution or storage?
- What is the final storage destination of the data?
- At what security level will the data be stored?
- Will stored data be retained in it be encrypted format?
- Will stored data be treated as “US Only?”
- Who will have access to collected data?
- How long will the data be retained after the event?

**A21.8. Imagery Review.** Any imagery of the NTTR, range equipment, or range operations taken by any NTTR user will normally be reviewed prior to release and dissemination. Users will provide access to the media in question, and related storage devices, upon request. Review will be accomplished by cognizant security officials from the organization requiring the review (USAF, DOE, NNSS/NFO, NNSA/SFO, etc.).

A21.8.1. Review will normally take place at the HQ NTTR building. Alternate locations may be used, if required, at the direction of NTTR/SD. The user will deliver the imagery in question, related storage devices, and any required cabling (RCATM, High Definition Media Interface (HDMI), etc.) necessary to display the images, to the review location as soon as possible.

A21.8.2. In the event that imagery of concern is not available in the Nellis AFB area (i.e., an off-station aircraft has already returned to its home base), NTTR/SD or other cognizant security official will contact the parent unit with guidance for securing the imagery until such time as a review can be accomplished.

A21.8.3. Inadvertent/unauthorized imaging of the NTTR must be immediately reported to NTTR Range Monitoring, NTTR Duty Officer and NTTR/SD. NTTR/SD will coordinate with the appropriate agencies to take appropriate action.

## Attachment 22

## AIR COMBAT MANUEVERING INSTRUMENTATION

**A22.1. NTTR Air Combat Training System (NACTS).** NACTS provides high-fidelity, real-time monitoring of aircraft position, flight dynamics, weapon status, and limited firing parameters. By using computerized displays and direct radio communication, qualified ground personnel can monitor missions in real-time and record all flight data and communications in support of the mission debrief. NACTS provides the same capability for up to 100 aircraft for air-to-air, air-to-ground, and ground-to-air weapons employment. NACTS consists of:

A22.1.1. Airborne Instrumentation (AI) Pods. The P5 Combat Training System (CTS) AI pod is currently employed by NTTR.

A22.1.1.1. AI pods are constructed using an AIM-9 Sidewinder body (minus canards and fins). Pods incorporate a global positioning satellite receiver to provide accurate time, space, and position information (TSPI).

A22.1.1.2. AI pods may be requested by Nellis AFB-based military aircraft operating away from the NTTR. The process for requesting P5 pods is the requester must submit a Pod-only Program Introduction Document (PID) through the NTTR Projects Office six weeks prior to the requested date. Availability is based on projected NTTR pod requirements under the JTECH contract. AI Pods are assigned though ACC to NTTR to support JTECH contract requirements and are maintained by contractors. Each MDS must have a “cleared-to-fly” pod certification letter, for each airframe, before use.

A22.1.1.3. AI pod data is collected via P5 Remote Range Units (RRUs) and is transmitted to Nellis AFB for processing. Any support outside of the NTTR (i.e. R-2508, W-291, UTTR, etc.) is scheduled through the Business Management Process with DOXP/Projects Office. This affords everyone the opportunity to discuss requirements and manage expectations. Additionally, for operations occurring in R-2508, live/post mission products are not consistent nor reliable due to lack of ground equipment physically located within that airspace. To ensure real-time presentation and post-mission data products meet mission goals, support for events occurring in R-2508 require specific coordination prior to scheduling. Units requesting P5 Support within R-2508 must confirm if they require live mission viewing from an operations location and if they are requesting a post-mission P5 ICADS recording.

A22.1.1.4. It is imperative for Unit Ops Desks working with the P5 system to email or call in their PSIDs, Callsigns, and Tail numbers for loading in the P5 stack NLT aircrew step. All MALD information should be emailed to P5 office NLT 24 hours prior.

A22.1.3. Processed AI pod data is converted into a usable display and debriefing formats via the P5 Live Monitor System (LMS).

A22.1.4. ICADS is a Windows, PC-based aircrew debriefing system and is located at several flying units on Nellis AFB. ICADS provides enhanced visual and debriefing formats that can be tailored for different debrief styles by aircrew members.

**A22.2. NACTS Range Training Officer (RTO)**. Units using NACTS may appoint a qualified RTO (IAW unit-established standards) for each mission. The RTO acts as the primary interface between NACTS and the flight. As such, the RTO must be familiar with both NACTS operations and unit training objectives. RTOs will not act as a Range Safety Officer or Range Control Officer.

A22.2.1. RTO uses the NACTS display and available communication systems to provide air combat maneuvering (ACM) support requested by the flight lead, such as real-time flight data, system dropouts, and rejoin information.

A22.2.2. The presence of an RTO does not diminish the authority or responsibilities of the airborne flight lead or primary instructor. Aircrews are responsible for accomplishing the mission, adhering to training rules, and maintaining flight safety.

A22.2.3. Normally, the flight lead or designated representative will brief the RTO on mission objectives before the flight and will provide mission data to the RTO not less than 30 minutes before range time. In the absence of the RTO, the flight lead will pass this information to the Display and Debriefing Subsystem technician. The flight lead will pass tail numbers, AI pod numbers, and armament station to the technicians at least 15 minutes prior to range time start.

## Attachment 23

## RESOURCE MANAGEMENT

**A23.1. Requirements.** NTTR Resource Management ensures range user resource requirements are provided to the maximum extent possible in a way that optimizes coordinated range user activities. HQ NTTR manages the following resource activities:

**A23.2. CSE**. NTTR Resource Managers utilize the CSE as a common system to fully integrate and manage range scheduling, business management processes and resource management processes in order to meet customer needs, prioritize decision-making, and forecast requirements.

**A23.3. Requirement Submission**. All units must submit their NTTR resource support requirements accordingly through the NTTR Scheduling and BMP processes.

**A23.4. Adjudication.** The NTTR Resource Management Adjudication Process is identical to the NTTR Range Scheduling Adjudication Process and provides a standardized framework for thorough evaluation and administration of range user resource requirements for all air and ground activities conducted on the range whenever a resource conflict exists.

**A23.5. Conflicts**. The NTTR Range Resource Manager is responsible for the initial administration of the resource management adjudication process and is the first step for resolving resource conflicts.

**A23.6. Timeline.** The NTTR Resource Management timeline is organized into three progressive phases: planning, processing, and execution.

A23.6.1. The NTTR resource management planning phase is a collaborative and integrated undertaking that is led by the NTTR Range Resource Manager. The NTTR resource management planning phase provides a method for consolidating long-term NTTR resource issues and requirements from all range users. The planning phase ranges from 12 months to 90 days prior to the month of execution.

A23.6.2. The NTTR resource management processing phase translates long-term requirements into an executable support schedule.

A23.6.2.1. The NTTR resource management processing phase begins once the initial set of mission requirements are reflected on the shell. NTTR Range Resource Manager will collaborate with NTTR/RSO and NTTR contractor leadership to identify proposed NTTR resource support windows based on range user requirements. The NTTR Range Resource Manager will annotate proposed support windows on the margins of the shell and release the shell to 57 OSS/OSO no later than 60 days prior to the month of execution.

A23.6.2.2. Once 57 OSS/OSO receives the shell, every attempt will be made to collectively submit Nellis, Creech, and other ACC range requirements within the proposed NTTR resource support windows during the 57 WG Range Integration meeting. Once Nellis, Creech, and other ACC range requirements are reflected on the spreadsheet, the NTTR Range Resource Manager will identify NTTR resource support shortfalls and begin the resource adjudication process for all user resource requirements that are outside of the proposed NTTR resource support windows.

A23.6.2.3. When all NTTR range users and associated project managers have adjusted the spreadsheets and coordinated individual range requirements through the NTTR Range War meeting, the NTTR Range Resource Manager will continue to adjudicate all range user resource requirement shortfalls and attempt to resolve potential resource shortfalls prior to the NTTR Pencil meeting.

A23.6.2.4. The NTTR Range Resource Manager will annotate actual range resource support windows on margins of the final version of the post-range war spreadsheets and distribute to all range users immediately following the NTTR Range War meeting.

A23.6.3. NTTR Resource Management meeting (21-day). The NTTR Resource Management meeting is hosted by NTTR Range Resource Manager and conducted in the HQ NTTR building approximately three weeks prior to the week of execution at 0800L each Tuesday. The purpose of the Resource Management meeting is for all NTTR resource owners, associated project managers to conduct a three-week breakdown of upcoming NTTR range user resource requirements and take necessary actions to provide NTTR range resource support for all NTTR range user requirements to the maximum extent possible. Select NTTR range users may attend when necessary.

## Attachment 24

## RANGE USE ONLY (RUO) ASSETS PROGRAM

**A24.1. RUO Designation**. The Nevada Test and Training Range utilizes Defense Logistics Agency disposition services program to obtain materials identified for destruction to be utilized as target assets. Range Services Support II Targets Team will then evaluate the items/materials for suitability and safety to use in convoy operations, target movements to include surrogates and threat emitters, or other similar target-related functions, with final approval by the NTTR Mission Support Director or designated representative. Any assets to be entered and operated under the RUO program are only authorized on-range and will not be used on public roads. These assets will not be registered or plated, however, the asset must be marked with yellow lettering indicating the asset is "range-use-only" and assigned a unique registration number (e.g., RSST xxxxx). Any asset that requires movement on public roads must be transported as cargo (e.g., loaded onto a trailer).

**A24.2 Tracking of RUO Assets.** RUO assets must be identified and tracked separately from standard target assets through a government approved database.

**A24.3 Repairs to RUO Assets**. Only minor maintenance is authorized on RUO assets. Operators will source replacement parts and components from other target assets and salvaged vehicles to the maximum extent possible. Written approval is required from the Government Transportation representative for all repairs exceeding $1,500 excluding labor costs. When asset becomes unsuitable or unsafe for use, the Range Services Support II Targets Team will notify NTTR/DOS and return asset back into the standard target inventory.

## Attachment 25

## NON-PARTICIPATING GROUND PARTY INTERACTION RULES DURING RANGE EVENTS

**A25.1. Purpose**. This attachment serves as a framework to define how primary participants in range exercises may interact with associated and non-associated personnel.

**A25.2. Definitions.**

A25.2.1. Primary Participant is defined as a person that is part of the organization that is the primary focus of training for the event. Usually a military member.

A25.2.2. Associated Personnel is locally defined as one or more persons whose mission duties require presence on the range in association with a range event. This is broken down further into Physical Participants and Non-Physical Participants.

A25.2.3. Physical Participants are personnel participating in a range event that are capable of being interacted with physically (See Para 25.2.2.) during the event.

A25.2.4. Physical Interaction is defined as interaction with a person(s) that involves changes to their physical behavior. This can include any form of touching, binding, requiring a person to perform actions outside of their required duties, and in any way putting a person into harm's way in any form. For example, having a vehicle driver get out of their vehicle and lie on the ground would be considered Physical Interaction even if this individual was not touched or bound and was just commanded to do so.

A25.2.5. Non-Physical Participants are personnel participating in a range event that are NOT to be physically interacted with. These are typically personnel participating in an event as convoy drivers, or threat system operators, directly participating in the event. White force personnel are included in this definition. These personnel may be directed to do something or to change their actions (depending on the event scenario and ROEs) but cannot be touched or put in harms way at any point. For example a convoy vehicle driver that is not listed as a Physical Participant, may be directed to pull off the road and simulate being eliminated from the scenario, but shall not be removed from their vehicle and placed in zip-ties or handcuffs.

A25.2.6. Non-Associated Personnel is locally defined as one or more persons whose mission duties require presence on the range during but not associated with an executing range event. Personnel such as range maintenance, NTTR Targets, threat system operators, et. Are examples of some of the personnel that may be on range during an event but not working in association with the event.

**A25.3. Non-Military Personnel**. Unless specifically hired/contracted to participate as Physical Participants, non-military personnel on the range are not to be touched or put in harm's way at any time. Since some personnel are incorporated into events as Physical Participants, it is vitally necessary that methods are established to positively identify the Physical Participants from Non-Physical Participants, and to provide physical separation to protect all personnel who are not Physical Participants. An example is to require the Physical Participants to wear yellow reflective vests to be easily identified from the rest of the personnel on the range and maintaining a 300-meter stand-off from all other personnel.

**A25.5. Radiation**. All devices/equipment emitting any form of radiation (RF, LASER, etc.) must be properly placarded and safeguarded from unknowing/unassociated passerby's, to prevent them from inadvertently entering the hazard zone of the device/equipment.

**A25.6. Plan and Approval.** It is left to the coordinating unit to determine the specific method(s), procedure(s), and stand-off distance(s) to provide the greatest flexibility to the unit to ensure maximum training realism while still protecting the Non-Physical Participants and all Non-Associated Personnel from inadvertent harm. This detailed plan must be reviewed and approved by NTTR Safety and NTTR Director of Operations or their representatives before event execution.

A25.6.1. If no plan is established for the event, ALL non-military personnel shall be avoided by 300 meters when weapons (with simulated ammunition) are being used, and shall be avoided by 100 meters if no weapons are in use. All radiation emitters being operated by personnel other than the NTTR shall avoid operations within 300 meters of any personnel on the range. WARNING: Weapon use with live ammunition is not covered by this attachment and will require additional consideration by NTTR/SE and NTTR/DO

**A25.7. Protection**. Determining the method(s)/procedure(s) of protecting Non-Physical Participants and Non-Associated Personnel is the responsibility of the coordinating range user (fiscal and operational) and must be coordinated with and approved by NTTR/DO and NTTR/SE prior to event execution. The specific method(s)/procedure(s) is not directed in this attachment to provide the maximum flexibility for the event to provide the most realistic training while still maintaining the safety of all personnel. This attachment shall be addressed for ALL range events involving ground parties of any kind.

**A25.8. Procedures.** The methods/procedures developed for the event will be coordinated/approved in association with the NTTR Business Management Process (BMP) and its timeline, and a Memorandum of Agreement (MOA) will be used as the coordination and approval mechanism for compliance with this attachment. The unsigned MOA will be submitted to the NTTR Project Officer for the event no later than the Initial Planning Team (IPT) meeting, roughly about a week after the Program Introduction Document (PID) is submitted. This MOA is required at the IPT to facilitate the initial dialog necessary for coordination and approval at the NTTR.

A25.8.1. The MOA shall include at a minimum include the information in table A25.1:

**Table A25.1. Ground Party Interaction MOA Information.**

| |
|---|
| • Physical Participants – the organization(s)/company(ies) participating as Physical Participants in the event.<br>• Identification method – to identify who is a Physical Participant vs a Non-Physical Participant/Non-Associated Personnel.<br>• Detail how Non-Associated Personnel will be kept safe from event operations.<br>• Stand-off Distance from each type of person. Physical/Non-Physical Participant, Associated/Non-Associated Personnel.<br>• How personnel will be notified of these measures/ROEs.<br>• This includes Non-Associated Personnel that may be on the range(s) during event execution.<br>• Identify any form of White Force controller/observers that may be included in ground events.<br>• If weapons are used, the type of ammunition (simulated only) and its required stand-off distance.<br>• References to the specific manual/instructions/directives that define the stand-off distance are required.<br>• Identify all radiation emitters (RF, LASER, etc.) if any are to be used (non-NTTR operated only, NTTR-operated emitters already meet the requirements/intent of this attachment).<br>• References to the specific manual/instructions/directives that define the hazard distance(s) and required placarding, if any.<br>• Detail how the operators will ensure the required stand-off distance and/or placards will be in place prior to any operations of the emitter.<br>• Any other deconfliction/ID method to be used, in detail. |

## Attachment 26

## CLOSE AIR SUPPORT (CAS) AND COMBAT SEARCH AND RESCUE (CSAR) OPERATIONS ON THE NTTR

**A26.1. Guidance**. Most CAS and CSAR operations are covered under JFIRE, JPUB 3-09.3 and DAFMAN 11-214. This appendix serves in addition to those documents, and is intended only to apply to CAS and CSAR missions employing ordnance on the NTTR.

**A26.2. Definitions:**

A26.2.1 Associated Ground Party: Locally defined as one or more persons whose mission duties require presence on a numbered range with the TAC Team, or whose capabilities are required by the CAS or CSAR mission.

A26.2.2. Cold Range: Locally defined as times before and after the event and other times as necessary when aircraft and ground munitions are "safed". Only during these cold range times may TAC personnel transit the ranges required for the event.

A26.2.3. Dry Operations: Locally defined as operations that do not require ordnance.

A26.2.4. Ground Mission Commander: Locally defined as a member of the TAC Team who is responsible for all operational planning, validating, and execution phases of the HSMT mission. The GMC will maintain "2-way communication" with all ground parties and aircraft and will be co-located at the HSMT control site.

A26.2.5 Hot Range: Locally defined as those times when a TAC Team assumes control of all ranges required for an event. The ranges will remain hot until suspension of all munitions and/or munitions are safed.

A26.2.6. Non-associated Ground Party: Locally defined as ground personnel who are not directly involved with the CAS or CSAR mission. Non-associated ground parties are not allowed on a "hot range".

A26.2.7. Positive Control: Locally defined as the ability to instantly communicate instructions visibly or audibly.

A26.2.8. TAC: Locally defined as a Terminal Attack Controller (TAC) who is a currently qualified Joint Terminal Attack Controller (JTAC) or Forward Air Controller (Airborne) [FAC (A)], or service equivalent. For CSAR missions only, the TAC may also be a call-for-fire qualified individual IAW DAFMAN 11-214.

A26.2.9. TAC Team: Locally defined as one or more TACs. At least one member of this team must be in continuous, uninterrupted communication with a team member and all aircraft being controlled.

A26.2.10. Threat System: Locally defined as a manned vehicle or site whose purpose is to simulate the presence of enemy surface-to-air combat capability. Threat systems are always non-associated ground parties.

**A26.3. TAC responsibilities.**

A26.3.1. The TAC is responsible for mission planning to include knowing what ranges and airspace are required for the execution of the mission. Under no circumstances will the TAC or an associated ground party enter a hot range that the TAC does not control, nor allow a non-associated ground party (or TAC) to enter hot ranges under their control. Associated ground parties must be under positive control of the TAC at all times.

A26.3.2. When the range is "hot" the TAC is responsible for the air and ground parties operating on that range. The TAC must ensure all personnel adhere to Minimum Safe Distances listed in the JFIRE.

A26.3.3. When employing ground-to-ground ordnance with the CAS mission, the TAC team that owns the range is responsible for making all "calls for fire".

**A26.4. Range Control.** Unless otherwise informed by the TAC, ranges are defined to be hot during CAS or CSAR scheduled range periods, in compliance with range restrictions. Blackjack will not authorize non-associated ground parties onto CAS or CSAR scheduled ranges without notification that the range is cold.

## Attachment 27

## STRANGER AIRCRAFT PROCEDURES

### A27.1. Stranger Aircraft Procedures.

A27.1.1 If apparent unscheduled STRANGER is encountered, immediately ensure safe flight deconfliction then notify other scheduled participants via safety of flight frequencies. Once deconfliction is accomplished, notify Blackjack on their primary frequency with the number, type, location, altitude, direction of travel, estimated airspeed, winds aloft, and any relevant remarks. Use the codewords in table A27.1 to pass type:

**Table A27.1. Stranger Traffic Codewords.**

| Aircraft Category | Codeword |
|---|---|
| Multi-Copter UAS | Mosquito |
| Winged UAS | Cicada |
| Parachute | Squirrel |
| Balloon (With Payload) | Hippo |
| Sky Trash (No Payload) | Raccoon |
| Other/Unknown | Chupacabra |
| GA, Airline, Heli, etc | No codeword req'd |

A27.1.2 Example communication: "BLACKJACK, FLANKER, STRANGER traffic, 1x Hippo, Center of 75E, 25,000, TRACK East, SLOW, wind 300/100, passed within 100 feet."

A27.1.3 After notification, Blackjack will execute the Intruder Notification Checklist (not required for "sky trash"). After landing aircrew will contact 57 WG/SE and debrief unit intel as required.